

Wave Optics

Question1

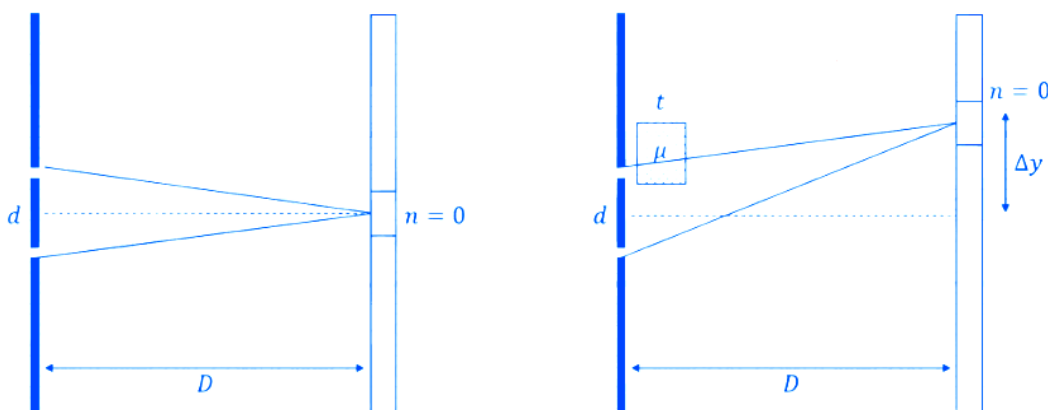
In a Young's double slit experiment set up, the two slits are kept 0.4 mm apart and screen is placed at 1 m from slits. If a thin transparent sheet of thickness $20 \mu\text{m}$ is introduced in front of one of the slits then center bright fringe shifts by 20 mm on the screen.

The refractive index of transparent sheet is given by $\frac{\alpha}{10}$, where α is ____.

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Answer: 14

Solution:



When a transparent sheet of thickness t and refractive index μ is placed in front of one slit, the optical path length of the light from that slit increases.

Extra path length introduced is $\Delta x = (\mu - 1)t \dots (i)$

In YDSE, the path length at a point is given as $\Delta x = d \sin \theta \Rightarrow \sin \theta = \frac{\Delta x}{d} \dots (ii)$

For small angle, $\sin \theta \approx \tan \theta \approx \theta$ (in radians).

Also, $\tan \theta = \frac{\Delta y}{D} \dots (iii)$

From (i), (ii) and (iii);

$$\frac{\Delta y}{D} = \frac{\Delta x}{d}$$

$$\Rightarrow \Delta y = \frac{D\Delta x}{d} = \frac{D(\mu-1)t}{d}$$

$$\Rightarrow \Delta y = \frac{D(\mu-1)t}{d} \Rightarrow (\mu-1) = \frac{\Delta y \cdot d}{D \cdot t}$$

$$\Rightarrow \mu = \frac{\Delta y \cdot d}{D \cdot t} + 1$$

Where :

- D = distance between the slits and the screen = 1 m.
- d = distance between the two slits = 0.4 mm = 0.4×10^{-3} m.
- Sheet thickness is t = 20μ m = 20×10^{-6} m
- Fringe shift is $\Delta y = 20$ mm = 20×10^{-3} m

Substituting these values,

$$\mu = \frac{(20 \times 10^{-3}) \times (0.4 \times 10^{-3})}{1 \times (20 \times 10^{-6})} + 1$$

$$\Rightarrow \mu = \frac{8 \times 10^{-6}}{20 \times 10^{-6}} + 1$$

$$\Rightarrow \mu = 0.4 + 1 = 1.4 = \frac{14}{10}$$

$$\Rightarrow \frac{14}{10} = \frac{\alpha}{10} \Rightarrow \alpha = 14$$

Therefore, the value of α is 14 .

Question2

In two separate Young's double-slit experimental set-ups and two monochromatic light sources of different wavelengths are used to get fringes of equal width. The ratios of the slits separations and that of the wavelengths of light used are 2 : 1 and 1 : 2 respectively. The corresponding ratio of the distances between the slits and the respective screens (D_1/D_2) is ____.

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Answer: 4

Solution:

In two Young's Double Slit Experiments (YDSE) the fringe widths are equal ($\beta_1 = \beta_2$).

Ratio of slit separations (d) is $\frac{d_1}{d_2} = \frac{2}{1} = 2$

Ratio of wavelengths (λ) is $\frac{\lambda_1}{\lambda_2} = \frac{1}{2}$

The formula for fringe width (β) in a YDSE setup is:

$$\beta = \frac{\lambda D}{d}$$

where:

- λ is the wavelength of light.
- D is the distance between the slits and the screen.
- d is the separation between the slits.

$$\beta_1 = \beta_2 \Rightarrow \frac{\lambda_1 D_1}{d_1} = \frac{\lambda_2 D_2}{d_2}$$

$$\Rightarrow \frac{D_1}{D_2} = \left(\frac{\lambda_2}{\lambda_1} \right) \times \left(\frac{d_1}{d_2} \right)$$

Since $\frac{\lambda_1}{\lambda_2} = \frac{1}{2}$, so, $\frac{\lambda_2}{\lambda_1} = 2$

Putting the values into the equation :

$$\frac{D_1}{D_2} = (2) \times (2) = 4$$

The corresponding ratio of the distances between the slits and the respective screens is 4 : 1.

Hence, the correct answer is **4**.

Question3

A beam of light consisting of wavelengths 650 nm and 550 nm illuminates the Young's double slits with separation of 2 mm such that the interference fringes are formed on a screen, placed at a distance of 1.2 m from the slits. The least distance of a point from the central maximum, where the bright fringes due to both the wavelengths coincide, is _____ $\times 10^{-5}$ m.

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Answer: 429

Solution:

The wavelength of first light is $\lambda_1 = 650 \text{ nm} = 650 \times 10^{-9} \text{ m}$

The wavelength of second light is $\lambda_2 = 550 \text{ nm} = 550 \times 10^{-9} \text{ m}$

Slit separation is $d = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

Screen distance is $D = 1.2 \text{ m}$

For bright fringes of different wavelengths to coincide at the same distance (y) from the central maximum, their positions must be equal :

$$y = \frac{n_1 \lambda_1 D}{d} = \frac{n_2 \lambda_2 D}{d}$$

$$n_1 \lambda_1 = n_2 \lambda_2$$

Where n_1 and n_2 are the orders of the bright fringes for λ_1 and λ_2 respectively.

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{550}{650} = \frac{11}{13}$$

Since we are looking for the least distance, we take the smallest integers that satisfy this ratio $n_1 = 11$ and $n_2 = 13$.

This means the 11th bright fringe of 650 nm coincides with the 13th bright fringe of 550 nm .

Using $n_1 = 11$ and $\lambda_1 = 650 \text{ nm}$:

$$\Rightarrow y = \frac{n_1 \lambda_1 D}{d}$$

$$\Rightarrow y = \frac{11 \times (650 \times 10^{-9}) \times 1.2}{2 \times 10^{-3}} \text{ m} = 429 \times 10^{-5} \text{ m}$$

The least distance from the central maximum where the bright fringes coincide is $429 \times 10^{-5} \text{ m}$.

Hence, the correct answer is 429 .

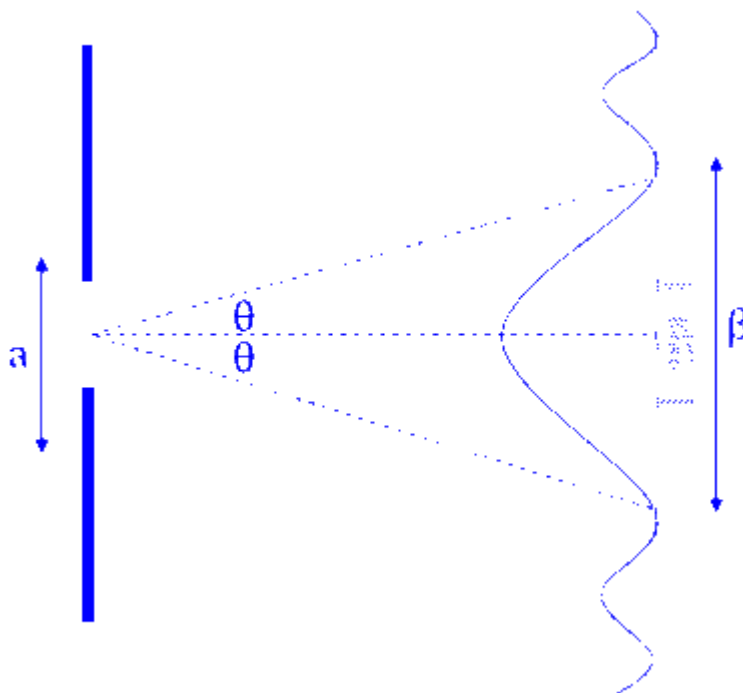
Question4

In single slit diffraction pattern, the wavelength of light used is 628 nm and slit width is 0.2 mm, the angular width of central maximum is $\alpha \times 10^{-2}$ degrees. The value of α is _____.

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Answer: 36

Solution:



In single slit diffraction, the condition for the n^{th} minimum is:

$$a \sin \theta = n\lambda$$

Where, λ is the wavelength of incident light on slit width of a .

For the first minimum ($n = 1$),

Assuming the angle θ is small such that $\sin \theta \approx \theta$ (in radians):

$$\theta = \frac{\lambda}{a}$$

The central maximum extends from the first minimum on one side to the first minimum on the other side.

Therefore, the total angular width (2θ) in radians is:

$$2\theta = \frac{2\lambda}{a}$$

To convert the angular width from radians to degrees, we multiply by $\frac{180}{\pi}$:

$$\text{Width in degrees} = \left(\frac{2\lambda}{a} \right) \times \frac{180}{\pi}$$

The wavelength of light is $\lambda = 628 \text{ nm} = 628 \times 10^{-9} \text{ m}$

The slit width is $a = 0.2 \text{ mm} = 0.2 \times 10^{-3} \text{ m}$

So,

$$\beta = \frac{2 \times 628 \times 10^{-9}}{0.2 \times 10^{-3}} \times \frac{180}{3.14}$$

$$\beta = \frac{1256 \times 10^{-6}}{0.2} \times 57.32$$

$$\beta = 6280 \times 10^{-6} \times 57.32 \approx 0.36 \text{ degrees}$$

$$= \alpha \times 10^{-2} \text{ degrees} = 0.36 \text{ degrees}$$

$$36 \times 10^{-2} = \alpha \times 10^{-2}$$

$$\alpha = 36$$

Therefore, the value of α is 36. Thus, the correct answer is 36.

Question5

In a Young's double slit experiment, the intensity at some point on the screen is found to be $\frac{3}{4}$ times of the maximum of the interference pattern. The path difference between the interfering waves at this point is $\frac{\lambda}{x}$ where λ is wavelength of the incident light. The value of x is _____.

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Answer: 6

Solution:

The resultant intensity I at a point where two waves of equal intensity I_0 interfere is given by:

$$I = I_{\max} \cos^2 \left(\frac{\phi}{2} \right)$$

Where $I_{\max}$ is the maximum intensity ($4I_0$) and ϕ is the phase difference between the waves.

The intensity at some point on screen is given as $I = \frac{3}{4} I_{\max}$:

$$\frac{3}{4} I_{\max} = I_{\max} \cos^2 \left(\frac{\phi}{2} \right)$$

$$\cos^2\left(\frac{\phi}{2}\right) = \frac{3}{4}$$

$$\cos\left(\frac{\phi}{2}\right) = \frac{\sqrt{3}}{2}$$

$$\frac{\phi}{2} = \frac{\pi}{6}$$

$$\phi = \frac{\pi}{3}$$

The phase difference ϕ is related to the path difference Δp by the formula:

$$\phi = \frac{2\pi}{\lambda} \Delta p$$

Substituting the value of ϕ :

$$\frac{\pi}{3} = \frac{2\pi}{\lambda} \Delta p$$

$$\Delta p = \frac{\pi}{3} \times \frac{\lambda}{2\pi}$$

$$\Delta p = \frac{\lambda}{6} = \frac{\lambda}{x} \implies x = 6$$

Therefore, the value of x is 6. So, the correct answer is 6.

Question6

In a double slit experiment, when one of the slits is covered by a transparent mica sheet of refractive index 1.56 , the central fringe shifts to the position of 7th bright fringe, obtained with both slits uncovered. If the light source wavelength is 450 nm , the thickness of mica sheet is $\alpha \times 10^{-9}$ m. The value of α is

_____ .

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Answer: 5625

Solution:

When a transparent sheet of thickness t and refractive index μ is placed in the path of one of the beams, the light travels a distance t in the medium instead of air.

The optical path length in the sheet is μt .

The extra path difference introduced (Δp) is:

$$\Delta p = \mu t - t = (\mu - 1)t$$

A bright fringe occurs when the path difference is an integer multiple of the wavelength (λ) :

$$\Delta P = m\lambda$$

Where m is the order of the fringe.

Since the central fringe (where path difference was zero) shifts to the position of the 7th bright fringe ($m = 7$), the introduced path difference must be equal to 7λ :

$$(\mu - 1)t = 7\lambda$$

The refractive index of mica, is $\mu = 1.56$.

The order of fringe shift, is $m = 7$.

The wavelength of light is $\lambda = 450 \text{ nm} = 450 \times 10^{-9} \text{ m}$.

Substituting the values into the formula:

$$(1.56 - 1)t = 7 \times 450 \times 10^{-9}$$

$$\Rightarrow 0.56t = 3150 \times 10^{-9}$$

$$\Rightarrow t = \frac{3150 \times 10^{-9}}{0.56}$$

$$\Rightarrow t = 5625 \times 10^{-9} \text{ m} = \alpha \times 10^{-9} \text{ m}$$

$$\Rightarrow \alpha = 5625$$

Therefore, the value of α is 5625 .

Question7

An unpolarized light of intensity I_0 passes through polarizer and then through a certain optically active solution and finally it goes to analyser. If the angle between analyser and polariser is 0° and intensity of light emerged from analyser is $\frac{3}{8} I_0$, the angle of rotation of the light by the solution with respect to analyser is ____ degrees.

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Answer: 30

Solution:

When unpolarized light of intensity I_0 passes through a polarizer, the intensity of the transmitted polarized light (I_1) is always half of the initial intensity.

$$I_1 = \frac{I_0}{2}$$

An optically active solution rotates the plane of polarization by a certain angle. Let the angle of rotation is θ . After passing through the solution, the light is still plane-polarized, but its axis has shifted by θ .

Using Malus's Law when completely plane-polarized light is incident on an analyzer, the intensity I of the light transmitted by the analyzer is directly proportional to the square of the cosine of the angle between the transmission axes of the analyzer and the polarizer.

$$I = I_{\text{incident}} \cos^2(\phi)$$

Here, ϕ is the final angle between the light's polarization plane (rotated by the solution) and the analyzer's axis.

The initial intensity after polarizer is $I_1 = \frac{I_0}{2}$ and the final intensity after analyser is $I = \frac{3}{8} I_0$.

Since the analyzer and polarizer were originally at 0° , the angle ϕ in Malus's law is the rotation angle θ caused by the solution.

Substituting the values into Malus's Law:

$$\frac{3}{8} I_0 = \left(\frac{I_0}{2} \right) \cos^2 \theta$$

$$\Rightarrow \frac{3/8}{1/2} = \cos^2(\theta)$$

$$\Rightarrow \frac{3}{4} = \cos^2(\theta)$$

$$\Rightarrow \cos(\theta) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = 30^\circ$$

Therefore, the angle of rotation is 30 degrees.

Question8

Some distant star is to be observed by some telescope of diameter of objective lens a , at an angular resolution of 3.0×10^{-7} radian. If the wavelength of light from the star reaching the telescope is 500 nm , the minimum diameter of the objective lens of the telescope is _____ cm. (nearest interger)

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Answer: 203

Solution:

According to Rayleigh criterion the angular resolution (θ) of a telescope is the smallest angle between two distant objects that can be seen as separate points. For a circular aperture, the formula is:

$$\theta = \frac{1.22\lambda}{a}$$

Where θ = Angular resolution in radians, λ = Wavelength of the light and a = Diameter of the objective lens.

We need to find the diameter of the objective lens,

$$a = \frac{1.22\lambda}{\theta}$$

The wavelength of light is $\lambda = 500 \text{ nm} = 500 \times 10^{-9} \text{ m}$.

The angular resolution of telescope is $\theta = 3.0 \times 10^{-7} \text{ rad}$.

Substituting the values,

$$a = \frac{1.22 \times (500 \times 10^{-9})}{3.0 \times 10^{-7}}$$

$$\Rightarrow a = \frac{610 \times 10^{-9}}{3.0 \times 10^{-7}}$$

$$\Rightarrow a = \frac{610}{3} \times 10^{-2} \text{ m}$$

$$\Rightarrow a \approx 203.33 \times 10^{-2} \text{ m}$$

$$\Rightarrow a = 2.0333 \text{ m} \times 100$$

$$\Rightarrow a = 203.33 \text{ cm}$$

Therefore, the minimum diameter of the objective lens is 203 cm .

Question9

In a double slit experiment the distance between the slits is 0.1 cm and the screen is placed at 50 cm from the slits plane. When one slit is covered with a transparent sheet having thickness t and refractive index $n(= 1.5)$, the central fringe shifts by 0.2 cm . The value of t is _____ cm.

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Options:

A.

$$6.0 \times 10^{-3}$$

B.

$$8 \times 10^{-4}$$

C.

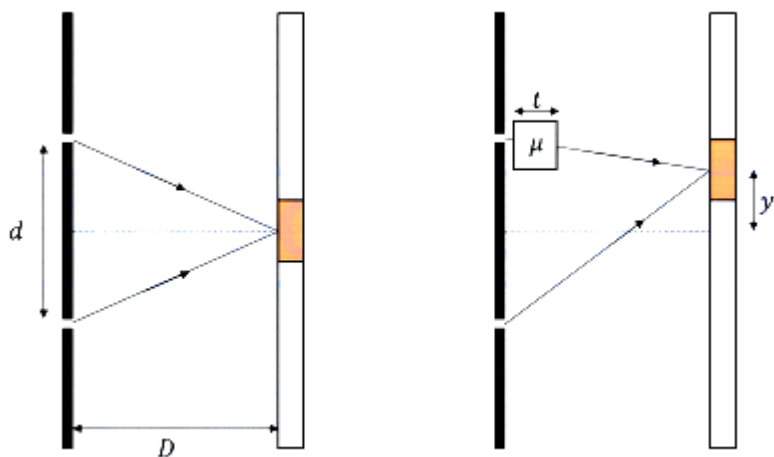
$$5.0 \times 10^{-3}$$

D.

$$5.6 \times 10^{-4}$$

Answer: B

Solution:



In a Young's Double Slit Experiment (YDSE), the position of the fringes is determined by the path difference between the two waves. When a transparent sheet of thickness t and refractive index n is placed in front of one slit, the light travels through the medium instead of air for that distance.

The optical path through the sheet is nt , while the path in air for the same distance would be t . Thus, an additional path difference (Δx) is introduced :

$$\Delta x_{\text{optical}} = nt - t = (n - 1)t$$

The central fringe which originally occurs at the centre where the geometric path difference is zero, shifts to a new position Δy where the net path difference is zero.

The geometric path difference for a point at distance y from the center is :

$$\Delta x_{\text{geo}} = \frac{d \cdot y}{D}$$

Where d is the distance between slits and D is the distance to the screen.

So, at the position of central maximum net path difference must be zero, $\Delta x_{\text{optical}} + \Delta x_{\text{geo}} = 0$

$$\frac{d \cdot y}{D} = (n - 1)t$$

$$\Rightarrow t = \frac{y \cdot d}{D(n-1)}$$

Putting the values,

- Distance between slits, $d = 0.1 \text{ cm}$
- Distance to screen $D = 50 \text{ cm}$
- Shift of central fringe $y = 0.2 \text{ cm}$
- Refractive index $n = 1.5$

$$t = \frac{0.2 \times 0.1}{50(1.5-1)}$$

$$\Rightarrow t = \frac{0.02}{25}$$

$$\Rightarrow t = 8 \times 10^{-4} \text{ cm}$$

Therefore, the thickness of the sheet to be $8 \times 10^{-4} \text{ cm}$.

Hence, the correct option is (B).

Question10

Given below are two statements :

Statement I : In a Young's double slit experiment, the angular separation of fringes will increase as the screen is moved away from the plane of the slits

Statement II : In a Young's double slit experiment, the angular separation of fringes will increase when monochromatic source is replaced by another monochromatic source of higher wavelength

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both **Statement I** and **Statement II** are true

B.

Statement I is true but **Statement II** is false

C.

Statement I is false but **Statement II** is true

D.

Both **Statement I** and **Statement II** are false

Answer: C

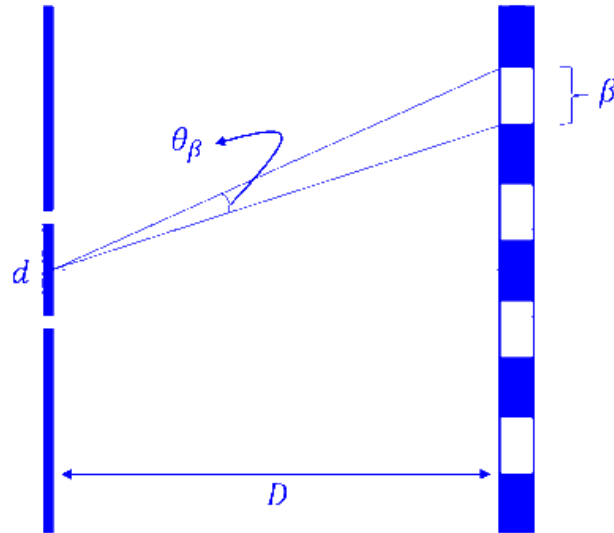
Solution:

In YDSE, the linear position of the n^{th} bright fringe on the screen is given by :

$$y_n = \frac{n\lambda D}{d}$$

Where :

- λ = wavelength of light used.
- D = distance between the slits and the screen.
- d = distance between the two slits.



The linear fringe width (β), which is the distance between two consecutive bright or dark fringes, is :

$$\beta = y_{n+1} - y_n = \frac{\lambda D}{d}$$

The angular separation (or angular fringe width) θ_β is the angle subtended by the fringe width at the slits. For small angles :

$$\theta_\beta \approx \frac{\beta}{D}$$

Substituting the value of β :

$$\theta_\beta = \frac{\left(\frac{\lambda D}{d}\right)}{D} = \frac{\lambda}{d}$$

Statement I says that angular separation (θ_β) increases as the screen is moved away (D increases).

From the formula $\theta_\beta = \lambda/d$, we can see that θ_β depends only on the wavelength (λ) and the slit separation (d).

$$\theta_\beta \propto \lambda \dots \text{(i)}, \quad \theta_\beta \propto \frac{1}{d} \dots \text{(ii)}$$

It is independent of the distance to the screen (D).

Therefore, Statement I is false.

Statement II says that θ increases when a source of higher wavelength is used.

From the formula $\theta_\beta \propto \lambda$, i.e., if λ increases, θ must also increase.

Therefore, Statement II is true.

Therefore, Statement I is false and Statement II is true.

Hence, the correct option is (C).

Question11

Which of the following are true for a single slit diffraction?

- A. Width of central maxima increases with increase in wavelength keeping slit width constant.**
- B. Width of central maxima increases with decrease in wavelength keeping slit width constant.**
- C. Width of central maxima increases with decrease in slit width at constant wavelength.**
- D. Width of central maxima increases with increase in slit width at constant wavelength.**
- E. Brightness of central maxima increases for decrease in wavelength at constant slit width.**

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Options:

A.

B, D only

B.

A, D only

C.

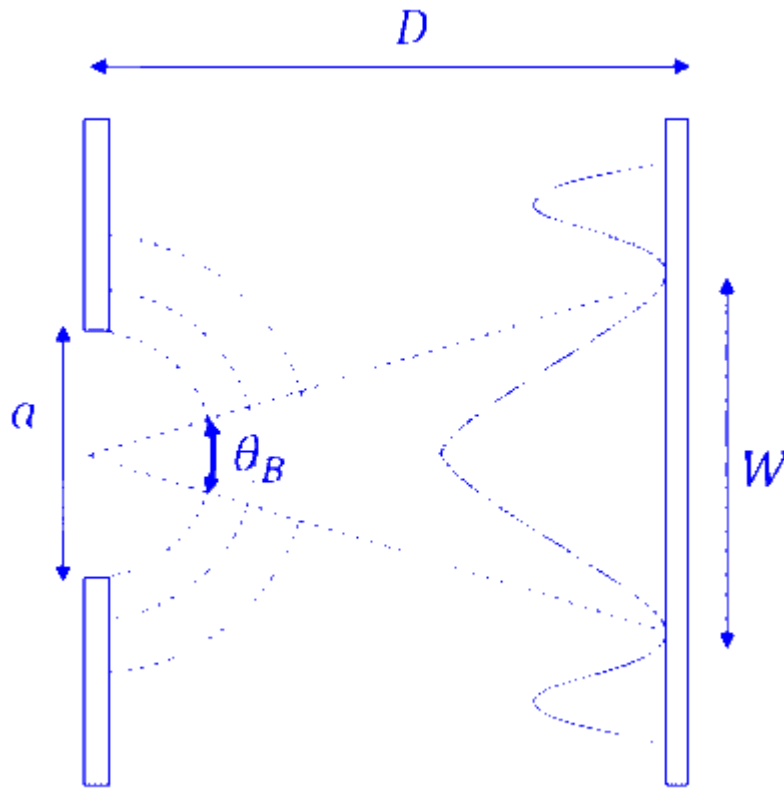
A, C, E only

D.

B, C only

Answer: C

Solution:



In a single slit diffraction experiment the linear width (W) of the central maximum on a screen at distance D is given by the formula:

$$W = \frac{2\lambda D}{a}$$

Where :

- 1. λ is the wavelength of the light.
- 2. a is the width of the slit.

From the formula, the width of the central maximum is directly proportional to the wavelength ($W \propto \lambda$) when the slit width (a) is kept constant.

This means if the wavelength increases, the width of the central maximum also increases. Therefore, Statement A is true and Statement B is false.

From the formula, the width of the central maximum is inversely proportional to the slit width ($W \propto \frac{1}{a}$) when the wavelength (λ) is kept constant.

This means if the slit width decreases, the width of the central maximum increases. Therefore, Statement C is true and Statement D is false.

The peak intensity (brightness) of the central maximum depends on how the superposition of waves on screen occurs. For a given incident intensity, the power passing through the slit is constant for a fixed slit width. When the wavelength decreases, the central maximum becomes narrower (the angular width decreases). Because the exact same amount of energy is now concentrated into a smaller area, the peak brightness of the central maximum increases. Therefore, Statement E is true.

Hence, the true statements are **A**, **C**, and **E**.

Question12

The wavelength of light, while it is passing through water is 540 nm . The refractive index of water is $4/3$. The wavelength of the same light when it is passing through a transparent medium having refractive index of $3/2$ is _____ nm.

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Options:

A.

540

B.

840

C.

480

D.

380

Answer: C

Solution:

When light travels from one medium to another, its frequency remains constant, but its velocity and wavelength change.

The absolute refractive index is defined as:

$$\mu = \frac{c}{v} = \frac{\lambda_{\text{vacuum}}}{\lambda_{\text{medium}}}$$

Therefore, for any two media, the product of the refractive index and the wavelength is constant:

$$\mu_1 \lambda_1 = \mu_2 \lambda_2$$

Wavelength in water, $\lambda_1 = 540 \text{ nm}$

Refractive index of water, $\mu_1 = 4/3$

Refractive index of the second medium, $\mu_2 = 3/2$

Using the relation $\mu_1 \lambda_1 = \mu_2 \lambda_2$,

$$\lambda_2 = \frac{\mu_1 \lambda_1}{\mu_2}$$

$$\Rightarrow \lambda_2 = \frac{\frac{4}{3} \times 540}{3/2} = \frac{4 \times 540 \times 2}{3 \times 3} = 480 \text{ nm}$$

Therefore, the wavelength of the light in the second transparent medium is 480 nm .

Hence, the correct option is (C).

Question13

When an unpolarized light falls at a particular angle on a glass plate (placed in air), it is observed that the reflected beam is linearly polarized. The angle of refracted beam with respect to the normal is _____ .

($\tan^{-1}(1.52) = 57.7^\circ$, refractive indices of air and glass are 1.00 and 1.52, respectively.)

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Options:

A.

36.3°

B.

39.6°

C.

42.6°

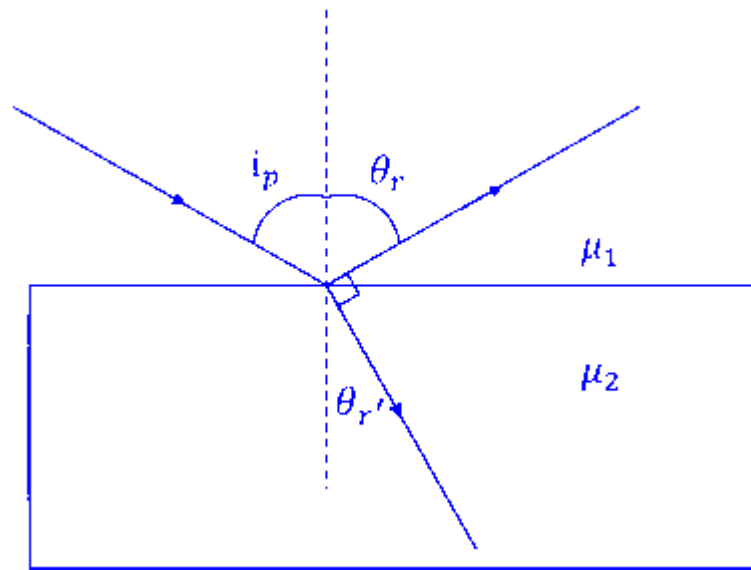
D.

32.3°

Answer: D

Solution:

When unpolarized light strikes an interface at a specific angle called the Brewster angle (i_B or i_p), the reflected light is completely linearly polarized. At this specific angle, the reflected ray and the refracted ray are perpendicular (90°) to each other.



- 1. i_p = Angle of incidence (polarizing angle)
- 2. θ_r = Angle of reflection
- 3. $\theta_{r'}$ = Angle of refraction

Since the angle of reflection equals the angle of incidence :

$$\theta_r = i_p$$

Since the angle between the reflected and refracted ray is 90° :

$$i_p + 90^\circ + \theta_{r'} = 180^\circ$$

$$\Rightarrow \theta_{r'} = 90^\circ - i_p$$

Applying Snell's Law

$$\mu_1 \sin(i_p) = \mu_2 \sin(\theta_{r'})$$

$$\Rightarrow \mu_1 \sin(i_p) = \mu_2 \sin(90^\circ - i_p)$$

$$\Rightarrow \mu_1 \sin(i_p) = \mu_2 \cos(i_p)$$

$$\Rightarrow \tan(i_p) = \frac{\mu_2}{\mu_1}$$

If the first medium is air ($\mu_1 \approx 1$), the formula simplifies to :

$$\tan(i_p) = \mu$$

Substituting the values:

$$\tan(i_p) = \frac{1.52}{1.00} = 1.52$$

$$\Rightarrow i_p = \tan^{-1}(1.52) = 57.7^\circ$$

As established by the geometry of Brewster's condition :

$$i_p + \theta_{r'} = 90^\circ$$

$$\Rightarrow \theta_{r'} = 90^\circ - i_p$$

$$\Rightarrow \theta_{r'} = 90^\circ - 57.7^\circ$$

$$\theta_{r'} = 32.3^\circ$$

$\Rightarrow$ The angle of the refracted beam with respect to the normal is 32.3° . Hence, the correct option is (D).

Question14

In the Young's double slit experiment the intensity produced by each one of the individual slits is I_0 . The distance between two slits is 2 mm . The distance of screen from slits is 10 m . The wavelength of light is $6000 \text{ \AA}$. The intensity of light on the screen in front of one of the slits is _____

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Options:

A.

$$\frac{I_0}{2}$$

B.

$$I_0$$

C.

$$2I_0$$

D.

$$4I_0$$

Answer: B

Solution:

The intensity from each slit is equal, $I_1 = I_2 = I_0$

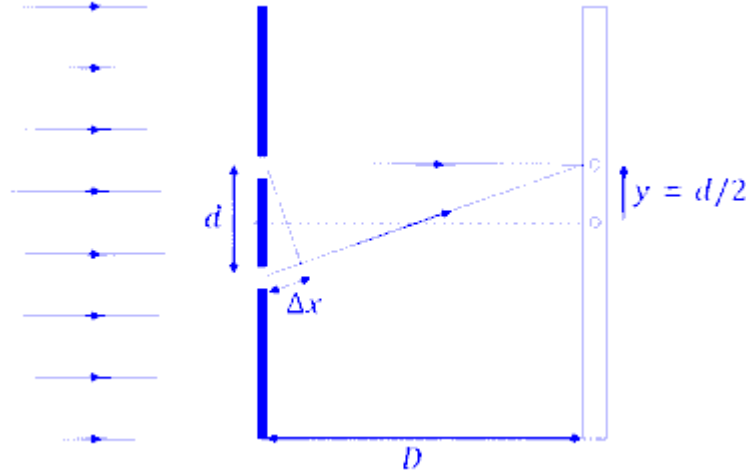
The slit width is equal to distance between slits $d = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

Distance of screen from slits $D = 10 \text{ m}$

Wavelength of light $\lambda = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m}$

The distance from the central maximum to a point directly in front of one slit is :

$$y = \frac{d}{2}$$



The path difference for a point at height y on the screen is given by the formula :

$$\Delta x = \frac{dy}{D}$$

$$\Rightarrow \Delta x = \frac{d(\frac{d}{2})}{D} = \frac{d^2}{2D}$$

$$\Delta x = \frac{(2 \times 10^{-3})^2}{2 \times 10} = \frac{4 \times 10^{-6}}{20} = 2 \times 10^{-7} \text{ m}$$

The relation between path difference and phase difference is :

$$\phi = \frac{2\pi}{\lambda} \times \Delta x$$

$$\Rightarrow \phi = \frac{2\pi}{6 \times 10^{-7}} \times (2 \times 10^{-7}) = \frac{4\pi}{6} = \frac{2\pi}{3}$$

The formula for resultant intensity when two waves of equal intensity I_0 interfere is :

$$I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$\Rightarrow I = 4I_0 \cos^2 \left(\frac{\frac{2\pi}{3}}{2} \right) = 4I_0 \cos^2 \left(\frac{\pi}{3} \right)$$

$$\Rightarrow I = 4I_0 \left(\frac{1}{2} \right)^2 = 4I_0 \times \frac{1}{4}$$

$$\Rightarrow I = I_0$$

The intensity of light on the screen directly in front of one of the slits is equal to the intensity of a single slit, i.e., I_0 .

Hence, the correct option is (B).

Question15

Given below are two statements :

Statement I : A plane wave after passing through prism remains as plane wave but passing through small pin hole may become spherical wave.

Statement II : The curvature of a spherical wave emerging from a slit will increase for increasing slit width.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are false

B.

Both Statement I and Statement II are true

C.

Statement I is true but Statement II is false

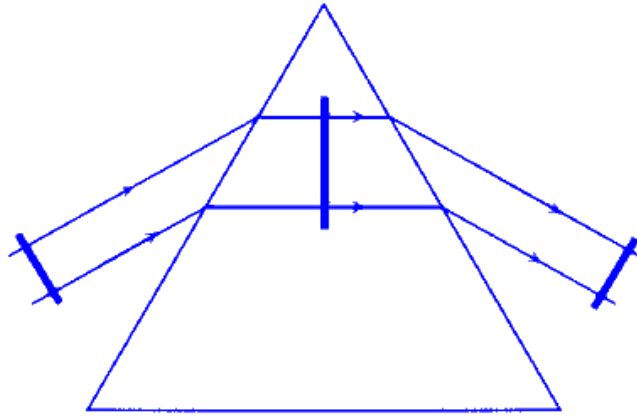
D.

Statement I is false but Statement II is true

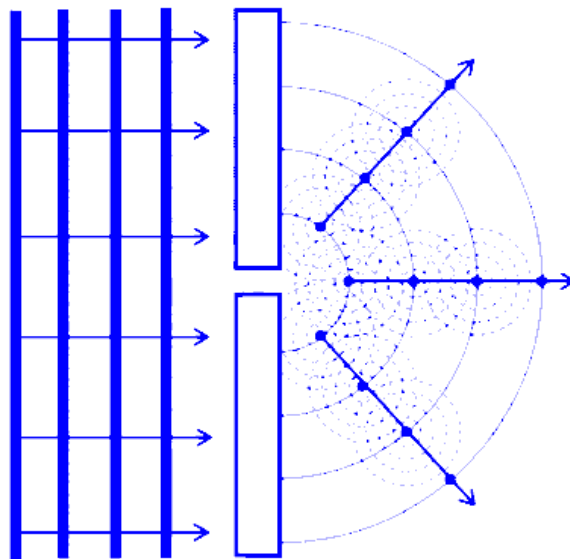
Answer: C

Solution:

When a plane wave-front (parallel rays) enters a prism, all parts of the wave-front travel at the same speed through the medium. While the wave-front is tilted due to refraction, it remains flat and parallel, thus remaining a plane wave.



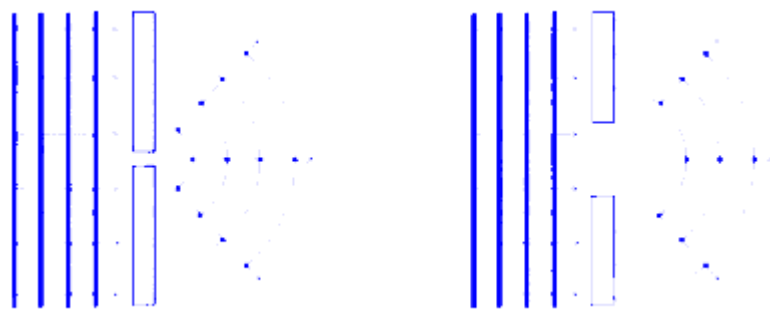
According to Huygens' Principle, every point on a wave-front acts as a source of secondary spherical wavelets. When a plane wave encounters a very small opening (pinhole) comparable to the wavelength of light, it undergoes diffraction. The emerging light spreads out in all directions, forming a spherical wave-front.



Hence, the statement I is true.

When light passes through a slit, the degree of spreading (diffraction) depends on the slit width (d) relative to the wavelength (λ).

Smaller slit widths lead to greater diffraction and more pronounced curvature. As the slit width increases, the light behaves more like a straight beam, meaning the wave-front becomes flatter.



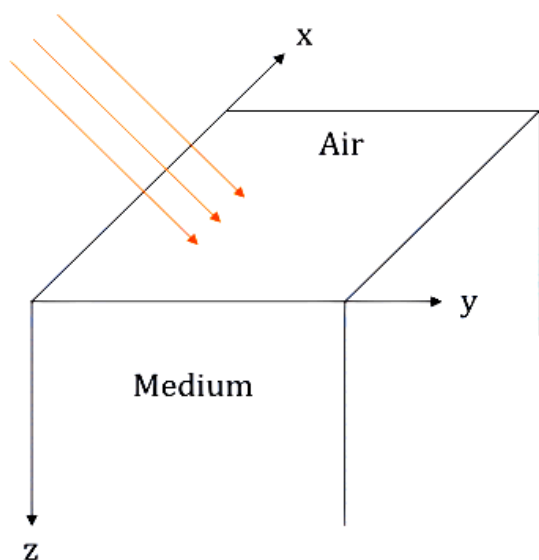
Curvature is the reciprocal of the radius of the wave-front, i.e., $\propto \frac{1}{r}$. So, a flatter wave has less curvature. Therefore, increasing the slit width decreases the curvature, it does not increase it.

Hence, the statement II is false.

Since Statement I is true and Statement II is false, so the correct option is (C).

Question16

An unpolarized light is incident on the plane interface of air-dielectric medium shown in figure. If the incident angle is equal to Brewster angle, identify the expression representing reflected wave.



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Options:

A.

$$(E_x \hat{i} + E_y \hat{j}) \sin(kx - kz - \omega t)$$

B.

$$(E_x \hat{i} + E_z \hat{k}) \sin(kx + ky - \omega t)$$

C.

$$(E_x \hat{j} + E_y \hat{k}) \sin(ky + kz - \omega t)$$

D.

$$(E_x \hat{i} + E_y \hat{j} + E_z \hat{k}) \sin(kx + ky - kz - \omega t)$$

Answer: A

Solution:

Brewster's Law states that when unpolarized light strikes a transparent medium at a specific angle called the Brewster angle (θ_p), the reflected light becomes completely plane-polarized. At this angle, the reflected ray and the refracted ray are perpendicular to each other.

For light incident at the Brewster angle, the reflected wave is polarized with its electric field vector (E) oscillating perpendicular to the plane of incidence. The plane of incidence is defined by the incident ray and the normal to the surface.

From the diagram, the interface lies in the xy -plane ($z = 0$). The incident light is traveling in the xz -plane

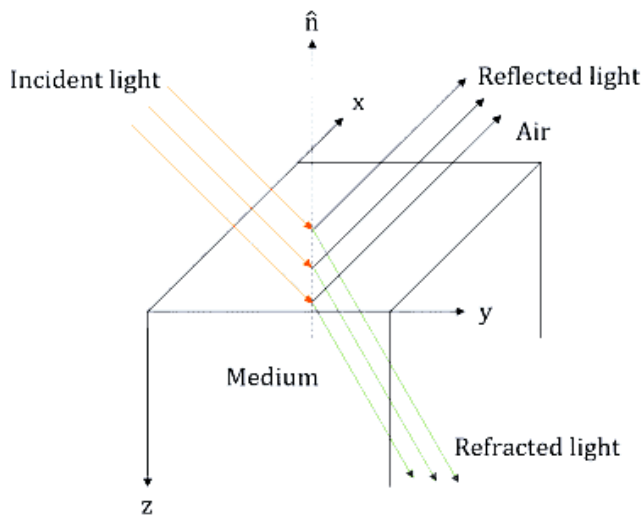
The normal to the surface is the z -axis.

Since the plane of incidence is the xz -plane, the electric field of the reflected light must be perpendicular to this plane, which means it must oscillate along the y -axis (direction $\hat{j}$).

A wave equation for an electromagnetic wave is generally represented as

$$\vec{E} = \vec{E}_0 \sin(\vec{k} \cdot \vec{r} - \omega t)$$

The reflected wave will be traveling back into the air (positive x and negative z direction relative to the interface).



So, in reflected rays the electric field should only have components perpendicular to the plane of incidence.

So, from the options, we need an expression where the electric field components perpendicular to the plane of incident rays.

Option (A) describes a wave with components E_x and E_y .

Therefore, the correct option is (A).

Question17

A telescope with objective diameter R is used to observe a distant star emitting light of wavelength 500 nm , at a resolution of 5×10^{-7} radian. The value of R is

_____ cm .

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

61

B.

122

C.

244

D.

305

Answer: B

Solution:

The ability of an optical instrument to distinguish between two closely spaced objects is defined by its resolving power.

According to the Rayleigh criterion, for a circular aperture of diameter R , the minimum angular separation ($\Delta\theta$) that can be resolved is given by the formula :

$$\Delta\theta = \frac{1.22\lambda}{R}$$

Where $\Delta\theta$ is the angular resolution in radians, λ is the wavelength of the incident light and R is the diameter of the objective lens or mirror.

We are given the wavelength of light as $\lambda = 500 \text{ nm} = 500 \times 10^{-9} \text{ m} = 5 \times 10^{-7} \text{ m}$

And angular resolution as $\Delta\theta = 5 \times 10^{-7} \text{ rad}$

$$R = \frac{1.22\lambda}{\Delta\theta}$$

Substituting the values :

$$R = \frac{1.22 \times (5 \times 10^{-7})}{5 \times 10^{-7}}$$

$$\Rightarrow R = 1.22 \text{ meters}$$

$$\Rightarrow R = 1.22 \times 100 \text{ cm} = 122 \text{ cm}$$

Therefore, the value of R is 122 cm .

Thus, the correct option is (B).

Question18

A slit of width a is illuminated by light of wavelength λ . The linear separation between 1st and 3rd minima in the diffraction pattern produced on a screen placed at a distance D from the slit system is _____ .

Options:

A.

$$\frac{D\lambda}{a}$$

B.

$$1.5 \frac{D\lambda}{a}$$

C.

$$2 \frac{D\lambda}{a}$$

D.

$$3 \frac{D\lambda}{a}$$

Answer: C

Solution:

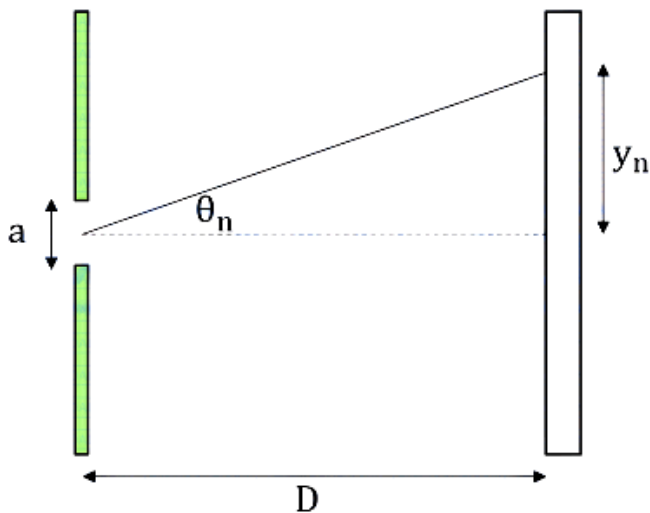
For a slit of width a , the condition for the n^{th} order minimum is that the path difference between the extreme rays is an integral multiple of the wavelength (λ):

$$a \sin \theta_n = n\lambda$$

Where $n = 1, 2, 3, \dots$

For small angles of diffraction (θ_n), we can approximate $\sin \theta_n \approx \theta_n$. Thus: $\theta_n \approx \frac{n\lambda}{a}$

The linear distance (y_n) of the n^{th} minimum from the central maximum on a screen at distance D is given by :



$$y_n = D \tan \theta_n \approx D\theta_n$$

Substituting the value of θ_n :

$$y_n = \frac{nD\lambda}{a}$$

We need to find the linear separation between the 1st minimum and the 3rd minimum on the same side of the central maximum.

$$\text{Position of 1}^{\text{st}} \text{ minimum (} n = 1 \text{) : } y_1 = \frac{1 \cdot D\lambda}{a}$$

$$\text{Position of 3}^{\text{rd}} \text{ minimum (} n = 3 \text{) : } y_3 = \frac{3 \cdot D\lambda}{a}$$

The linear separation (Δy) is :

$$\Delta y = y_3 - y_1$$

$$\Rightarrow \Delta y = \frac{3D\lambda}{a} - \frac{D\lambda}{a}$$

$$\Rightarrow \Delta y = \frac{2D\lambda}{a}$$

Therefore, the correct option is (C).

Question19

An unpolarized light of certain intensity passes through a combination of two polarizers whose transmission axes are at 30° and 90° , respectively, with respect to the horizontal axis. A third polarizer with its transmission axis at 60° with the horizontal axis is placed between the two existing polarizers. The ratio of the output intensities with and without the third polarizer is ____ .

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

$\frac{3}{4}$

B.

$\frac{4}{3}$

C.

9/4

D.

4/9

Answer: C

Solution:

When polarized light of intensity I_{in} passes through a polarizer, the intensity of the transmitted light (I_{out}) is given by Malus's Law

$$I_{\text{out}} = I_{\text{in}} \cos^2 \theta$$

Where θ is the angle between the polarization direction of the incident light and the transmission axis of the polarizer.

The initial intensity of unpolarised light = I_0 .

After the first polarizer (at 30°), the light becomes polarized with intensity $I_1 = \frac{1}{2}I_0$.

This light then hits the second polarizer (at 90°). The angle between them is $\theta = 90^\circ - 30^\circ = 60^\circ$.

So, the final intensity without third polarizer is,

$$I_{\text{without}} = I_2 = I_1 \cos^2 60^\circ = \left(\frac{1}{2}I_0\right)\left(\frac{1}{2}\right)^2 = \frac{1}{2}I_0 \times \frac{1}{4} = \frac{I_0}{8}$$

The intensity after the first polarizer (at 30°) is $I_1 = \frac{1}{2}I_0$.

This light hits the middle polarizer (at 60°). The angle is $\theta_{12} = 60^\circ - 30^\circ = 30^\circ$.

So, intensity after middle polarizer (I_3):

$$I_3 = I_1 \cos^2 30^\circ = \left(\frac{1}{2}I_0\right)\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{2}I_0 \times \frac{3}{4} = \frac{3I_0}{8}$$

This light hits the final polarizer (at 90°). The angle is $\theta_{23} = 90^\circ - 60^\circ = 30^\circ$.

So, final intensity after the final polarizer is

$$I_{\text{with}} = I_3 \cos^2 30^\circ = \left(\frac{3I_0}{8}\right)\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3I_0}{8} \times \frac{3}{4} = \frac{9I_0}{32}$$

So, the ratio of the output intensities with and without the third polarizer is,

$$\frac{I_{\text{with}}}{I_{\text{without}}} = \frac{\left(\frac{9I_0}{32}\right)}{\left(\frac{I_0}{8}\right)} = \frac{9}{4}$$

Therefore, the ratio of the output intensities with and without the third polarizer is 9/4.

Thus, the correct option is (C).

Question20

In Young's double slit experiment, the fringe width of the interference pattern produced on the screen is $2.4\mu\text{ m}$. If the experiment is carried out in another medium having refractive index 1.2 , the fringe width will be _____ μm .

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

1.2

B.

2

C.

2.4

D.

2.88

Answer: B

Solution:

In Young's Double Slit Experiment (YDSE), the fringe width (β) is given by :

$$\beta = \frac{\lambda D}{d}$$

Where, λ is the wavelength of light used, D is the distance between the slits and the screen, and d is the distance between the two slits.

When the experiment is moved from air/vacuum to a medium with a refractive index μ , the speed of light decreases, which in turn decreases the wavelength. The new wavelength λ' is:

$$\lambda' = \frac{\lambda}{\mu}$$

Since D and d are constant for the experimental setup, the new fringe width β' in the medium is :

$$\beta' = \frac{\lambda' D}{d} = \frac{\left(\frac{\lambda}{\mu}\right) D}{d} = \frac{1}{\mu} \left(\frac{\lambda D}{d}\right)$$

Therefore $\beta' = \frac{\beta}{\mu}$

The original fringe width (when setup is in air) is $\beta = 2.4\mu\text{ m}$.

The refractive index of the medium is $\mu = 1.2$.

Putting the values,

$$\beta' = \frac{2.4}{1.2} = 2.0\mu\text{ m}$$

Therefore, the correct option is (B).

Question21

The maximum intensity in a Young's double slit experiment is I_0 . Distance between the slits (d) is 5λ , where λ is the wavelength of light used. The intensity of the fringe, exactly opposite to one of the slits on the screen, placed at $D = 10d$ is ____ .

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

$$\frac{I_0}{4}$$

B.

$$\frac{I_0}{2}$$

C.

$$I_0$$

D.

$$\frac{3I_0}{4}$$

Answer: B

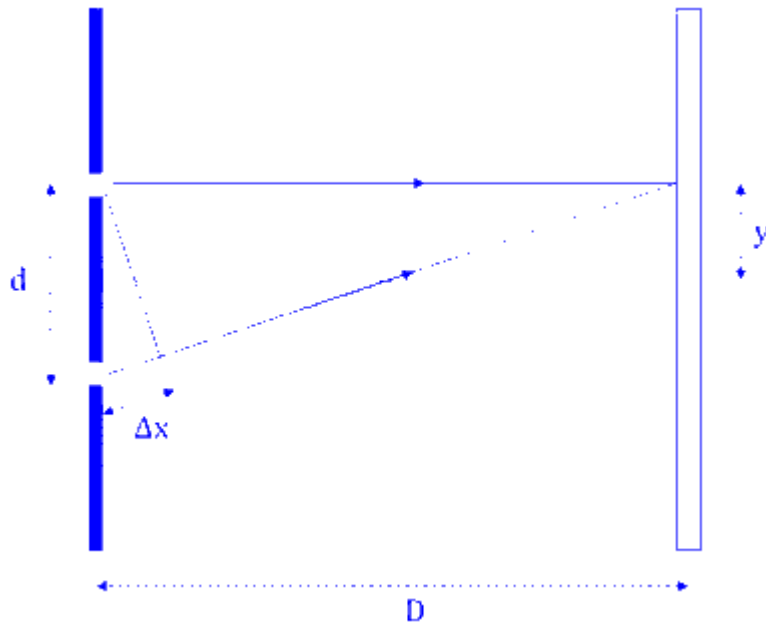
Solution:

A point exactly opposite to one of the slits is at a distance $y = \frac{d}{2}$ from the central maximum.

The slit separation is given as $d = 5\lambda$

The distance between slits and screen is $D = 10d = 50\lambda$

And the vertical distance of required point from the central point is $y = \frac{d}{2} = 2.5\lambda$



For a point on the screen, path difference is approximately:

$$\Delta x = y \cdot \frac{d}{D}$$

Substituting the given values:

$$\Delta x = \left(\frac{d}{2}\right) \cdot \frac{d}{10d} = \frac{d}{20}$$

Substituting $d = 5\lambda$:

$$\Delta x = \frac{5\lambda}{20} = \frac{\lambda}{4}$$

The relation between phase difference and path difference is :

$$\phi = \frac{2\pi}{\lambda} \cdot \Delta x$$

$$\Rightarrow \phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2} = 90^\circ$$

The intensity I at a point with phase difference ϕ is given by:

$$I = I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

Where I_0 is the maximum intensity of light at the central maxima.

Substituting $\phi = \frac{\pi}{2}$:

$$I = I_0 \cos^2 \left(\frac{\pi/2}{2} \right) = I_0 \cos^2 \left(\frac{\pi}{4} \right)$$

$$\Rightarrow I = I_0 \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{I_0}{2}$$

Therefore, the intensity of the fringe, exactly opposite to one of the slits on the screen is $\frac{I_0}{2}$.

Thus, the correct option is (B).

Question22

In interference experiment the path difference between two interfering waves at a point A on the screen is $\lambda/3$, where λ is the wavelength of these waves, and at another point B the path difference is $\lambda/6$. The ratio of intensities at points A and B is ____ .

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

3

B.

4

C.

1/3

D.

1/4

Answer: C

Solution:

The relation between phase difference (ϕ) and path difference (Δx) is:

$$\phi = \frac{2\pi}{\lambda} \Delta x$$

For two waves of equal intensity I_0 , the resultant intensity I at a point is:

$$I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

At point A the path difference is $\Delta x_A = \frac{\lambda}{3}$.

$$\phi_A = \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$$

So, the resultant intensity at A is,

$$I_A = 4I_0 \cos^2 \left(\frac{2\pi/3}{2} \right) = 4I_0 \cos^2 \left(\frac{\pi}{3} \right) = 4I_0 \left(\frac{1}{2} \right)^2 = I_0$$

At point B the path difference is $\Delta x_B = \frac{\lambda}{6}$.

$$\phi_B = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}$$

So, the resultant intensity at B is,

$$I_B = 4I_0 \cos^2 \left(\frac{\pi/3}{2} \right) = 4I_0 \cos^2 \left(\frac{\pi}{6} \right) = 4I_0 \left(\frac{\sqrt{3}}{2} \right)^2 = 3I_0$$

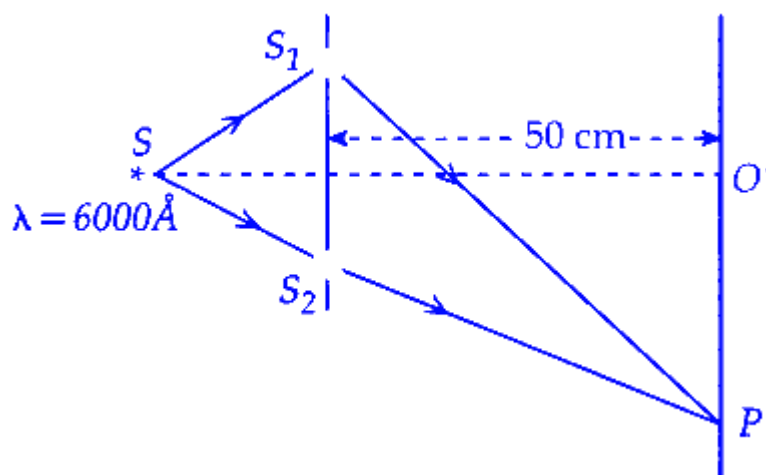
So, the ratio of intensities at A and B is;

$$\frac{I_A}{I_B} = \frac{I_0}{3I_0} = \frac{1}{3}$$

Thus, the correct option is (C).

Question23

In a Young double slit experiment, the wavelength of incident light is $6000\text{\AA}$, the separation between slits S_1 and S_2 is 5 cm and the distance between slits plane and screen is 50 cm, as shown in the figure below. If the resultant intensity at P is equal to the intensity due to individual slits, the path difference between interfering waves is ____ $\text{\AA}$.



JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

4000

B.

3000

C.

2000

D.

1000

Answer: C

Solution:

When two waves of intensities I_1 and I_2 interfere with a phase difference ϕ , the resultant intensity I_R is given by:

$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

Since the slits are identical, the intensities due to individual slits are equal ($I_1 = I_2 = I_0$). The formula simplifies to:

$$I_R = I_0 + I_0 + 2\sqrt{I_0^2} \cos \phi$$

$$\Rightarrow I_R = 2I_0(1 + \cos \phi)$$

Using the trigonometric identity $1 + \cos \phi = 2 \cos^2 \left(\frac{\phi}{2} \right)$

$$I_R = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

It is given that the resultant intensity at point P is equal to the intensity of an individual slit:

$$I_R = I_0$$

Substituting this into equation:

$$I_0 = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$\Rightarrow \cos^2 \left(\frac{\phi}{2} \right) = \frac{1}{4}$$

$$\Rightarrow \cos \left(\frac{\phi}{2} \right) = \pm \frac{1}{2}$$

$$\text{For } \cos \left(\frac{\phi}{2} \right) = \frac{1}{2} \Rightarrow \frac{\phi}{2} = \frac{\pi}{3} \Rightarrow \phi = \frac{2\pi}{3}$$

$$\text{For } \cos\left(\frac{\phi}{2}\right) = -\frac{1}{2} \Rightarrow \frac{\phi}{2} = \pi - \frac{\pi}{3} \Rightarrow \frac{\phi}{2} = \frac{2\pi}{3} \Rightarrow \phi = \frac{4\pi}{3}$$

As we take phase difference in the range $0 \leq \phi \leq \pi$, therefore the required phase difference between interfering waves will be $\phi = \frac{2\pi}{3}$.

For other values of ϕ which satisfy the condition $\cos^2\left(\frac{\phi}{2}\right) = \frac{1}{4}$, the path difference will be same. So, we take the principle solution in the range $0 \leq \phi \leq \pi$.

The phase difference is related to the path difference by the formula:

$$\phi = \frac{2\pi}{\lambda}(\Delta x)$$

$$\Rightarrow \Delta x = \frac{\phi \cdot \lambda}{2\pi}$$

The wavelength of light is given as $\lambda = 6000\text{\AA}$ and $\phi = \frac{2\pi}{3}$:

$$\Delta x = \frac{\left(\frac{2\pi}{3}\right) \cdot 6000\text{\AA}}{2\pi}$$

$$\Rightarrow \Delta x = \frac{6000}{3}\text{\AA}$$

$$\Rightarrow \Delta x = 2000\text{\AA}$$

Therefore, the path difference between interfering waves is $2000\text{\AA}$.

Thus, the correct option is (C).

Question24

A double slit interference experiment performed with a light of wavelength 600 nm forms an interference fringe pattern on a screen with 10 th bright fringe having its centre at a distance of 10 mm from the central maximum. Distance of the centre of the same 10 th bright fringe from the central maximum when the source of light is replaced by another source of wavelength 660 nm would be _____ mm .

JEE Main 2025 (Online) 28th January Morning Shift

Answer: 11

Solution:

In Young's double slit experiment, the distance of n^{th} maxima from central maxima is given by,

$$Y = \frac{n\lambda D}{d}$$

Here, n , D & d are same for both the cases.

So, $Y \propto \lambda$

$$\Rightarrow \frac{Y_1}{Y_2} = \frac{\lambda_1}{\lambda_2} \Rightarrow \frac{10}{Y_2} = \frac{600}{660} = \frac{1}{11}$$

$$\Rightarrow Y_2 = 11 \text{ mm.}$$

Question25

A thin transparent film with refractive index 1.4 , is held on circular ring of radius 1.8 cm . The fluid in the film evaporates such that transmission through the film at wavelength 560 nm goes to a minimum every 12 seconds. Assuming that the film is flat on its two sides, the rate of evaporation is _____ $\pi \times 10^{-13} \text{ m}^3/\text{s}$.

JEE Main 2025 (Online) 28th January Evening Shift

Answer: 54

Solution:

Maxima condition

$$2\mu t = n\lambda \Rightarrow t = \frac{n\lambda}{2\mu} \Rightarrow t = \frac{\lambda}{2\mu}, \frac{2\lambda}{2\mu}, \dots$$

Minima condition $2\mu t = (2n - 1)\lambda/2$

$$\Rightarrow t = \frac{(2n - 1)\lambda}{4\mu} \Rightarrow t = \frac{\lambda}{4\mu}, \frac{3\lambda}{4\mu}, \dots$$

$$\Delta t = \frac{2\lambda}{4\mu}$$

$$\text{Rate of evaporation} = \frac{A(\Delta t)}{\text{time}} = 54 \times 10^{-13} \text{ m}^3/\text{s}$$

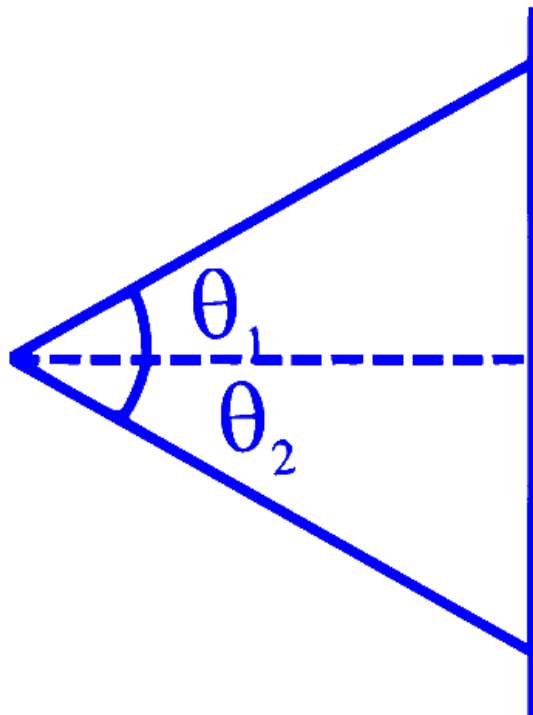
Question26

If the measured angular separation between the second minimum to the left of the central maximum and the third minimum to the right of the central maximum is 30° in a single slit diffraction pattern recorded using 628 nm light, then the width of the slit is _____ μ m.

JEE Main 2025 (Online) 2nd April Morning Shift

Answer: 6

Solution:



$$\theta_1 = \sin^{-1} \left(\frac{2\lambda}{a} \right)$$

$$\theta_2 = \sin^{-1} \left(\frac{3\lambda}{a} \right)$$

$$\therefore \theta_1 + \theta_2 = 30^\circ$$

$$\Rightarrow \sin^{-1} \left(\frac{2\lambda}{a} \right) + \sin^{-1} \left(\frac{3\lambda}{a} \right) = \frac{\pi}{6}$$

$$\Rightarrow \frac{2\lambda}{a} \sqrt{1 - \left(\frac{3\lambda}{a} \right)^2} + \frac{3\lambda}{a} \sqrt{1 + \left(\frac{2\lambda}{a} \right)^2} = \sin \frac{\pi}{6}$$

Here $\lambda = 628 \text{ nm}$

After solving

$$A = 6.07 \mu \text{ m}$$

Approximate Method :

$$\theta = \theta_1 + \theta_2$$

$$\Rightarrow \frac{\pi}{6} = \frac{2\lambda}{a} + \frac{3\lambda}{a}$$

$$\Rightarrow \frac{\pi}{6} = \frac{5}{a}(628 \text{ nm})$$

$$\Rightarrow a = 6 \mu \text{ m}$$

Question27

Two coherent monochromatic light beams of intensities $4I$ and $9I$ are superimposed. The difference between the maximum and minimum intensities in the resulting interference pattern is xI . The value of x is _____ .

JEE Main 2025 (Online) 3rd April Morning Shift

Answer: 24

Solution:

Calculate Maximum Intensity ($I_{\max}$):

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

Substituting the values, we have:

$$I_{\max} = (\sqrt{4I} + \sqrt{9I})^2$$

$$I_{\max} = (2\sqrt{I} + 3\sqrt{I})^2 = (5\sqrt{I})^2 = 25I$$

Calculate Minimum Intensity ($I_{\min}$):

$$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

Substituting the values, we have:

$$I_{\min} = (\sqrt{4I} - \sqrt{9I})^2$$

$$I_{\min} = (2\sqrt{I} - 3\sqrt{I})^2 = (-1\sqrt{I})^2 = I$$

Calculate the Difference Between Maximum and Minimum Intensities:

$$I_{\max} - I_{\min} = 25I - I = 24I$$

Thus, the value of x , which represents the difference between the maximum and minimum intensities, is 24.

Question28

In a Young's double slit experiment, two slits are located 1.5 mm apart. The distance of screen from slits is 2 m and the wavelength of the source is 400 nm . If the 20 maxima of the double slit pattern are contained within the central maximum of the single slit diffraction pattern, then the width of each slit is $x \times 10^{-3}$ cm, where x -value is _____ .

JEE Main 2025 (Online) 4th April Evening Shift

Answer: 15

Solution:

Width of 20 maxima of double slit = width of central maxima of single slit

$$\frac{20\lambda D}{d} = \frac{2\lambda D}{a}$$

$$\frac{10}{d} = \frac{1}{a}$$

$$a = \frac{d}{10} = \frac{1.5 \times 10^{-1}}{10} \text{ cm} = 15 \times 10^{-3} \text{ cm}$$

Value of x is 15

Answer is 15

Question 29

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion-(A) : If Young's double slit experiment is performed in an optically denser medium than air, then the consecutive fringes come closer.

Reason-(R) : The speed of light reduces in an optically denser medium than air while its frequency does not change.

In the light of the above statements, choose the most appropriate answer from the options given below :

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

- A. (A) is false but (R) is true
- B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
- C. Both (A) and (R) are true and (R) is the correct explanation of (A)
- D. (A) is true but (R) is false

Answer: C

Solution:

We know, in YDSE,

$$\beta \text{ (fringe width)} = \frac{\lambda D}{d}$$

In denser medium, $\lambda \downarrow \Rightarrow \beta \downarrow$

$\Rightarrow$ fringes come closer.

$$\text{Also, } \mu = \frac{c}{v} \Rightarrow v = \frac{c}{\mu}$$

$\Rightarrow \mu \uparrow \Rightarrow v \downarrow \Rightarrow$ speed of light reduces.

frequency remains same,

$$\Rightarrow \mu = \frac{\lambda_{vac.} f}{\lambda_{med.} f} \Rightarrow \lambda_{med} = \frac{\lambda_{vac.}}{\mu} \text{ (as } v = \lambda f \text{)}$$

Hence, option C is correct.

Question30

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In Young's double slit experiment, the fringes produced by red light are closer as compared to those produced by blue light.

Reason (R) : The fringe width is directly proportional to the wavelength of light.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

- A. (A) is true but (R) is false
- B. Both (A) and (R) are true and (R) is the correct explanation of (A)
- C. (A) is false but (R) is true
- D. Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

Answer: C

Solution:

In Young's double-slit experiment, the width of the fringes, denoted as β , can be calculated using the formula:

$$\beta = \frac{\lambda D}{d}$$

In this formula:

λ represents the wavelength of the light used.

D is the distance from the slits to the screen.

d is the distance between the slits.

According to this relationship, the fringe width β is directly proportional to the wavelength λ . This means that as the wavelength increases, the fringe width also increases.

When comparing red and blue light:

Red light has a longer wavelength (λ_R) compared to blue light (λ_b).

Therefore, the fringe width β_R for red light is greater than the fringe width β_b for blue light, as $\lambda_R > \lambda_b$.

This shows that with red light, the fringes are actually farther apart compared to the fringes produced by blue light. Thus, the assertion that red light fringes are closer than those produced by blue light is false, while the reason about fringe width being proportional to the wavelength is true.

Question31

A transparent film of refractive index, 2.0 is coated on a glass slab of refractive index, 1.45. What is the minimum thickness of transparent film to be coated for the maximum transmission of Green light of wavelength 550 nm . [Assume that the light is incident nearly perpendicular to the glass surface.]

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. 68.7 nm

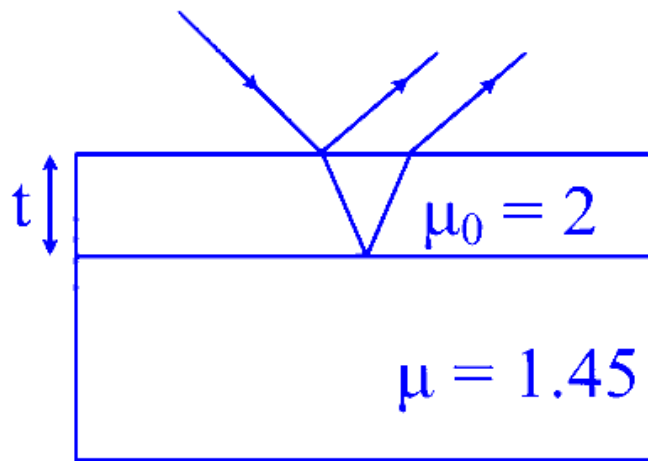
B. 137.5 nm

C. 94.8 nm

D. 275 nm

Answer: B

Solution:



For transmitted green light to be maxima, reflected green should be minima.

$$\Delta P = 2\mu_0 t = n\lambda$$

$$\Rightarrow t = \frac{n\lambda}{2\mu_0} \therefore t_{\min} = \frac{\lambda}{2\mu_0} = \frac{550}{2 \times 2} = 137.5$$

Question32

The width of one of the two slits in Young's double slit experiment is d while that of the other slit is $x d$. If the ratio of the maximum to the minimum intensity in the interference pattern on the screen is $9 : 4$ then what is the value of x ?(Assume that the field strength varies according to the slit width.)

JEE Main 2025 (Online) 23rd January Evening Shift

Options:

- A. 2
- B. 3
- C. 4
- D. 5

Answer: D

Solution:

Let the amplitude from the slit of width d be proportional to E and from the slit of width xd be proportional to xE (with $x > 1$, as is evident from the given options).

For two coherent waves with amplitudes E_1 and E_2 , the resultant intensity when added in phase (maximum) and out of phase (minimum) is given by

$$I_{\max} \propto (E_1 + E_2)^2 \quad \text{and} \quad I_{\min} \propto (E_2 - E_1)^2.$$

Here, setting

$$E_1 = E \text{ (from the narrower slit of width } d) \text{ and}$$

$$E_2 = xE \text{ (from the wider slit of width } xd),$$

we have

$$I_{\max} \propto (E + xE)^2 = E^2(1 + x)^2,$$

and since $x > 1$ the subtraction in the destructive case gives

$$I_{\min} \propto (xE - E)^2 = E^2(x - 1)^2.$$

The ratio of maximum to minimum intensity is therefore

$$\frac{I_{\max}}{I_{\min}} = \frac{(1+x)^2}{(x-1)^2}.$$

We are given that

$$\frac{(1+x)^2}{(x-1)^2} = \frac{9}{4}.$$

Taking the square root of both sides (noting that all quantities are positive) leads to

$$\frac{1+x}{x-1} = \frac{3}{2}.$$

To solve for x , cross-multiply:

$$2(1 + x) = 3(x - 1).$$

Expanding both sides:

$$2 + 2x = 3x - 3.$$

Rearrange the equation to isolate x :

$$2 + 2x + 3 = 3x \quad \implies \quad 5 + 2x = 3x,$$

which implies

$$x = 5.$$

Thus, the value of x is

$$\boxed{5}.$$

Question33

The Young's double slit interference experiment is performed using light consisting of 480 nm and 600 nm wavelengths to form interference patterns. The least number of the bright fringes of 480 nm light that are required for the first coincidence with the bright fringes formed by 600 nm light is

JEE Main 2025 (Online) 24th January Morning Shift

Options:

A. 8

B. 4

C. 5

D. 6

Answer: C

Solution:

$$d \sin \theta = m\lambda$$

For two wavelengths, $\lambda_1 = 600$ nm and $\lambda_2 = 480$ nm, the bright fringes coincide when the conditions

$$m\lambda_2 = n\lambda_1$$

are simultaneously satisfied for integers m and n , representing the bright fringe order for each wavelength.

Rearranging the equation gives:

$$\frac{m}{n} = \frac{\lambda_1}{\lambda_2} = \frac{600}{480} = \frac{5}{4}$$

Thus, the smallest positive integers that satisfy this ratio are:

$$m = 5 \quad \text{and} \quad n = 4.$$

This means the first coincidence of the bright fringes occurs at the 5th bright fringe for the 480 nm light.

Question34

Young's double slit interference apparatus is immersed in a liquid of refractive index 1.44. It has slit separation of 1.5 mm . The slits are illuminated by a parallel beam of light whose wavelength in air is 690 nm . The fringe-width on a screen placed behind the plane of slits at a distance of 0.72 m , will be:

JEE Main 2025 (Online) 24th January Evening Shift

Options:

A. 0.63 mm

B. 0.33 mm

C. 0.46 mm

D. 0.23 mm

Answer: D

Solution:

In Young's double-slit experiment, the fringe width (β) can be calculated using the formula:

$$\beta = \frac{\lambda' D}{d}$$

where:

λ' is the wavelength of light in the medium,

D is the distance between the slits and the screen,

d is the separation between the slits.

The wavelength in the medium (λ') is given by:

$$\lambda' = \frac{\lambda_0}{n}$$

where:

$\lambda_0 = 690 \text{ nm}$ is the wavelength of light in air,

$n = 1.44$ is the refractive index of the liquid.

Substituting the values,

$$\lambda' = \frac{690 \text{ nm}}{1.44} \approx 479.17 \text{ nm}$$

Now, convert λ' to meters:

$$\lambda' = 479.17 \text{ nm} = 479.17 \times 10^{-9} \text{ m}$$

Next, use the values for D and d :

$$D = 0.72 \text{ m}$$

$$d = 1.5 \text{ mm} = 1.5 \times 10^{-3} \text{ m}$$

Calculating the fringe width:

$$\beta = \frac{479.17 \times 10^{-9} \text{ m} \cdot 0.72 \text{ m}}{1.5 \times 10^{-3} \text{ m}}$$

$$\beta \approx \frac{344.2 \times 10^{-9} \text{ m}}{1.5 \times 10^{-3} \text{ m}}$$

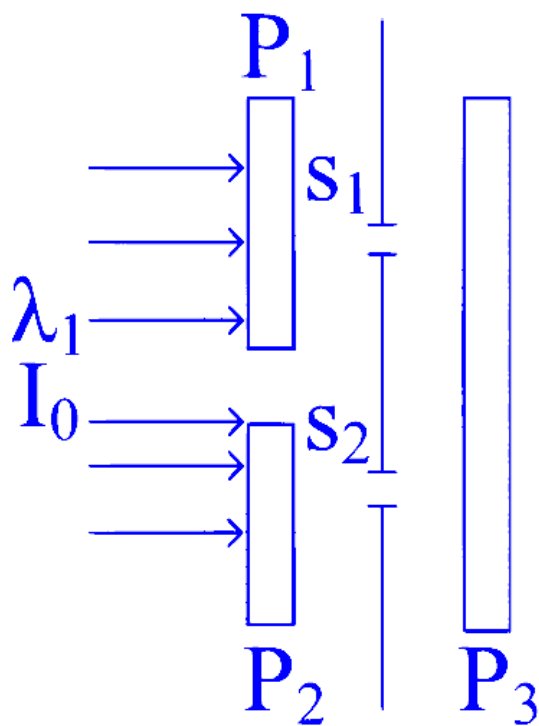
$$\beta \approx 0.229 \times 10^{-3} \text{ m} = 0.229 \text{ mm}$$

The closest answer choice is:

Option D: 0.23 mm

Question 35

In a Young's double slit experiment, three polarizers are kept as shown in the figure. The transmission axes of P_1 and P_2 are orthogonal to each other. The polarizer P_3 covers both the slits with its transmission axis at 45° to those of P_1 and P_2 . An unpolarized light of wavelength λ and intensity I_0 is incident on P_1 and P_2 . The intensity at a point after P_3 where the path difference between the light waves from s_1 and s_2 is $\frac{\lambda}{3}$, is



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Options:

A. I_0

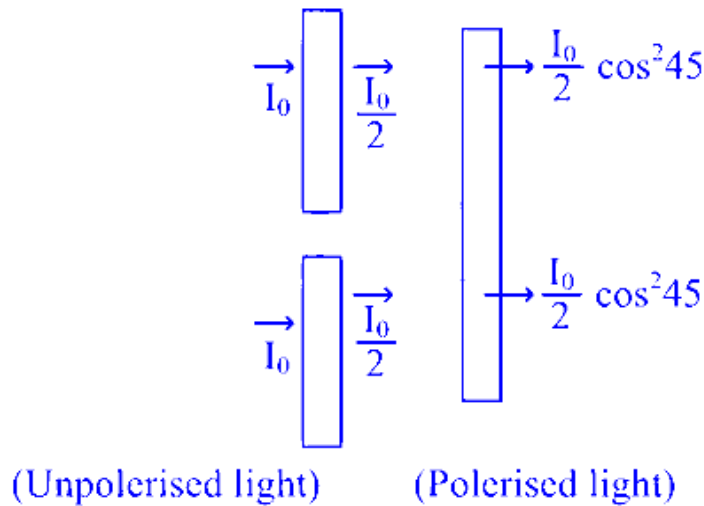
B. $\frac{I_0}{3}$

C. $\frac{I_0}{4}$

D. $\frac{I_0}{2}$

Answer: C

Solution:



after passing through third polariser, Intensity of both the waves must be $\frac{I_0}{4}$

now, at a point where path diff is $\frac{\lambda}{3}$, phase difference

$$\Delta\phi = 2\pi \left(\frac{\Delta x}{\lambda} \right) = \frac{2\pi}{3}$$

$$\begin{aligned} \therefore I_{\text{res}} &= \sqrt{\left(\frac{I_0}{4}\right)^2 + \left(\frac{I_0}{4}\right)^2 + 2\left(\frac{I_0}{4}\right)^2 \cos \frac{2\pi}{3}} \\ &= \frac{I_0}{4} \end{aligned}$$

Question36

At the interface between two materials having refractive indices n_1 and n_2 , the critical angle for reflection of an em wave is θ_{1C} . The n_2 material is replaced by another material having refractive index n_3 such that the critical angle at the interface between n_1 and n_3 materials is θ_{2C} . If $n_3 > n_2 > n_1$; $\frac{n_2}{n_3} = \frac{2}{5}$ and $\sin \theta_{2C} - \sin \theta_{1C} = \frac{1}{2}$, then θ_{1C} is :

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Options:

A.

$$\sin^{-1} \left(\frac{1}{6n_1} \right)$$

B.

$$\sin^{-1} \left(\frac{1}{3n_1} \right)$$

C.

$$\sin^{-1} \left(\frac{5}{6n_1} \right)$$

D.

$$\sin^{-1} \left(\frac{2}{3n_1} \right)$$

Answer: A

Solution:

$$\sin \theta_{1C} = \frac{n_1}{n_2}$$

$$\sin \theta_{2C} = \frac{n_1}{n_3}$$

$$\sin \theta_{2C} - \sin \theta_{1C} = \frac{1}{2}$$

$$n_1 \frac{n_2}{n_3} - \frac{n_1}{n_2} = \frac{1}{2}$$

$$n_1 \frac{n_2}{n_3} - n_1 = \frac{n_2}{2}$$

$$n_1 \left(\frac{2}{5} - 1 \right) = \frac{n_2}{2}$$

$$\frac{n_1}{n_2} = \frac{-5}{6}$$

$$= \sin^{-1} \left(-\frac{5}{6} \right)$$

Question37

A light wave is propagating with plane wave fronts of the type $x + y + z = \text{constant}$. Th angle made by the direction of wave propagation with the x -axis is :

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. $\cos^{-1}(2/3)$

B. $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

C. $\cos^{-1}\left(\frac{1}{3}\right)$

D. $\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Answer: B

Solution:

The light wave is propagating with plane wave fronts described by the equation $x + y + z = \text{constant}$. The wave propagates in a direction perpendicular to these wave fronts. This direction is symmetric with respect to the x , y , and z axes.

Since the wave is symmetric about these axes, the angle it makes with each axis is the same. Let these angles be α , β , and γ , which are the angles made by the light with the x , y , and z axes, respectively.

This implies:

$$\cos \alpha = \cos \beta = \cos \gamma$$

According to the sum of the squares of the direction cosines:

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Substituting the equality of direction cosines, we get:

$$3 \cos^2 \alpha = 1$$

$$\cos^2 \alpha = \frac{1}{3}$$

Thus, the angle α is given by:

$$\alpha = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

Question38

A monochromatic light of frequency 5×10^{14} Hz travelling through air, is incident on a medium of refractive index ' 2 '. Wavelength of the refracted light will be :

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

- A. 400 nm
- B. 300 nm
- C. 600 nm
- D. 500 nm

Answer: B

Solution:

To find the wavelength of the refracted light in a medium with a refractive index of 2, we can use the relationship between frequency, wavelength, and the speed of light. The frequency (f) of the light remains the same when it enters the medium.

The formula relating frequency, wavelength, and velocity is:

$$v = f\lambda$$

Where:

v is the speed of light in the medium,

f is the frequency of the light,

λ is the wavelength of the light.

The wavelength of light in the medium (λ_{medium}) can be found using the refractive index (μ) and the wavelength in a vacuum (λ_{vacuum}) as follows:

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{\mu}$$

Given:

Frequency of the light, $f = 5 \times 10^{14}$ Hz

Speed of light in a vacuum, $v_{\text{vacuum}} = 3 \times 10^8$ m/s

Refractive index of the medium, $\mu = 2$

First, calculate λ_{vacuum} :

$$\lambda_{\text{vacuum}} = \frac{v_{\text{vacuum}}}{f} = \frac{3 \times 10^8}{5 \times 10^{14}}$$

Now calculate λ_{medium} :

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{2} = \frac{3 \times 10^8}{2 \times 5 \times 10^{14}}$$

$$\lambda_{\text{medium}} = 0.3 \times 10^{-6} \text{ m}$$

Converting to nanometers:

$$0.3 \times 10^{-6} \text{ m} = 300 \text{ nm}$$

Therefore, the wavelength of the refracted light in the medium is **300 nm**.

Question 39

Width of one of the two slits in a Young's double slit interference experiment is half of the other slit. The ratio of the maximum to the minimum intensity in the interference pattern is :

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Options:

A. $3 : 1$

B. $(2\sqrt{2} + 1) : (2\sqrt{2} - 1)$

C. $9 : 1$

D. $(3 + 2\sqrt{2}) : (3 - 2\sqrt{2})$

Answer: D

Solution:

In Young's double-slit interference experiment, the width of one slit is half that of the other. The intensity, I , is proportional to the slit width. This gives us:

$$I_1 = I_0,$$

$$I_2 = 2I_0.$$

The maximum intensity, I_{max} , occurs when the amplitudes add constructively:

$$I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2.$$

The minimum intensity, I_{min} , occurs when the amplitudes interfere destructively:

$$I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2.$$

Now, using $I_1 = I_0$ and $I_2 = 2I_0$, we find:

$$\frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{2}+1)^2}{(\sqrt{2}-1)^2} = \frac{3+2\sqrt{2}}{3-2\sqrt{2}}.$$

Question 40

Two monochromatic light beams have intensities in the ratio 1:9. An interference pattern is obtained by these beams. The ratio of the intensities of maximum to minimum is

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. 8 : 1

B. 4 : 1

C. 3 : 1

D. 9 : 1

Answer: B

Solution:

To find the ratio of the intensities of the maximum to the minimum in an interference pattern formed by two monochromatic light beams with intensity ratios of 1:9, you use the formula:

$$\frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2}$$

Given that the intensity ratio is 1:9, we have $I_1 = 1$ and $I_2 = 9$.

Calculate $\sqrt{I_1}$ and $\sqrt{I_2}$:

$$\sqrt{I_1} = \sqrt{1} = 1$$

$$\sqrt{I_2} = \sqrt{9} = 3$$

Substitute these values into the formula:

$$\frac{I_{\max}}{I_{\min}} = \frac{(1+3)^2}{(1-3)^2} = \frac{4^2}{2^2} = \frac{16}{4} = 4$$

Thus, the ratio of the intensities of the maximum to the minimum is 4:1.

Question41

In a Young's double slit experiment, the slits are separated by 0.2 mm . If the slits separation is increased to 0.4 mm , the percentage change of the fringe width is :

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. 25%

B. 50%

C. 0%

D. 100%

Answer: B

Solution:

In Young's double slit experiment, the fringe width β is given by:

$$\beta = \frac{\lambda D}{d}$$

where:

λ is the wavelength of light,

D is the distance from the slits to the screen,

d is the separation between the slits.

Here's how the change affects the fringe width:

Initial Situation:

When $d = 0.2$ mm, the fringe width is:

$$\beta_{\text{initial}} = \frac{\lambda D}{0.2 \text{ mm}}$$

After Increasing Slit Separation:

When d is increased to 0.4 mm:

$$\beta_{\text{new}} = \frac{\lambda D}{0.4 \text{ mm}}$$

Comparing the Two Fringe Widths:

Notice that:

$$\beta_{\text{new}} = \frac{1}{2} \times \frac{\lambda D}{0.2 \text{ mm}} = \frac{1}{2} \beta_{\text{initial}}$$

This means the fringe width is halved, which is a reduction of 50%.

Thus, the fringe width decreases by 50% when the slit separation is increased from 0.2 mm to 0.4 mm.

The correct answer is Option B: 50%.

Question 42

Two polarisers P_1 and P_2 are placed in such a way that the intensity of the transmitted light will be zero. A third polariser P_3 is inserted in between P_1 and P_2 , at particular angle between P_2 and P_3 . The transmitted intensity of the light passing through all three polarisers is maximum. The angle between the polarisers P_2 and P_3 is :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. $\pi/6$

B. $\frac{\pi}{3}$

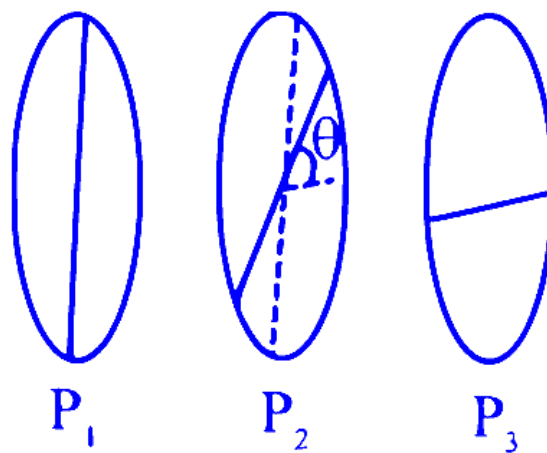
C. $\frac{\pi}{4}$

D. $\pi/8$

Answer: C

Solution:

Through P_2 $I_1 = I_0 \sin^2 \left(\frac{\pi}{2} - \theta \right)$



$$I_1 = I_0 \cos^2 \theta$$

$$\text{Through } P_3 I_{\text{net}} = (I_0 \cos^2 \theta) \sin^2 \theta$$

$$I_{\text{net}} = \frac{I_0}{4} [\sin(2\theta)]^2 \text{ for max } I_{\text{net}} \quad \theta = 45^\circ$$

$$\text{So angle between } P_2 \text{ and } P_3 = \frac{\pi}{4}$$

Correct Ans. (1)

Question43

Two plane polarized light waves combine at a certain point whose electric field components are

$$E_1 = E_0 \sin \omega t$$

$$E_2 = E_0 \sin \left(\omega t + \frac{\pi}{3} \right)$$

Find the amplitude of the resultant wave.

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. $1.7 E_0$

B. E_0

C. $0.9 E_0$

D. $3.4 E_0$

Answer: A

Solution:

To find the amplitude of the resultant wave for the given electric field components, we use the formula for the resultant amplitude of two combining waves. These components are:

$$E_1 = E_0 \sin \omega t$$

$$E_2 = E_0 \sin \left(\omega t + \frac{\pi}{3} \right)$$

The general formula for the amplitude E of two superimposed waves E_1 and E_2 with a phase difference is:

$$E = \sqrt{E_1^2 + E_2^2 + 2E_1E_2 \cos \phi}$$

where ϕ is the phase difference between the two waves. In this scenario, $\phi = \frac{\pi}{3}$.

Substituting the given expressions and phase difference into the formula, we get:

$$E = \sqrt{(E_0)^2 + (E_0)^2 + 2(E_0)(E_0) \cos \left(\frac{\pi}{3} \right)}$$

Given that $\cos \left(\frac{\pi}{3} \right) = \frac{1}{2}$, this simplifies to:

$$E = \sqrt{2E_0^2 + E_0^2} = \sqrt{3}E_0$$

Thus, the amplitude of the resultant wave is $\sqrt{3}E_0 \approx 1.73E_0$.

Question44

In a Young's double slit experiment, the source is white light. One of the slits is covered by red filter and another by a green filter. In this case:

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

there shall be alternate interference fringes of red and green.

B.

there shall be an interference pattern for red distinct from that for green.

C.

there shall be an interference pattern, where each fringe's pattern center is green and outer edges is red.

D.

there shall be no interference fringes.

Answer: D

Solution:

When white light is used in Young's double slit experiment and one slit is covered by a red filter and the other by a green filter, several effects occur:

Different Fringe Widths: Each color will produce fringes of different widths due to their varying wavelengths. This means that red and green will not overlap perfectly.

Color Overlap: As the fringe patterns develop, some regions will have overlap due to the differing fringe spacings of red and green light. This overlap will create areas where the colors mix, potentially leading to an appearance of intermediate colors.

Overall, because each color produces its own pattern and these do not align, a clear interference pattern as seen with monochromatic light (a single color) will not be formed. Instead, there will be a complex pattern where the fringes do not distinctly appear as solid bands of red or green due to the overlapping and different fringe widths.

Question45

**When a polaroid sheet is rotated between two crossed polaroids then the transmitted light intensity will be maximum for a rotation of :
[27-Jan-2024 Shift 2]**

Options:

A. 60°

B. 30°

C. 90°

D. 45°

Answer: D

Solution:

Let I_0 be intensity of unpolarised light incident on first polaroid.

I_1 = Intensity of light transmitted from 1st polaroid

$$= \frac{I_0}{2}$$

θ be the angle between 1st and 2nd polaroid

ϕ be the angle between 2nd and 3rd polaroid

$\theta + \phi = 90^\circ$ (as 1st and 3rd polaroid are crossed)

$$\phi = 90^\circ - \theta$$

I_2 = Intensity from 2nd polaroid

$$I_2 = I_1 \cos^2 \theta = \frac{I_0}{2} \cos^2 \theta$$

I_3 = Intensity from 3rd polaroid

$$I_3 = I_2 \cos^2 \phi$$

$$I_3 = I_1 \cos^2 \theta \cos^2 \phi$$

$$I_3 = \frac{I_0}{2} \cos^2 \theta \cos^2 \phi$$

$$\phi = 90 - \theta$$

$$I_3 = \frac{I_0}{2} \cos^2 \theta \sin^2 \theta$$

$$I_3 = \frac{I_0}{2} \left[\frac{2 \sin \theta \cos \theta}{2} \right]^2$$

$$I_3 = \frac{I_0}{8} \sin^2 2\theta$$

I_3 will be maximum when $\sin 2\theta = 1$

$$2\theta = 90^\circ$$

$$\theta = 45^\circ$$

Question46

A parallel beam of monochromatic light of wavelength $5000\text{\AA}$ is incident normally on a single narrow slit of width 0.001 mm . The light is focused by convex lens on screen, placed on its focal plane. The first minima will be formed for the angle of diffraction of _____ (degree).

[27-Jan-2024 Shift 2]

Answer: 30

Solution:

For first minima

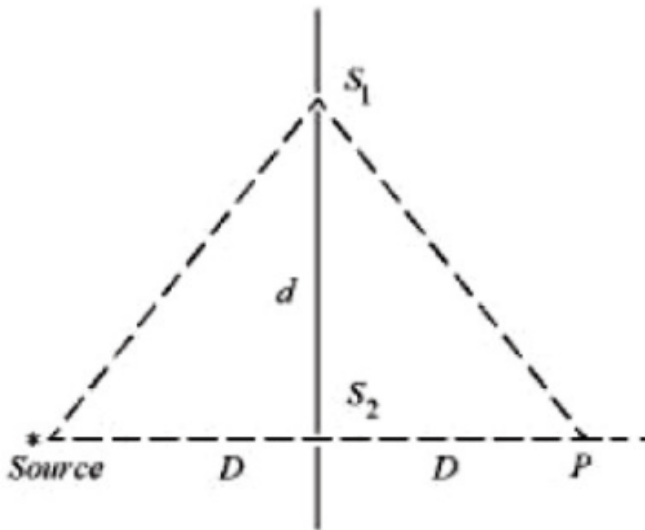
$$a \sin \theta = \lambda$$

$$\Rightarrow \sin \theta = \frac{\lambda}{a} = \frac{5000 \times 10^{-10}}{1 \times 10^{-6}} = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

Question47

In a double slit experiment shown in figure, when light of wavelength 400 nm is used, dark fringe is observed at P. If $D = 0.2\text{ m}$. the minimum distance between the slits S_1 and S_2 is _____ mm.



[29-Jan-2024 Shift 1]

Answer: 0.20

Solution:

Path difference for minima at P

$$2\sqrt{D^2 + d^2} - 2D = \frac{\lambda}{2}$$

$$\therefore \sqrt{D^2 + d^2} - D = \frac{\lambda}{4}$$

$$\therefore \sqrt{D^2 + d^2} = \frac{\lambda}{4} + D$$

$$\Rightarrow D^2 + d^2 = D^2 + \frac{\lambda^2}{16} + \frac{D\lambda}{2}$$

$$\Rightarrow d^2 = \frac{D\lambda}{2} + \frac{\lambda^2}{16}$$

$$\Rightarrow d^2 = \frac{0.2 \times 400 \times 10^{-9}}{2} + \frac{4 \times 10^{-14}}{4}$$

$$\Rightarrow d^2 \approx 400 \times 10^{-10}$$

$$\therefore d = 20 \times 10^{-5}$$

$$\Rightarrow d = 0.20 \text{ mm}$$

Question48

In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $\frac{7\lambda}{4}$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is :

[29-Jan-2024 Shift 2]

Options:

A. $1/2$

B. $3/4$

C. $1/3$

D. $1/4$

Answer: A

Solution:

$$\Delta x = \frac{7\lambda}{4}$$

$$\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} \times \frac{7\lambda}{4} = \frac{7\pi}{2}$$

$$I = I_{\max} \cos^2\left(\frac{\phi}{2}\right)$$

$$\frac{I}{I_{\max}} = \cos^2\left(\frac{\phi}{2}\right) = \cos^2\left(\frac{7\pi}{2 \times 2}\right) = \cos^2\left(\frac{7\pi}{4}\right)$$

$$= \cos^2\left(2\pi - \frac{\pi}{4}\right)$$

$$= \cos^2 \frac{\pi}{4}$$

$$= \frac{1}{2}$$

Question 49

In a single slit diffraction pattern, a light of wavelength $6000\text{\AA}$ is used. The distance between the first and third minima in the diffraction pattern is found to be 3 mm when the screen is placed 50 cm away from slits. The width of the slit is $\times 10^{-4}\text{m}$
[29-Jan-2024 Shift 2]

Answer: 2

Solution:

For n^{th} minima

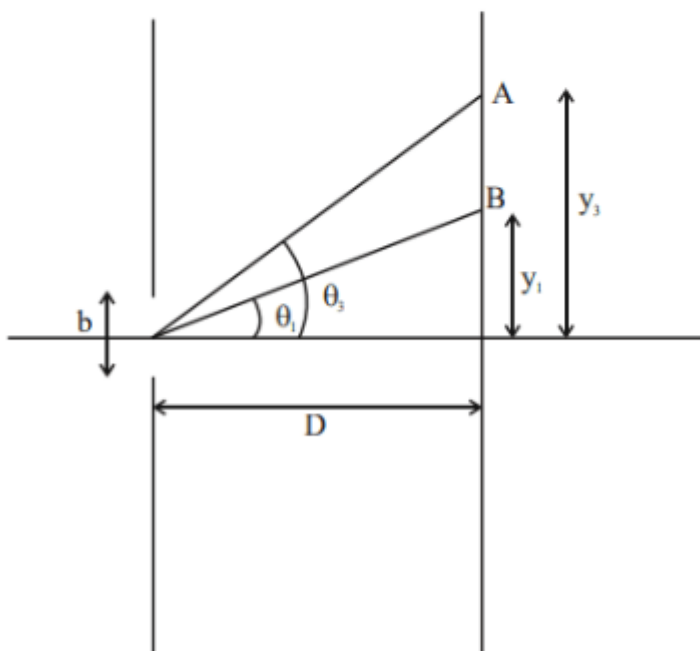
$$b \sin \theta = n\lambda$$

(λ is small so $\sin \theta$ is small, hence $\sin \theta \simeq \tan \theta$)

$$b \tan \theta = n\lambda$$

$$b \frac{y}{D} = n\lambda$$

$$\Rightarrow y_n = \frac{n\lambda D}{b} \text{ (Position of } n^{\text{th}} \text{ minima)}$$



B $\rightarrow$ 1st minima, A $\rightarrow$ 3rd minima

$$y_3 = \frac{3\lambda D}{b}, y_1 = \frac{\lambda D}{b}$$

$$\Delta y = y_3 - y_1 = \frac{2\lambda D}{b}$$

$$3 \times 10^{-3} = \frac{2 \times 6000 \times 10^{-10} \times 0.5}{b}$$

$$b = \frac{2 \times 6000 \times 10^{-10} \times 0.5}{3 \times 10^{-3}}$$

$$b = 2 \times 10^{-4} \text{ m}$$

$$x = 2$$

Question50

The diffraction pattern of a light of wavelength 400 nm diffracting from a slit of width 0.2 mm is focused on the focal plane of a convex lens of focal length 100 cm. The width of the 1st secondary maxima

will be :

[30-Jan-2024 Shift 1]

Options:

A. 2 mm

B. 2 cm

C. 0.02 mm

D. 0.2 mm

Answer: A

Solution:

$$\text{Width of 1}^{\text{st}} \text{ secondary maxima} = \frac{\lambda}{a} \cdot D$$

Here

$$a = 0.2 \times 10^{-3} \text{ m}$$

$$\lambda = 400 \times 10^{-9} \text{ m}$$

$$D = 100 \times 10^{-2}$$

Width of 1st secondary maxima

$$= \frac{400 \times 10^{-9}}{0.2 \times 10^{-3}} \times 100 \times 10^{-2}$$

$$= 2 \text{ mm}$$

Question51

A beam of unpolarised light of intensity I_0 is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of 45° relative to that of A.

The intensity of emergent light is :

[30-Jan-2024 Shift 2]

Options:

A. $I_0/4$

B. I_0

C. $I_0/2$

D. $I_0/8$

Answer: A

Solution:

Intensity of emergent light

$$= \frac{I_0}{2} \cos^2 45^\circ = \frac{I_0}{4}$$

Question52

Two waves of intensity ratio 1 : 9 cross each other at a point. The resultant intensities at the point, when (a) Waves are incoherent is I_1 (b) Waves are coherent is I_2 and differ in phase by 60° . If $\frac{I_1}{I_2} = \frac{10}{x}$ then $x = \underline{\hspace{2cm}}$
[31-Jan-2024 Shift 1]

Answer: 13

Solution:

For incoherent wave $I_1 = I_A + I_B \Rightarrow I_1 = I_0 + 9I_0$

$$I_1 = 10I_0$$

For coherent wave $I_2 = I_A + I_B + 2\sqrt{I_A I_B} \cos 60^\circ$

$$I_2 = I_0 + 9I_0 + 2\sqrt{9I_0^2} \cdot \frac{1}{2} = 13I_0$$

$$\frac{I_1}{I_2} = \frac{10}{13}$$

Question 53

When unpolarized light is incident at an angle of 60° on a transparent medium from air. The reflected ray is completely polarized. The angle of refraction in the medium is
[31-Jan-2024 Shift 2]

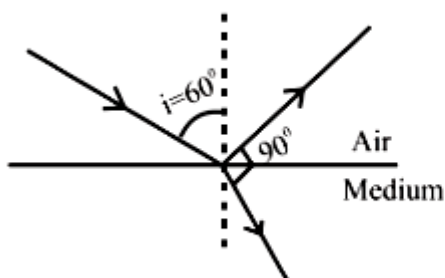
Options:

- A. 30°
- B. 60°
- C. 90°
- D. 45°

Answer: A

Solution:

By Brewster's law



At complete reflection refracted ray and reflected ray are perpendicular.

Question54

A monochromatic light of wavelength $6000\text{\AA}$ is incident on the single slit of width 0.01 mm . If the diffraction pattern is formed at the focus of the convex lens of focal length 20 cm , the linear width of the central maximum is :
[1-Feb-2024 Shift 1]

Options:

- A. 60 mm
- B. 24 mm
- C. 120 mm
- D. 12 mm

Answer: B

Solution:

Linear width

$$W = \frac{2\lambda d}{a} = \frac{2 \times 6 \times 10^{-7} \times 0.2}{1 \times 10^{-5}} \\ = 2.4 \times 10^{-2} = 24\text{ mm}$$

Question55

A microwave of wavelength 2.0 cm falls normally on a slit of width 4.0 cm . The angular spread of the central maxima of the diffraction pattern obtained on a screen 1.5 m away from the slit, will be:
[1-Feb-2024 Shift 2]

Options:

- A. 30°

B. 15°

C. 60°

D. 45°

Answer: C

Solution:

For first minima $a \sin \theta = \lambda$

$$\sin \theta = \frac{\lambda}{a} = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\text{Angular spread} = 60^\circ$$

Question 56

In Young's double slit experiment, monochromatic light of wavelength $5000\text{\AA}$ is used. The slits are 1.0 mm apart and screen is placed at 1.0 m away from slits. The distance from the centre of the screen where intensity becomes half of the maximum intensity for the first time is _____ $\times 10^{-6}\text{ m}$.

[1-Feb-2024 Shift 2]

Answer: 125

Solution:

Let intensity of light on screen due to each slit is I_0

So intensity at centre of screen is $4I_0$

Intensity at distance y from centre-

$$I = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \phi$$

$$I_{\max} = 4I_0$$

$$\frac{I_{\max}}{2} = 2I_0 = 2I_0 + 2I_0 \cos \phi$$

$$\cos \phi = 0$$

$$\phi = \frac{\pi}{2}$$

$$K\Delta x = \frac{\pi}{2}$$

$$\frac{2\pi}{\lambda} d \sin \theta = \frac{\pi}{2}$$

$$\frac{2}{\lambda} d \times \frac{y}{D} = \frac{1}{2}$$

$$y = \frac{\lambda D}{4d} = \frac{5 \times 10^{-7} \times 1}{4 \times 10^{-3}}$$

$$= 125 \times 10^{-6}$$

$$= 125$$

Question 57

Given below are two statements :

Statement I : If the Brewster's angle for the light propagating from air to glass is θ_B , then Brewster's angle for the light propagating from glass to air is $\frac{\pi}{2} - \theta_B$.

Statement II : The Brewster's angle for the light propagating from glass to air is $\tan^{-1}(\mu_g)$ where μ_g is the refractive index of glass.

In the light of the above statements, choose the correct answer from the options given below :

[24-Jan-2023 Shift 1]

Options:

A. Both Statements I and Statement II are true.

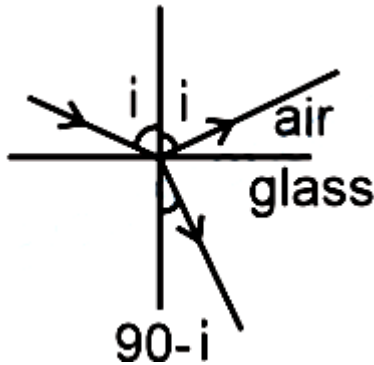
B. Statement I is true but Statement II is false.

C. Both Statement I and Statement II are false.

D. Statement I is false but Statement II is true.

Answer: B

Solution:



$$\mu_a \sin i_1 = \mu_g \sin (90 - i_1)$$

$$\tan i_1 = \frac{\mu_g}{\mu_a}$$

$$\text{When going from glass to air } \tan i_2 = \frac{\mu_a}{\mu_g} = \cot i_1$$

Hence

$$i_2 = \frac{\pi}{2} - i_1$$

Question58

In Young's double slits experiment, the position of 5th bright fringe from the central maximum is 5 cm. The distance between slits and screen is 1m and wavelength of used monochromatic light is 600 nm. The separation between the slits is:

[25-Jan-2023 Shift 1]

Options:

A. 60 μ m

B. 48 μ m

C. 12 μ m

D. $36\mu\text{m}$

Answer: A

Solution:

Given

$$D = 1\text{m}$$

$$\lambda = 600 \times 10^{-9}\text{m}$$

$$n = 5$$

$$\text{As } y_{\text{nth}} = \frac{n\lambda D}{d}$$

$$\Rightarrow \frac{5 \times 600 \times 10^{-9} \times 1}{d} = 5 \times 10^{-2}$$

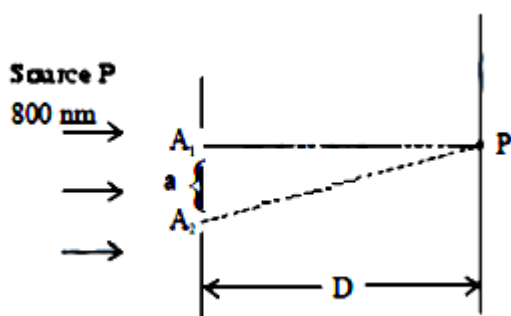
$$\Rightarrow d = \frac{5 \times 600 \times 10^{-9} \times 1}{5 \times 10^{-2}} = 60 \times 10^{-6}\text{m}$$

$$\Rightarrow d = 60\mu\text{m}$$

Question59

In a Young's double slit experiment, two slits are illuminated with a light of wavelength 800 nm . The line joining A_1P is perpendicular to A_1A_2 as shown in the figure. If the first minimum is detected at P, the value of slits separation 'a' will be :

The distance of screen from slits $D = 5\text{ cm}$



[29-Jan-2023 Shift 1]

Options:

A. 0.4 mm

B. 0.5 mm

C. 0.2 mm

D. 0.1 mm

Answer: C

Solution:

$$A_2P - A_1P = \frac{\lambda}{2} \text{ (Condition of minima)}$$

$$\sqrt{D^2 + a^2} - D = \frac{\lambda}{2}$$

$$D \left(1 + \frac{a^2}{D^2} \right)^{1/2} - D = \frac{\lambda}{2}$$

$$D \left(1 + \frac{1}{2} \times \frac{a^2}{D^2} \right) - D = \frac{\lambda}{2}$$

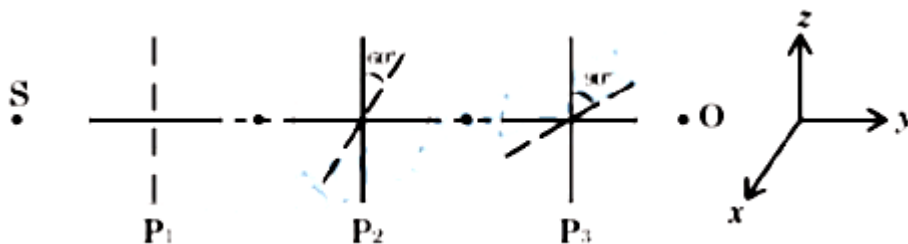
$$\frac{a^2}{2D} = \frac{\lambda}{2} \Rightarrow a = \sqrt{\lambda \cdot D}$$

$$= \sqrt{800 \times 10^{-6} \times 50}$$

$$a = 0.2 \text{ mm}$$

Question60

As shown in figures, three identical polaroids P_1 , P_2 and P_3 are placed one after another. The pass axis of P_2 and P_3 are inclined at angle of 60° and 90° with respect to axis of P_1 . The source S has an intensity of $256 \frac{W}{m^2}$. The intensity of light at point O is ----- $\frac{W}{m^2}$.



[29-Jan-2023 Shift 1]

Answer: 24

Solution:

By first polaroid P1 intensity will be halved then P2 and P3 will make intensity $\cos^2(60^\circ)$ and $\cos^2(30^\circ)$ times respectively.

$$\text{Intensity out} = \frac{256}{2} \times \frac{1}{4} \times \left(\frac{\sqrt{3}}{2} \right)^2 = \frac{256 \times 3}{2 \times 4 \times 4} = 24$$

Question61

Unpolarised light is incident on the boundary between two dielectric media, whose dielectric constants are 2.8 (medium -1) and 6.8 (medium 2), respectively. To satisfy the condition, so that the reflected and refracted rays are perpendicular to each other, the

angle of incidence should be $\tan^{-1} \left(1 + \frac{10}{\theta} \right)^{\frac{1}{2}}$ the value of θ is _____.

(Given for dielectric media, $\mu_r = 1$)

[29-Jan-2023 Shift 2]

Answer: 7

Solution:

$$\mu_1 = \sqrt{2.8 \times 1} = \sqrt{2.8}$$

$$\mu_2 = \sqrt{6.8 \times 1} = \sqrt{6.8}$$

$$\mu_1 \sin i = \mu_2 \cos i \quad \tan i = \frac{\mu_2}{\mu_1} = \sqrt{\frac{6.8}{2.8}}$$

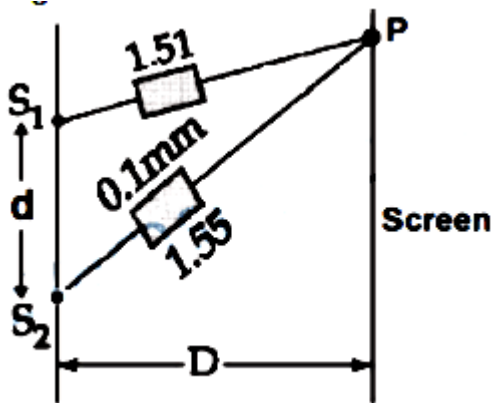
$$\tan i = \left(\frac{2.8+4}{2.8} \right)^{1/2} \quad i = \tan^{-1} \left(1 + \frac{10}{7} \right)^{1/2}$$

$$\theta = 7 \quad \text{Ans.}$$

Question62

In Young's double slit experiment, two slits S_1 and S_2 are ' d ' distance apart and the separation from slits to screen is D (as shown in figure). Now if two transparent slabs of equal thickness 0.1 mm

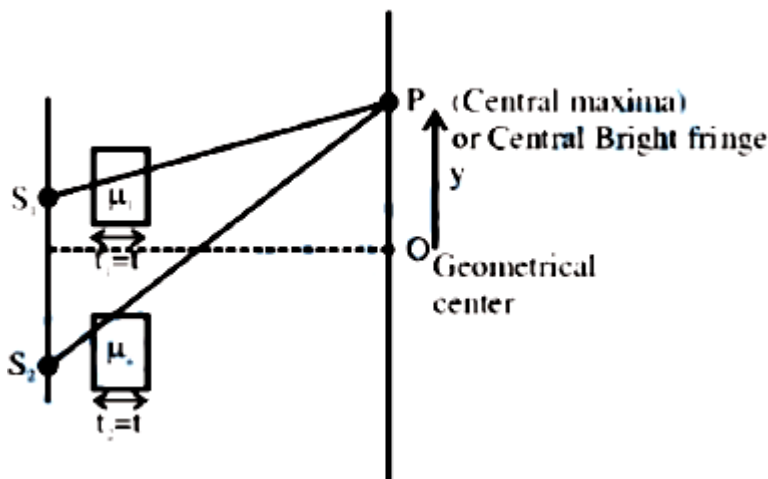
but refractive index 1.51 and 1.55 are introduced in the path of beam ($\lambda = 4000\text{\AA}$) from S_1 and S_2 respectively. The central bright fringe spot will shift by _____ number of fringes.



[30-Jan-2023 Shift 1]

Answer: 10

Solution:



Path difference at P be Δx

$$\begin{aligned}\Delta x &= (\mu_2 - \mu_1)t \\ &= (1.55 - 1.51)0.1 \text{ mm} \\ &= 0.04 \times 10^{-4}\end{aligned}$$

$$\Delta x = 4 \times 10^{-6} = 4\mu\text{m}$$

$$y = \frac{\Delta x D}{d} = 4 \times 10^{-6} \frac{D}{d}$$

y is the distance of central maxima from geometric center
fringe width

$$(\beta) = \frac{\lambda D}{d} = 4 \times 10^{-6} \text{ m} \frac{D}{d} = 4\mu\text{m} \frac{D}{d}$$

Central bright fringe spot will shift by 'x'

$$\begin{aligned}\text{Number of shift} &= \frac{y}{\beta} \\ &= \frac{4 \times 10^{-6} D/d}{4 \times 10^{-7} D/d} = 10 \text{ Ans}\end{aligned}$$

Question63

In a Young's double slit experiment, the intensities at two points, for the path difference $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$ (λ being the wavelength of light used) are I_1 and I_2 respectively. If I_0 denotes the intensity produced by each one of the individual slits, then $\frac{I_1 + I_2}{I_0} = \dots\dots$

[30-Jan-2023 Shift 2]

Answer: 3

Solution:

$$\begin{aligned}I &= 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right) \\ I_1 &= 4I_0 \cos^2\left(\frac{\pi}{4}\right) = 2I_0 \\ I_2 &= 4I_0 \cos^2\left(\frac{2\pi}{3}\right) = I_0 \\ \Rightarrow \frac{I_1 + I_2}{I_0} &= 3\end{aligned}$$

Question64

Two polaroids A and B are placed in such a way that the pass-axis of polaroids are perpendicular to each other. Now, another polaroid C is placed between A and B bisecting angle between them. If intensity of unpolarised light is I_0 then intensity of transmitted light after passing through polaroid B will be :

[31-Jan-2023 Shift 1]

Options:

A. $\frac{I_0}{4}$

B. $\frac{I_0}{2}$

C. $\frac{I_0}{8}$

D. Zero

Answer: C

Solution:

$$I_A = \frac{I_0}{2}$$

$$I_C = \frac{I_0}{2} \cos^2 45 = \frac{I_0}{4}$$

$$I_B = I_C \cos^2 45 = \frac{I_0}{8}$$

Option 3.

Question65

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R

Assertion A: The beam of electrons shows wave nature and exhibit interference and diffraction.

Reason R : Davisson Germer Experimentally verified the wave nature of electrons.

In the light of the above statements. Choose the most appropriate answer from the options given below :

[31-Jan-2023 Shift 1]

Options:

A. A is correct but R is not correct

B. A is not correct but R is correct

C. Both A and R are correct but R is Not the correct explanation of A

D. Both A and R are correct and R is the correct explanation of A

Answer: D

Solution:

Conceptual

Question66

Two light waves of wavelengths 800 and 600 nm are used in Young's double slit experiment to obtain interference fringes on a screen placed 7m away from plane of slits. If the two slits are separated by 0.35 mm, then shortest distance from the central bright maximum to the point where the bright fringes of the two wavelength coincide will be _____ mm.

[31-Jan-2023 Shift 2]

Answer: 48

Solution:

$$\omega_1 = \frac{\lambda_1 D}{d} \text{ and } \omega_2 = \frac{\lambda_2 D}{d}$$

$$\omega_1 = 16 \text{ mm and } \omega_2 = 12 \text{ mm}$$

$$\text{So LCM}(\omega_1, \omega_2) = 48 \text{ mm}$$

So at 48 mm distance both bright fringes will be found.

Question67

' n ' polarizing sheets are arranged such that each makes an angle 45° with the proceeding sheet. An unpolarized light of intensity I is

incident into this arrangement. The output intensity is found to be $\frac{I}{64}$. The value of n will be:

[1-Feb-2023 Shift 1]

Options:

A. 3

B. 6

C. 5

D. 4

Answer: B

Solution:

After passing through first sheet

$$I_1 = \frac{I}{2}$$

After passing through second sheet

$$I_2 = I_1 \cos^2(45^\circ) = \frac{I}{4}$$

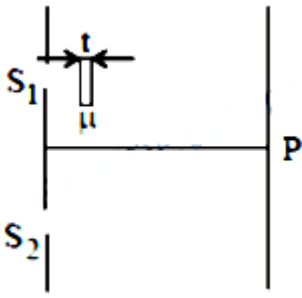
After passing through n^{th} sheet

$$I_n = \frac{I}{2^n} = \frac{I}{64}$$

$$n = 6$$

Question 68

As shown in the figure, in Young's double slit experiment, a thin plate of thickness $t = 10\mu\text{m}$ and refractive index $\mu = 1.2$ is inserted in front of slit S_1 . The experiment is conducted in air ($\mu = 1$) and uses a monochromatic light of wavelength $\lambda = 500\text{ nm}$. Due to the insertion of the plate, central maxima is shifted by a distance of $x\beta_0$. β_0 is the fringe-width before the insertion of the plate. The value of the x is _____.



[1-Feb-2023 Shift 2]

Answer: 4

Solution:

$$\begin{aligned}
 \text{Fringe shift} &= \frac{t(\mu - 1)}{\lambda} B \\
 &= \frac{10 \times 10^{-6}(1.2 - 1)}{5 \times 10^{-7}} B \\
 &= \frac{10^{-5} \times 0.2}{5 \times 10^{-7}} = 4
 \end{aligned}$$

Question69

A beam of light consisting of two wavelengths $7000\text{\AA}$ and $5500\text{\AA}$ is used to obtain interference pattern in Young's double slit experiment. The distance between the slits is 2.5 mm and the distance between the place of slits and the screen is 150 cm . The least distance from the central fringe, where the bright fringes due to both the wavelengths coincide, is $n \times 10^{-5}\text{ m}$. The value of n is

[6-Apr-2023 shift 2]

Answer: 462

Solution:

Let n_1 maxima of $7000\text{\AA}$ coincides with n_2 maxima of $5500\text{\AA}$

therefore $n_1\beta_1 = n_2\beta_2$

$$\text{or } \frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{5500}{7000} = \frac{11}{14}$$

therefore 11^{th} maxima of $7000\text{\AA}$ will coincide with 14^{th} maximum of $5500\text{\AA}$

To find the least distance of this

$$y = n_1\beta_1$$

$$\text{or } y = \frac{n_1\lambda_1 D}{d} = \frac{11 \times 7000 \times 10^{-10} \times 150 \times 10^{-2}}{2.5 \times 10^{-3}}$$

$$= \frac{11 \times 7 \times 5}{2.5} \times 10^{-5} \text{m}$$

$$\text{or } y = 462 \times 10^{-5} \text{m}$$

therefore $n = 462$

Question70

The width of fringe is 2 mm on the screen in a double slits experiment for the light of wavelength of 400 nm. The width of the fringe for the light of wavelength 600 nm will be:

[8-Apr-2023 shift 2]

Options:

A. 1.33 mm

B. 3 mm

C. 2 mm

D. 4 mm

Answer: B

Solution:

$$\beta = \frac{D\lambda}{d}$$

$$\Rightarrow \beta \propto \lambda$$

$$\frac{\beta_1}{\beta_2} = \frac{\lambda_1}{\lambda_2} \Rightarrow \frac{2}{\beta} = \frac{400}{600}$$

$$\beta = 3 \text{ mm}$$

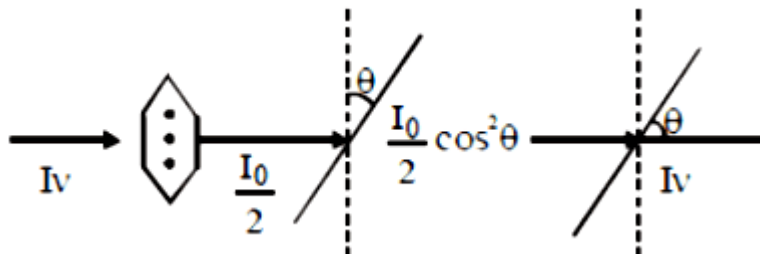
Question71

Unpolarised light of intensity 32 Wm^{-2} passes through the combination of three polaroids such that the pass axis of the last polaroid is perpendicular to that of the pass axis of first polaroid. If intensity of emerging light is 3 Wm^{-2} , then the angle between pass axis of first two polaroids is _____.

[10-Apr-2023 shift 1]

Answer: 30

Solution:



$$I_{\text{net}} = 3 = \frac{32}{8}(\sin 2\theta)^2 = \frac{I_0}{2}\cos^2\theta\sin^2\theta$$

$$\sin(2\theta) = \frac{\sqrt{3}}{2} \Rightarrow 2\theta = 60^\circ \& 120^\circ = \frac{I_0}{8}(\sin 2\theta)^2$$

$$\theta = 30^\circ \& 60^\circ$$

Question72

The ratio of intensities at two points P and Q on the screen in a Young's double slit experiment where phase difference between two waves of same amplitude are $\pi/3$ and $\pi/2$, respectively are

[10-Apr-2023 shift 2]

Options:

A. 3 : 2

B. 3 : 1

C. 2 : 3

D. 1 : 3

Answer: A

Solution:

$$I_{\text{res}} = 4I_o \cos^2\left(\frac{\theta}{2}\right)$$

$$\begin{aligned}\text{If } \theta &= \frac{\pi}{3}, I_{\text{res}} = 4I_o \cdot \cos^2\left[\frac{\pi}{6}\right] \\ &= 4I_o \cdot \left(\frac{\sqrt{3}}{2}\right)^2\end{aligned}$$

$$I_1 = (4I_o)\left(\frac{3}{4}\right) = 3I_o$$

$$\begin{aligned}\text{If } \theta &= \frac{\pi}{2}, I_{\text{res}} = 4I_o \cdot \cos^2\left(\frac{\pi}{2}\right) \\ &= 4I_o \left(\frac{1}{\sqrt{2}}\right)^2 \\ &= (4I_o)\left(\frac{1}{2}\right)\end{aligned}$$

$$I_2 = 2I_o$$

$$\frac{I_1}{I_2} = \frac{3}{2}$$

Question73

In a Young's double slits experiment, the ratio of amplitude of light coming from slits is 2 : 1. The ratio of the maximum to minimum intensity in the interference pattern is :

[13-Apr-2023 shift 2]

Options:

A. 9 : 1

B. 9 : 4

C. 2 : 1

D. 25 : 9

Answer: A

Solution:

Given that

$$\frac{A_1}{A_2} = \frac{2}{1}$$

$$\frac{I_{\max}}{I_{\min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{9}{1}$$

= 9 : 1

Question74

**A single slit of width a is illuminated by a monochromatic light of wavelength 600 nm. The value of ' a ' for which first minimum appears at $\theta = 30^\circ$ on the screen will be :
[15-Apr-2023 shift 1]**

Options:

A. $0.6\mu\text{m}$

B. $3\mu\text{m}$

C. $1.8\mu\text{m}$

D. $1.2\mu\text{m}$

Answer: D

Solution:

For minima, $a \sin \theta = \lambda$

$$a = \frac{\lambda}{\sin 30^\circ} = 2\lambda$$

$$a = 1200 \text{ nm}$$
$$a = 1.2 \mu\text{m}$$

Question 75

Sodium light of wavelengths 650 nm and 655 nm is used to study diffraction at a single slit of aperture 0.5 mm. The distance between the slit and the screen is 2.0m. The separation between the positions of the first maxima of diffraction pattern obtained in the two cases is $\underline{\hspace{1cm}} \times 10^{-5} \text{ m}$.
[24-Jun-2022-Shift-1]

Answer: 3

Solution:

Position of 1st maxima is $\frac{3}{2} \frac{\lambda D}{a}$

$$\Rightarrow \text{According to given values, required separation} = \frac{3}{2} \times (655 \text{ nm} - 650 \text{ nm}) \times \frac{2\text{m}}{0.5 \text{ mm}}$$

$$\Rightarrow \text{Required separation} = 3 \times 10^{-5} \text{ m}.$$

Question 76

The two light beams having intensities I and 9I interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be :
[25-Jun-2022-Shift-1]

Options:

A. 2I

B. α

C. 51

D. 71

Answer: B

Solution:

$$I_P = I + 9I + 2\sqrt{I \times 9I} \cos \frac{\pi}{2} = 10I$$

$$I_Q = I + 9I + 2\sqrt{I \times 9I} \cos \pi = 4I$$

$$\text{So, } I_P - I_Q = 6I$$

Question 77

The interference pattern is obtained with two coherent light sources of intensity ratio 4 : 1. And the ratio $\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}}$ is $\frac{5}{x}$. Then, the value of

x will be equal to :

[25-Jun-2022-Shift-2]

Options:

A. 3

B. 4

C. 2

D. 1

Answer: B

Solution:

$$\begin{aligned} \frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}} &= \frac{I_1 + I_2 + 2\sqrt{I_1 I_2} + I_1 + I_2 - 2\sqrt{I_1 I_2}}{I_1 + I_2 + 2\sqrt{I_1 I_2} - I_1 - I_2 + 2\sqrt{I_1 I_2}} \\ &= \frac{2(I_1 + I_2)}{4\sqrt{I_1 I_2}} \end{aligned}$$

$$= \frac{\left(\frac{I_1}{I_2} + 1\right)}{2 \sqrt{\frac{I_1}{I_2}}} = \frac{4+1}{2 \times 2} = \frac{5}{4}$$

So $x = 4$

Question78

In free space, an electromagnetic wave of 3 GHz frequency strikes over the edge of an object of size $\frac{\lambda}{100}$, where λ is the wavelength of the wave in free space. The phenomenon, which happens there will be :

[26-Jun-2022-Shift-1]

Options:

- A. Reflection
- B. Refraction
- C. Diffraction
- D. Scattering

Answer: D

Solution:

$$\frac{a}{\lambda} = \frac{1}{100}$$

For reflection size of obstacle must be much larger than wavelength, for diffraction size should be order of wavelength.

Since the object is of size $\frac{\lambda}{100}$, much smaller than wavelength, so scattering will occur.

Question79

For a specific wavelength 670 nm of light coming from a galaxy moving with velocity v , the observed wavelength is 670.7 nm. The

value of v is :

[26-Jun-2022-Shift-2]

Options:

A. $3 \times 10^8 \text{ms}^{-1}$

B. $3 \times 10^{10} \text{ms}^{-1}$

C. $3.13 \times 10^5 \text{ms}^{-1}$

D. $4.48 \times 10^5 \text{ms}^{-1}$

Answer: C

Solution:

$$\lambda_{\text{obs}} = \lambda_{\text{source}} \sqrt{\frac{1 + \frac{v}{C}}{1 - \frac{v}{C}}}$$

For $v \ll C$,

$$\frac{670.7}{670} = 1 + \frac{v}{C}$$

$$\Rightarrow v = \frac{0.7}{670} \times 3 \times 10^8 \text{m/s}$$

$$\Rightarrow v \simeq 3.13 \times 10^5 \text{m/s}$$

Question80

In Young's double slit experiment the two slits are 0.6 mm distance apart. Interference pattern is observed on a screen at a distance 80 cm from the slits. The first dark fringe is observed on the screen directly opposite to one of the slits. The wavelength of light will be _____ nm.

[27-Jun-2022-Shift-1]

Answer: 450

Solution:

$$y = \frac{d}{2}$$

$$\therefore \Delta x = y \frac{d}{D}$$

$$\Rightarrow \frac{d^2}{2D} = \frac{\lambda}{2}$$

$$\Rightarrow \lambda = \frac{(0.6 \times 10^{-3})^2}{0.8}$$
$$= 450 \text{ nm}$$

Question81

In Young's double slit experiment performed using a monochromatic light of wavelength λ , when a glass plate ($\mu = 1.5$) of thickness $x\lambda$ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be :
[28-Jun-2022-Shift-2]

Options:

A. 3

B. 2

C. 1.5

D. 0.5

Answer: B

Solution:

For the intensity to remain same the position must be of a maxima so path difference must be $n\lambda$ so

$$(1.5 - 1) \times \lambda = n\lambda$$

$$x = 2n (n = 0, 1, 2 \dots)$$

So, value of x will be

$$x = 0, 2, 4, 6 \dots$$

Question82

In a Young's double slit experiment, an angular width of the fringe is 0.35° on a screen placed at 2m away for particular wavelength of 450 nm. The angular width of the fringe, when whole system is immersed in a medium of refractive index $7/5$, is $\frac{1}{\alpha}$. The value of α is _____

[28-Jun-2022-Shift-2]

Answer: 4

Solution:

$$\text{Angular fringe width } \theta = \frac{\lambda}{D}$$

$$\text{So } \frac{\theta_1}{\lambda_1} = \frac{\theta_2}{\lambda_2}$$

$$\theta_2 = \frac{0.35^\circ}{450 \text{ nm}} \times \frac{450 \text{ nm}}{715} = 0.25^\circ = \frac{1}{4}$$

$$\text{So } \alpha = 4$$

Question83

Using Young's double slit experiment, a monochromatic light of wavelength $5000\text{\AA}$ produces fringes of fringe width 0.5 mm. If another monochromatic light of wavelength $6000\text{\AA}$ is used and the separation between the slits is doubled, then the new fringe width will be :

[29-Jun-2022-Shift-1]

Options:

A. 0.5 mm

B. 1.0 mm

C. 0.6 mm

D. 0.3 mm

Answer: D

Solution:

$$\text{Fringe width} = \frac{\lambda D}{d}$$

$$\Rightarrow \text{Fringe width} \propto \frac{\lambda}{d}$$

$$\Rightarrow \text{New fringe width} = 0.5 \text{ mm} \times \frac{1.2}{2} = 0.3 \text{ mm}$$

Question84

**In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-3} cm. If the distance between the slits is 1 mm, then the wavelength of the light will be _____ nm.
[29-Jun-2022-Shift-2]**

Answer: 600

Solution:

$$\text{Fringe width } \beta = \frac{\lambda D}{d}$$

$$\Rightarrow |d\beta| = \frac{\lambda}{d} |d(D)|$$

$$\Rightarrow 3 \times 10^{-3} \text{ cm} = \frac{\lambda}{1 \text{ mm}} (5 \times 10^{-2} \text{ m})$$

$$\Rightarrow \lambda = \frac{3 \times 10^{-8}}{5 \times 10^{-2}} \text{ m}$$

$$\Rightarrow \lambda = 600 \text{ nm}$$

Question85

In Young's double slit experiment, the fringe width is 12 mm. If the entire arrangement is placed in water of refractive index $\frac{4}{3}$, then the fringe width becomes (in mm):
[26-Jul-2022-Shift-1]

Options:

A. 16

B. 9

C. 48

D. 12

Answer: B

Solution:

$$\begin{aligned} B &= 12 \times 10^{-3} \\ \beta' &= \frac{\beta}{\mu} = \frac{12 \times 10^{-3}}{\frac{4}{3}} \\ &= 9 \times 10^{-3} \text{ m} = 9 \text{ mm} \end{aligned}$$

Question86

In Young's double slit experiment, the fringe width is 12 mm. If the entire arrangement is placed in water of refractive index $\frac{4}{3}$, then the fringe width becomes (in mm):
[26-Jul-2022-Shift-1]

Options:

A. 16

B. 9

C. 48

D. 12

Answer: B

Solution:

$$\begin{aligned} B &= 12 \times 10^{-3} \\ \beta' &= \frac{\beta}{\mu} = \frac{12 \times 10^{-3}}{\frac{4}{3}} \\ &= 9 \times 10^{-3} \text{ m} = 9 \text{ mm} \end{aligned}$$

Question87

Two coherent sources of light interfere. The intensity ratio of two sources is 1 : 4. For this interference pattern if the value of $\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}}$ is equal to $\frac{2\alpha + 1}{\beta + 3}$, then $\frac{\alpha}{\beta}$ will be :
[27-Jul-2022-Shift-2]

Options:

A. 1.5

B. 2

C. 0.5

D. 1

Answer: B

Solution:

$$\begin{aligned} \frac{I_1}{I_2} &= \frac{1}{4} \\ I_2 &= 4I_1 \\ I_{\max} &= I_1 + 4I_1 + 2\sqrt{I_1 4I_1} = 9I_1 \\ I_{\min} &= I_1 + 4I_1 - 2\sqrt{I_1 4I_1} = I_1 \\ \therefore \frac{9I_1 + I_1}{9I_1 - I_1} &= \frac{10}{8} = \frac{5}{4} = \frac{2\alpha + 1}{\beta + 1} \end{aligned}$$

$$\alpha = 2 \quad \beta = 1$$

$$\therefore \frac{\alpha}{\beta} = \frac{2}{1} = 2$$

Question88

In a Young's double slit experiment, a laser light of 560 nm produces an interference pattern with consecutive bright fringes' separation of 7.2 mm. Now another light is used to produce an interference pattern with consecutive bright fringes' separation of 8.1 mm. The wavelength of second light is _____ nm.
[28-Jul-2022-Shift-1]

Answer: 630

Solution:

$$\beta \propto \lambda$$

$$\lambda_2 = \frac{9}{8}\lambda_1$$

$$\therefore \beta_2 = \frac{9}{8}\beta_1 = \frac{9}{8} \times 560 = 630 \text{ nm.}$$

Question89

An unpolarised light beam of intensity $2I_0$ is passed through a polaroid P and then through another polaroid Q which is oriented in such a way that its passing axis makes an angle of 30° relative to that of P. The intensity of the emergent light is
[29-Jul-2022-Shift-2]

Options:

A. $\frac{I_0}{4}$

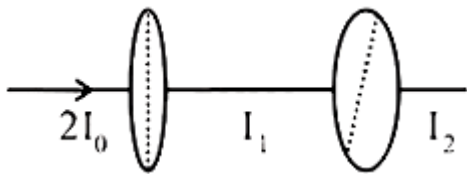
B. $\frac{I_0}{2}$

C. $\frac{3I_0}{4}$

D. $\frac{3I_0}{2}$

Answer: C

Solution:



$$I_1 = \frac{1}{2}(2I_0) = I_0$$

$$I_2 = I_1 \cos^2 30^\circ$$

$$= I_0 \cdot \frac{3}{4} = \frac{3I_0}{4}$$

Question90

Consider the diffraction pattern obtained from the sunlight incident on a pinhole of diameter $0.1\mu\text{m}$. If the diameter of the pinhole is slightly increased, it will affect the diffraction pattern such that
[25 Feb 2021 Shift 2]

Options:

- A. its size increases and intensity increases
- B. its size increases, but intensity decreases
- C. its size decreases, but intensity increases
- D. its size decreases and intensity decreases

Answer: C

Solution:

Given, diameter of pinhole, $a = 0.1 \mu\text{m} = 0.1 \times 10^{-6} \text{m}$

$\therefore$ Path difference $(\Delta x) = a \sin \phi = n\lambda \dots (i)$

where, ϕ is the phase difference and λ be the wavelength.

As, $I = 4I_0 \cos^2 \phi$

and $\sin \phi = \frac{n\lambda}{a}$ [from Eq. (i)]

If a increases $\leftrightarrow \sin \phi$ or ϕ decreases

As ϕ decreases $\leftrightarrow \cos \phi$ increases

$\therefore$ Intensity increases.

Hence, on decreasing diameter of pinhole, the size of diffraction pattern decreases and intensity increases.

Question91

In a Young's double slit experiment, the width of the one of the slit is three times the other slit. The amplitude of the light coming from a slit is proportional to the slit-width. Find the ratio of the maximum to the minimum intensity in the interference pattern.

[24feb2021shift1]

Options:

A. 1 : 4

B. 3 : 1

C. 4 : 1

D. 2 : 1

Answer: C

Solution:

Let the amplitude of light wave coming from the narrower slit be A_1 and amplitude of light wave from the wider slit be A_2 .

As amplitude $\propto$ Width of slit

$$\therefore A_2 = 3A_1$$

Maximum intensity occurs where constructive interference takes place and the minimum intensity where destructive interference takes place.

$$\therefore A_{\text{max}} = 3A_1 + A_1 = 4A_1 \text{ and}$$

$$A_{\text{min}} = 3A_1 - A_1 = 2A_1$$

∴ Intensity $I \propto A^2$

$$\therefore \frac{I_{\max}}{I_{\min}} = \frac{A_{\max}^2}{A_{\min}^2} = \left(\frac{4A_1}{2A_1} \right)^2 = 4 : 1$$

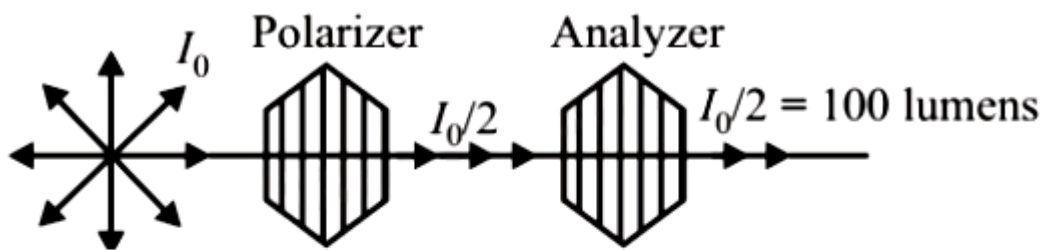
Question92

An unpolarized light beam is incident on the polarizer of a polarization experiment and the intensity of light beam emerging from the analyzer is measured as 100 Lumens. Now, if the analyzer is rotated around the horizontal axis (direction of light) by 30^{circ} in clockwise direction, the intensity of emerging light will be Lumens. [24feb2021shift1]

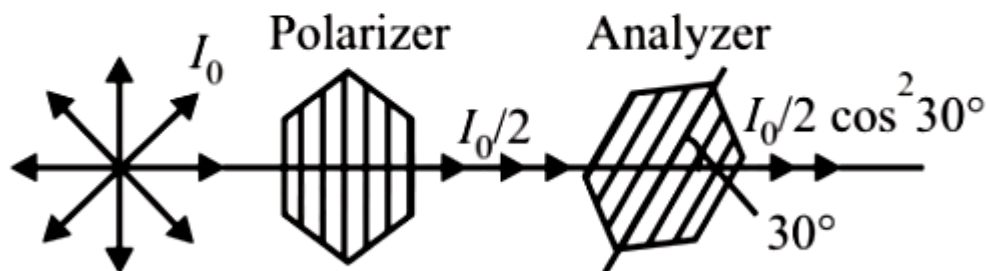
Answer: 75

Solution:

Assuming initially axis of polarizer and analyzer are parallel



When the angle between axis of polarizer and analyzer is 30° .



$$\text{Now emerging intensity} = \frac{I_0}{2} \cos^2 30^\circ$$

$$= 100 \left(\frac{\sqrt{3}}{2} \right)^2 = 100 \times \frac{3}{4} = 75$$

Question93

In a Young's double slit experiment, two slits are separated by 2mm and the screen is placed one metre away. When a light of wavelength 500nm is used, the fringe separation will be
[26 Feb 2021 Shift 1]

Options:

- A. 0.25mm
- B. 0.50mm
- C. 0.75mm
- D. 1mm

Answer: A

Solution:

Given, in YDSE, the separation between two slits, $d = 2\text{mm}$
 $= 2 \times 10^{-3}\text{m}$

Distance between slit and screen, $D = 1\text{m}$

Wavelength of light, $\lambda = 500\text{nm}$
 $= 500 \times 10^{-9}\text{m}$

Let B will be the fringe width

$$\therefore B = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 1}{2 \times 10^{-3}} = 250 \times 10^{-6}$$
$$= 0.25\text{mm}$$

Question94

Two coherent light sources having intensity in the ratio $2x$ produce an interference pattern. The ratio $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$ will be
[25 Feb 2021 Shift 1]

Options:

A. $\frac{2\sqrt{2x}}{x+1}$

B. $\frac{\sqrt{2x}}{2x+1}$

C. $\frac{\sqrt{2x}}{x+1}$

D. $\frac{2\sqrt{2x}}{2x+1}$

Answer: D

Solution:

Given, $\frac{I_1}{I_2} = 2x$

$\Rightarrow I_1 = 2I_2x$

As we know, and

$$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\therefore \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} \quad (\text{using dividendo rule})$$

$$= \frac{2\sqrt{2I_2x \cdot I_2}}{2I_2x + I_2} = \frac{2\sqrt{2x}}{2x + 1}$$

Question95

If the source of light used in a Young's double slit experiment is changed from red to violet, then
[24 Feb 2021 Shift 2]

Options:

- A. the consecutive fringe lines will come closer
- B. the central bright fringe will become a dark fringe
- C. the fringes will become brighter
- D. the intensity of minima will increase

Answer: A

Solution:

According to Young's double slit experiment, The distance of n th bright fringe from the centre,

$$y_n = \frac{n\lambda D}{d}$$

Since, $\lambda_{\text{violet}} < \lambda_{\text{red}}$

$$\therefore y_{\text{violet}} < y_{\text{red}}$$

$\therefore$ Consecutive fringe lines will come closer.

Question 96

In Young's double slit arrangement, slits are separated by a gap of 0.5mm, and the screen is placed at a distance of 0.5m from them. The distance between the first and the third bright fringe formed when the slits are illuminated by a monochromatic light of 5890Å is [18 Mar 2021 Shift 1]

Options:

A. $1178 \times 10^{-9} \text{m}$

B. $1178 \times 10^{-6} \text{m}$

C. $1178 \times 10^{-12} \text{m}$

D. $5890 \times 10^{-7} \text{m}$

Answer: B

Solution:

Given,

The distance of sources from the screen, $D = 0.5 \text{m}$

The distance between the slits, $d = 0.5 \text{mm} = 0.0005 \text{m}$

The wavelength of the monochromatic light,

$$\lambda = 5890 \text{Å} = 5890 \times 10^{-10} \text{m}$$

$$Y = 2\beta$$

$$\text{As we know, } \beta = \frac{\lambda D}{d}$$

Substituting the values in the above equation.

$$\begin{aligned} Y &= 2 \left(\frac{\lambda D}{d} \right) = 2 \left(\frac{5890 \times 10^{-10} \times 0.5}{0.0005} \right) \\ &= 1178 \times 10^{-6} \text{m} \end{aligned}$$

Question97

A fringe width of 6mm was produced for two slits separated by 1mm apart. The screen is placed 10m away. The wavelength of light used is xnm. The value of x to the nearest integer is

[16 Mar 2021 Shift 1]

Answer: 600

Solution:

Given,

Slit width, $d = 1\text{mm} = 10^{-3}\text{m}$

Fringe width, $\beta = 6\text{mm} = 6 \times 10^{-3}\text{m}$

Distance between the screen and the slit, $D = 10\text{m}$

Wavelength of light used $\lambda = ?$

$$\text{As, } \beta = \frac{D\lambda}{d}$$

$$\Rightarrow \lambda = \frac{\beta d}{D}$$

$$= \frac{6 \times 10^{-3} \times 10^{-3}}{10} = 6 \times 10^{-6} \times 10^{-1}$$

$$= 6 \times 10^{-7}\text{m} = 600 \times 10^{-9}\text{m}$$

$$= 600\text{nm}$$

Question98

A galaxy is moving away from the Earth at a speed of 286kms^{-1} . The shift in the wavelength of a red line at 630nm is $x \times 10^{-10}\text{m}$. The value of x to the nearest integer, is

(Take the value of speed of light c, as $3 \times 10^8\text{ms}^{-1}$)

[18 Mar 2021 Shift 2]

Answer: 6

Solution:

The shift in the wavelength of a red line,

$$v = c \frac{\Delta\lambda}{\lambda}$$

$$\Rightarrow \Delta\lambda = \frac{v}{c} \times \lambda$$

$$= \frac{286 \times 10^3}{3 \times 10^8} \times 630 \times 10^{-9}$$

$$= 6 \times 10^{-10} \text{ m}$$

Hence, the value of the x to the nearest integer is 6 .

Question99

In Young's double slit experiment, if the source of light changes from orange to blue then :

[27 Jul 2021 Shift 1]

Options:

- A. the central bright fringe will become a dark fringe.
- B. the distance between consecutive fringes will decrease.
- C. the distance between consecutive fringes will increase.
- D. the intensity of the minima will increase.

Answer: B

Solution:

Fringe width = $\lambda D/d$

as λ decreases , fringe width also decreases

Question100

In the Young's double slit experiment, the distance between the slits varies in time as $d(t) = d_0 + a_0 \sin \omega t$; where d_0 , ω and a_0 are constants. The difference between the largest fringe width and the

**smallest fringe width obtained over time is given as :
[25 Jul 2021 Shift 1]**

Options:

A. $\frac{2\lambda D(d_0)}{(d_0^2 - a_0^2)}$

B. $\frac{2\lambda D a_0}{(d_0^2 - a_0^2)}$

C. $\frac{\lambda D}{d_0^2} a_0$

D. $\frac{\lambda D}{d_0 + a_0}$

Answer: B

Solution:

Fringe Width, $\beta = \frac{\lambda D}{d}$

$\beta_{\max} \Rightarrow d_{\min}$ and $\beta_{\min} \Rightarrow d_{\max}$

$d = d_0 + a_0 \sin \omega t$

$d_{\max} = d_0 + a_0$ and $d_{\min} = d_0 - a_0$

$\therefore \beta_{\min} = \frac{\lambda D}{d_0 + a_0}$ and $\therefore \beta_{\max} = \frac{\lambda D}{d_0 - a_0}$

$\beta_{\max} - \beta_{\min} = \frac{\lambda D}{d_0 - a_0} - \frac{\lambda D}{d_0 + a_0} = \frac{2\lambda D a_0}{d_0^2 - a_0^2}$

Question101

With what speed should a galaxy move outward with respect to earth so that the sodium-D line at wavelength 5890 Å is observed at 5896 Å ?

[20 Jul 2021 Shift 2]

Options:

A. 306 km/sec

B. 322 km/sec

C. 296 km/sec

D. 336 km/sec

Answer: A

Solution:

$$f = f_0 \sqrt{\frac{1+\beta}{1-\beta}} \quad \beta = \frac{v}{c}$$

$$\frac{f}{f_0} \sqrt{\frac{1+\beta}{1-\beta}}$$

$$\left(1 + \frac{\Delta f}{f_0}\right)^2 = (1+\beta)(1-\beta)^{-1}$$

β is small compared to 1

$$\left(1 + \frac{2\Delta f}{f_0}\right) = (1+2\beta)$$

$$\beta = \frac{\Delta f}{f_0} = \frac{v}{c}$$

$$v = 6 \times \frac{c}{5890} = 305.6 \text{ km/s}$$

Question102

White light is passed through a double slit and interference is observed on a screen 1.5m away. The separation between the slits is 0.3 mm. The first violet and red fringes are formed 2.0 mm and 3.5 mm away from the central whitefringes. The difference in wavelengths of red and violet light is nm.

[26 Aug 2021 Shift 1]

Answer: 300

Solution:

In a Young's double slit experiment for interference pattern, the position of bright fringe is given by

$$y_n = \frac{n\lambda D}{d}$$

Here, $D = 1.5 \text{ m}$,
 $d = 0.3 \times 10^{-3} \text{ m}$
and $n = 1$

$$\lambda = \frac{y_n d}{D}$$

For first violet, $y_n = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

$$\therefore \lambda_{\text{violet}} = 2 \times 10^{-3} \frac{d}{D}$$

For first red $y_n = 3.5 \text{ mm} = 3.5 \times 10^{-3} \text{ m}$

$$\lambda_{\text{red}} = 3.5 \times 10^{-3} \frac{d}{D}$$

The difference in wavelengths of red and violet light is

$$\begin{aligned}\Delta\lambda &= \lambda_{\text{red}} - \lambda_{\text{violet}} \\ &= 3.5 \times 10^{-3} \frac{d}{D} - 2 \times 10^{-3} \frac{d}{D} \\ &= \frac{d}{D} (1.5) \times 10^{-3} \\ &= \frac{0.3 \times 10^{-3}}{1.5} \times 1.5 \times 10^{-3} \\ &= 0.3 \times 10^{-6} \\ &= 0.3 \times 10^{-6} \times 10^3 \times 10^{-3} \\ &= 300 \times 10^{-9} \\ &= 300 \text{ nm}\end{aligned}$$

Question103

A source of light is placed in front of a screen. Intensity of light on the screen is I . Two polaroids P_1 and P_2 are so placed in between the source of light and screen that the intensity of light on screen is $\frac{1}{2} I$. P_2 should be rotated by an angle of (degrees), so that the intensity of light on the screen becomes $\frac{3I}{8}$.

[26 Aug 2021 Shift 2]

Answer: 30

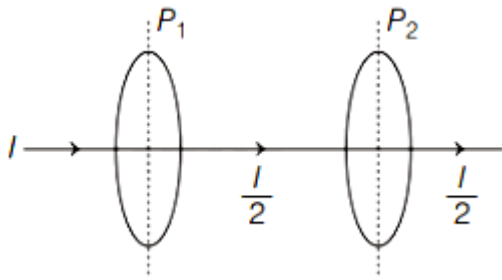
Solution:

Given, intensity of light falling on screen is I .

Intensity of light when P_1 and P_2 placed between screen and source is $I/2$.

As per Malus's law, the intensity of transmitted light when passing through a pair of polariser and analyser having angle ϕ between their axis is given as

$$I = I_0 \cos^2 \phi$$



Initially let us consider the angle between analyser and polariser is 0° . Then, the intensity of transmitted light is $I/2$. Thus, intensity of light become $I/2$ after transmitting through 1st polariser as shown in the figure.

If we rotate the polarise P_2 by angle ϕ , then the resulting intensity of light on screen will be $3I/8$.

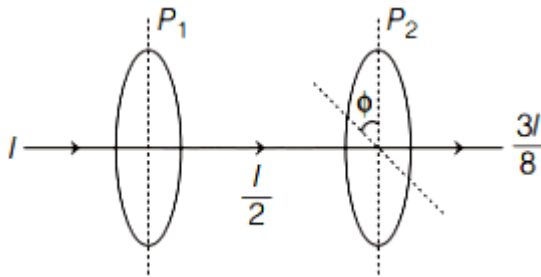
Now, for second condition, apply Malus's law,

$$\frac{3I}{8} = \frac{I}{2} \cos^2 \phi$$

$$\cos^2 \phi = \frac{3}{4}$$

$$\Rightarrow \cos \phi = \frac{\sqrt{3}}{2}$$

$$\phi = 30^\circ.$$



Thus, P_2 should be rotated by 30° to get intensity of light $3I/8$ on screen.

Question104

In a Young's double slit experiment, the slits are separated by 0.3 mm and the screen is 1.5m away from the plane of slits. Distance between fourth bright fringes on both sides of central bright is 2.4 cm. The frequency of light used is..... $\times 10^{14}$ Hz.

[31 Aug 2021 Shift 2]

Answer: 5

Solution:

Given, slit separation, $d = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$

Distance between slit and screen, $D = 1.5 \text{ m}$

Distance between 4th bright fringes $= 2Y_4 = 2.4 \text{ cm}$

Position of 4th fringe,

$$Y_4 = \frac{2.4}{2} = 1.2 \text{ cm} = 1.2 \times 10^{-2} \text{ m}$$

Using equation, $Y_n = \frac{n\lambda D}{d}$

For $n = 4$

$$1.2 \times 10^{-2} = \frac{4 \times \lambda \times 1.5}{0.3 \times 10^{-3}}$$

$\therefore$ Wavelength, $\lambda = 6 \times 10^{-7} \text{ m}$

Using $c = \lambda f$

$$\Rightarrow f = \frac{c}{\lambda} = \frac{3 \times 10^8}{6 \times 10^{-7}} = 5 \times 10^{14} \text{ Hz}$$

Hence, the frequency of light is $5 \times 10^{14} \text{ Hz}$.

So, correct answer is 5.

Question105

The light waves from two coherent sources have same intensity

$I_1 = I_2 = I_0$. In interference pattern the intensity of light at minima is zero. What will be the intensity of light at maxima?

[27 Aug 2021 Shift 2]

Options:

A. I_0

B. $2I_0$

C. $5I_0$

D. $4I_0$

Answer: D

Solution:

Given, intensity of two light waves

$$I_1 = I_2 = I_0$$

Since, $I_{\min} = 0$

Since, $I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$
 and $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$
 $= (2\sqrt{I_0})^2 = 4I_0$

Question 106

The width of one of the two slits in a Young's double slit experiment is three times the other slit. If the amplitude of the light coming from a slit is proportional to the slit-width, the ratio of minimum to maximum intensity in the interference pattern is $x : 4$ where x is

.....

[1 Sep 2021 Shift 2]

Answer: 1

Solution:

Given, $b_1 = 3b_2$

Here, b_1 = width of the one of the two slit

and b_2 = width of the other slit.

As we know that,

Intensity, $I \propto (\text{Amplitude})^2$

$$\Rightarrow \frac{I_1}{I_2} = \left(\frac{b_1}{b_2} \right)^2$$

$$\Rightarrow \frac{I_1}{I_2} = \left(\frac{3b_2}{b_2} \right)^2$$

As we know, the ratio of the minimum intensity to the maximum intensity in the interference pattern,

$$\frac{I_{\min}}{I_{\max}} = \left(\frac{\sqrt{I_1} - \sqrt{I_2}}{\sqrt{I_1} + \sqrt{I_2}} \right)^2$$

Substituting the values in the above equations, we get

$$\frac{I_{\min}}{I_{\max}} = \frac{(\sqrt{9I_2} - \sqrt{I_2})^2}{(\sqrt{9I_2} + \sqrt{I_2})^2} = \sqrt{\left(\frac{3-1}{3+1} \right)^2}$$

$$\frac{I_{\min}}{I_{\max}} = \frac{1}{4}$$

Comparing with, $\frac{I_{\min}}{I_{\max}} = \frac{x}{4}$

The value of $x = 1$.

Question107

In a Young's double slit experiment 15 fringes are observed on a small portion of the screen when light of wavelength 500 nm is used. Ten fringes are observed on the same section of the screen when another light source of wavelength λ is used. Then the value of λ is (in nm) _____.
[NA 9 Jan 2020 II]

Answer: 750

Solution:

Fringe width, $\beta = \frac{\lambda D}{d}$ where, λ = wavelength, D = distance of screen from slits, d = distance between slits

ATQ

$$15 \times \frac{\lambda_1 D}{d} = 10 \times \frac{\lambda_2 D}{d}$$

$$\Rightarrow 15\lambda_1 = 10\lambda_2$$

$$\Rightarrow \lambda_2 = 1.5\lambda_1 \quad 15\lambda_1 = 1.5 \times 500\text{nm}$$

$$\Rightarrow \lambda_2 = 750\text{nm}$$

Question108

In a double-slit experiment, at a certain point on the screen the path difference between the two interfering waves is $\frac{1}{8}$ th of a wavelength. The ratio of the intensity of light at that point to that at the centre of a bright fringe is:
[8 Jan 2020 II]

Options:

A. 0.853

B. 0.672

C. 0.568

D. 0.760

Answer: A

Solution:

Given, Path difference, $\Delta x = \frac{\lambda}{8}$

Phase differences, $\Delta\phi = \frac{2\pi}{\lambda}\Delta x$

$$= \frac{2\pi}{\lambda} \times \frac{\lambda}{8} = \frac{\pi}{4}$$

$$I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$$

$$\Rightarrow \frac{I}{I_0} = \cos^2\left(\frac{\frac{\pi}{4}}{2}\right) = \cos^2\left(\frac{\pi}{8}\right)$$

$$\Rightarrow \frac{I}{I_0} = 0.853$$

Question109

In a Young's double slit experiment, the separation between the slits is 0.15 mm. In the experiment, a source of light of wavelength 589 nm is used and the interference pattern is observed on a screen kept 1.5 m away. The separation between the successive bright fringes on the screen is:

[7 Jan 2020 II]

Options:

A. 6.9 mm

B. 3.9 mm

C. 5.9 mm

D. 4.9 mm

Answer: C

Solution:

Given, distance between screen and slits, $D = 1.5\text{m}$

Seperation between slits, $d = 0.15\text{mm}$

Wavelength of source of light, $\lambda = 589\text{nm}$

Fringe-width,

$$w = \frac{D}{d}\lambda = \frac{1.5}{0.15 \times 10^{-3}} \times 589 \times 10^{-9}\text{m}$$
$$= 589 \times 10^{-2}\text{mm} = 5.89\text{mm} \approx 5.9\text{mm}$$

Question110

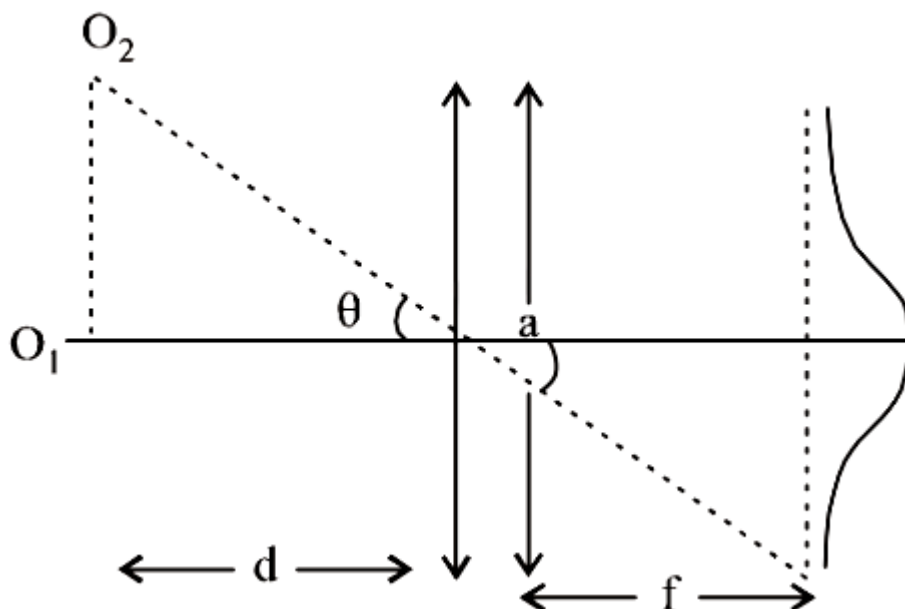
The aperture diameter of telescope is 5m. The separation between the moon and the earth is 4×10^5 km. With light of wavelength of $5500 \text{ \AA}$, the minimum separation between objects on the surface of moon, so that they are just resolved, is close to:
[9 Jan. 2020 I]

Options:

- A. 60 m
- B. 20 m
- C. 200 m
- D. 600 m

Answer: A

Solution:



Smallest angular separation between two distant objects here moon and earth,

$$\theta = 1.22 \frac{\lambda}{a}$$

a = aperture diameter of telescope

$$\text{Distance } O_1 O_2 = (\theta)d$$

Minimum separation between objects on the surface of moon, $= \left(1.22 \frac{\lambda}{a} \right) d$

$$= \frac{(1.22)(5500 \times 10^{-10}) \times 4 \times 10^5 \times 10^3}{5}$$

$$= 5368 \times 10^{-2} \text{m} = 53.68 \text{m} \approx 60 \text{m}$$

Question111

A polarizer - analyser set is adjusted such that the intensity of light coming out of the analyser is just 10% of the original intensity.

Assuming that the polarizer - analyser set does not absorb any light, the angle by which the analyser need to be rotated further to reduce the output intensity to be zero, is:

[7 Jan. 2020 I]

Options:

A. 71.6°

B. 18.4°

C. 90°

D. 45°

Answer: B

Solution:

According to question, the intensity of light coming out of the analyser is just 10% of the original intensity (I_0)

Using, $I = I_0 \cos^2 \theta$

$$\Rightarrow \frac{I_0}{10} = I_0 \cos^2 \theta \Rightarrow \frac{1}{10} = \cos^2 \theta$$

$$\Rightarrow \cos \theta = \frac{1}{\sqrt{10}} = 0.316 \Rightarrow \theta \approx 71.6^\circ$$

Therefore, the angle by which the analyser need to be rotated further to reduced the output intensity to be zero $\phi = 90^\circ - \theta = 90^\circ - 71.6^\circ = 18.4^\circ$

Question112

**In a double slit experiment, when a thin film of thickness t having refractive index μ is introduced in front of one of the slits, the maximum at the centre of the fringe pattern shifts by one fringe width. The value of t is (λ is the wavelength of the light used) :
[12 April 2020 I]**

Options:

A. $\frac{2\lambda}{(\mu - 1)}$

B. $\frac{\lambda}{2(\mu - 1)}$

C. $\frac{\lambda}{(\mu - 1)}$

D. $\frac{\lambda}{(2\mu - 1)}$

Answer: C

Solution:

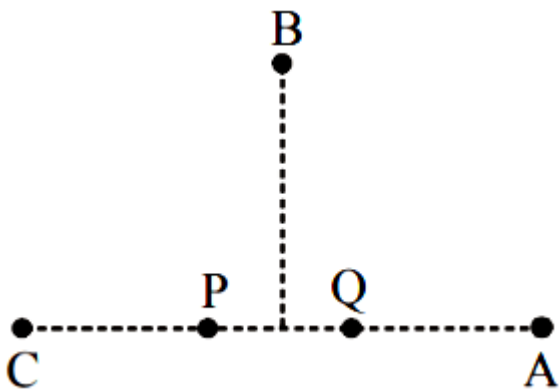
Given, $\Delta = \beta$

$$\text{or } \frac{D(\mu - 1)t}{d} = \frac{D\lambda}{d}$$

$$\therefore t = \frac{\lambda}{(\mu - 1)}$$

Question113

In the figure below, P and Q are two equally intense coherent sources emitting radiation of wavelength 20 m. The separation between P and Q is 5 m and the phase of P is ahead of that of Q by 90° . A, B and C are three distinct points of observation, each equidistant from the midpoint of PQ. The intensities of radiation at A, B, C will be in the ratio :



[Sep. 06, 2020 (I)]

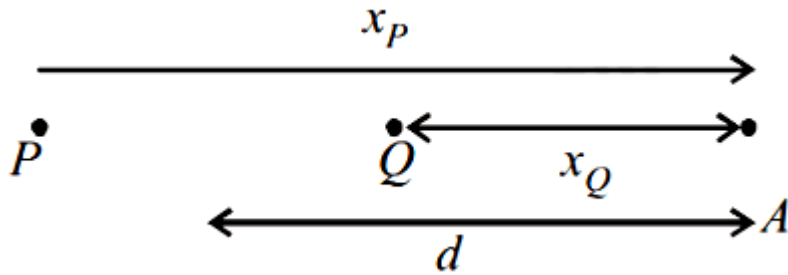
Options:

- A. 0 : 1 : 4
- B. 2 : 1 : 0
- C. 0 : 1 : 2
- D. 4 : 1 : 0

Answer: B

Solution:

For (A)



$$x_P - x_Q = (d + 2.5) - (d - 2.5) = 5\text{m}$$

Phase difference $\Delta\phi$ due to path difference

$$= \frac{2\pi}{\lambda}(\Delta x) = \frac{2\pi}{20}(5) = \frac{\pi}{2}$$

At A, Q is ahead of P by path, as wave emitted by Q reaches before wave emitted by P.

$$\therefore \text{Total phase difference at A } \frac{\pi}{2} - \frac{\pi}{2} = 0$$

(due to P being ahead of Q by 90°)

$$I_A = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\Delta\phi$$

$$= I + I + 2\sqrt{I}\sqrt{I}\cos(0) = 4I$$

For C

Path difference, $x_Q - x_P = 5\text{m}$

Phase difference $\Delta\phi$ due to path difference

$$= \frac{2\pi}{\lambda}(\Delta x) = \frac{2\pi}{20}(5) = \frac{\pi}{2}$$

$$\text{Total phase difference at C} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

$$I_{\text{net}} = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos(\Delta\phi)$$

$$= I + I + 2\sqrt{I}\sqrt{I}\cos(\pi) = 0$$

For B,

Path difference, $x_P - x_Q = 0$

$$\text{Phase difference, } \Delta\phi = \frac{\pi}{2}$$

(due to P being ahead of Q by 90°)

$$I_B = I + I + 2\sqrt{I}\sqrt{I}\cos\frac{\pi}{2} = 2I$$

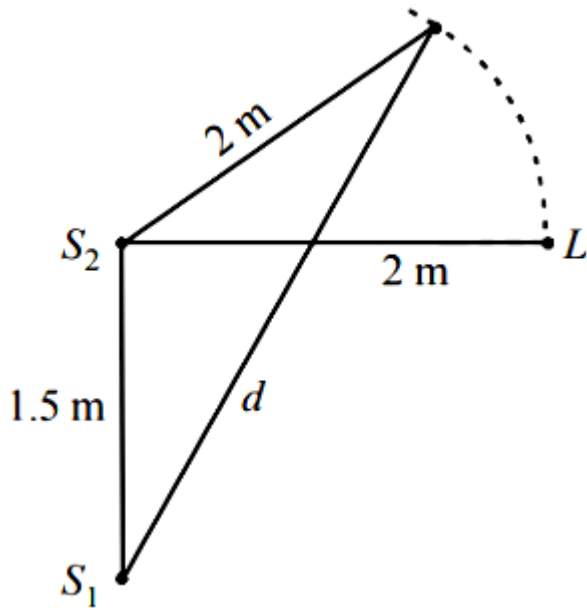
Therefore intensities of radiation at A, B and C will be in the ratio

$$I_A : I_B : I_C = 4I : 2I : 0 = 2 : 1 : 0$$

Question114

Two coherent sources of sound, S_1 and S_2 , produce sound waves of the same wavelength, $\lambda = 1\text{m}$, in phase. S_1 and S_2 are placed 1.5m apart (see fig.). A listener, located at L directly in front of S_2 finds that the intensity is at a minimum when he is 2m away from S_2 . The listener moves away from S_1 , keeping his distance from S_2 fixed. The

adjacent maximum of intensity is observed when the listener is at a distance d from S_1 . Then, d is:



[Sep. 05,2020 (II)]

Options:

- A. 12 m
- B. 5 m
- C. 2 m
- D. 3 m

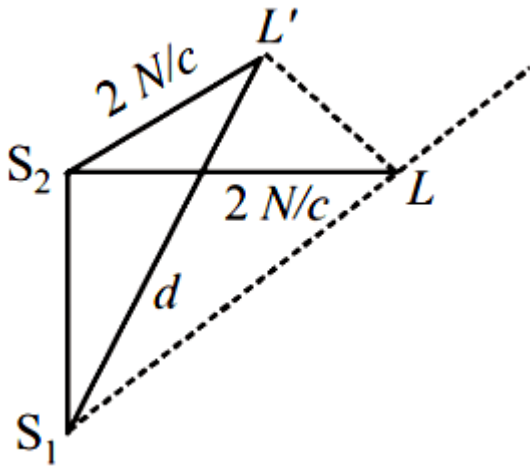
Answer: D

Solution:

Initially, $S_2L = 2\text{ m}$

$$S_1L = \sqrt{2^2 + \left(\frac{3}{2}\right)^2} = \frac{5}{2} = 2.5\text{ m}$$

$$\text{Path difference, } \Delta x = S_1L - S_2L = 0.5\text{ m} = \frac{\lambda}{2}$$



When the listner move from L, first maxima will appear if path difference is integral multiple of wavelength. For example

$$\Delta x = n\lambda = 1\lambda \quad (n = 1 \text{ for first maxima})$$

$$\therefore \Delta x = \lambda = S_1L' - S_2L$$

$$\Rightarrow 1 = d - 2 \Rightarrow d = 3\text{m}$$

Question115

Two light waves having the same wavelength λ in vacuum are in phase initially. Then the first wave travels a path L_1 through a medium of refractive index n_1 while the second wave travels a path of length L_2 through a medium of refractive index n_2 . After this the phase difference between the two waves is :

[Sep. 03,2020 (II)]

Options:

A. $\frac{2\pi}{\lambda} \left(\frac{L_2}{n_1} - \frac{L_1}{n_2} \right)$

B. $\frac{2\pi}{\lambda} \left(\frac{L_1}{n_1} - \frac{L_2}{n_2} \right)$

C. $\frac{2\pi}{\lambda} (n_1 L_1 - n_2 L_2)$

D. $\frac{2\pi}{\lambda} (n_2 L_1 - n_1 L_2)$

Answer: C

Solution:

The distance traversed by light in a medium of refractive index m in time t is given by

$$d = vt \dots\dots(i)$$

where v is velocity of light in the medium. The distance traversed by light in a vacuum in this time,

$$\Delta ct = c \times \frac{d}{v} \quad [\text{from equation (i)}]$$

$$= d \frac{c}{v} = \mu d \dots\dots(ii) \quad \left(\because \mu = \frac{c}{v} \right)$$

This distance is the equivalent distance in vacuum and is called optical path.

Optical path for first ray which travels a path L_1 through a medium of refractive index $n_1 = n_1 L_1$

Optical path for second ray which travels a path L_2 through a medium of refractive index $n_2 = n_2 L_2$

$$\text{Path difference} = n_1 L_1 - n_2 L_2$$

$$\text{Now, phase difference} = \frac{2\pi}{\lambda} \times \text{path difference} = \frac{2\pi}{\lambda} \times (n_1 L_1 - n_2 L_2)$$

Question 116

A young's double-slit experiment is performed using monochromatic light of wavelength λ . The intensity of light at a point on the screen, where the path difference is λ , is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{6}$ is given by $\frac{nK}{12}$, where n is an integer. The value of n is _____.
[NA Sep. 06, 2020 (II)]

Answer: 9

Solution:

In young's double slit experiment, intensity at a point is given by

$$I = I_0 \cos^2 \frac{\phi}{2} \dots\dots(i)$$

where, ϕ = phase difference,

$$\text{Using phase difference, } \phi = \frac{2\pi}{\lambda} \times \text{path difference}$$

$$\text{For path difference } \lambda, \text{ phase difference } \phi_1 = 2\pi$$

$$\text{For path difference, } \frac{\lambda}{6}, \text{ phase difference } \phi_2 = \frac{\pi}{3}$$

Using equation (i),

$$\frac{I_1}{I_2} = \frac{\cos^2\left(\frac{\phi_1}{2}\right)}{\cos^2\left(\frac{\phi_2}{2}\right)} = \frac{\cos^2\left(\frac{2\pi}{2}\right)}{\cos^2\left(\frac{\pi}{3}\right)}$$

$$\Rightarrow \frac{K}{I_2} = \frac{1}{\frac{3}{4}} = \frac{4}{3} \Rightarrow I_2 = \frac{3K}{4} = \frac{9K}{12}$$

$$\therefore n = 9$$

Question117

In a Young's double slit experiment, light of 500 nm is used to produce an interference pattern. When the distance between the slits is 0.05 mm, the angular width (in degree) of the fringes formed on the distance screen is close to :

[Sep. 03, 2020 (I)]

Options:

A. 0.17°

B. 0.57°

C. 1.7°

D. 0.07°

Answer: B

Solution:

Given : Wavelength of light, $\lambda = 500\text{nm}$
 Distance between the slits, $d = 0.05\text{mm}$
 Angular width of the fringe formed,

$$\theta = \frac{\lambda}{d} = \frac{500 \times 10^{-9}}{0.05 \times 10^{-3}} = 0.01 \text{ rad} = 0.57^\circ.$$

Question118

Interference fringes are observed on a screen by illuminating two thin slits 1 mm apart with a light source ($\lambda = 632.8 \text{ nm}$). The distance

between the screen and the slits is 100 cm. If a bright fringe is observed on a screen at a distance of 1.27 mm from the central bright fringe, then the path difference between the waves, which are reaching this point from the slits is close to :
[Sep. 02, 2020 (I)]

Options:

A. 1.27 μm

B. 2.87 nm

C. 2 nm

D. 2.05 μm

Answer: A

Solution:

Path difference, $\Delta P = d \sin \theta = d \theta$

d = distance between slits = 1mm = 10^{-3} mm

D = distance between the slits and screen = 100cm = 1m

y = distance between central bright fringe and observed fringe = 1.27mm

$$\therefore \Delta P = \frac{d y}{D} = \frac{10^{-3} \times 1.270\text{mm}}{1\text{m}} = 1.27\mu\text{m}$$

Question119

In a Young's double slit experiment, 16 fringes are observed in a certain segment of the screen when light of wavelength 700 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of the screen would be :
[Sep. 02, 2020 (II)]

Options:

A. 24

B. 30

C. 18

D. 28

Answer: D

Solution:

Let n_1 fringes are visible with light of wavelength λ_1 and n_2 with light of wavelength λ_2 . Then

$$\beta = \frac{n_1 D \lambda_1}{d} = \frac{n_2 D \lambda_2}{d} \left(\because \beta = \frac{n \lambda D}{d} \right)$$

$$\Rightarrow \frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2}$$

$$\Rightarrow n_2 = \frac{700}{400} \times 16 = 28$$

Question120

A beam of plane polarised light of large cross-sectional area and uniform intensity of 3.3 W m^{-2} falls normally on a polariser (cross sectional area 3 times 10^{-4} m^2) which rotates about its axis with an angular speed of 31.4 rad/s . The energy of light passing through the polariser per revolution, is close to :
[Sep. 04, 2020 (I)]

Options:

A. $1.0 \times 10^{-5} \text{ J}$

B. $1.0 \times 10^{-4} \text{ J}$

C. $1.5 \times 10^{-4} \text{ J}$

D. $5.0 \times 10^{-4} \text{ J}$

Answer: D

Solution:

Given:

Intensity, $I_0 = 3.3 \text{ W m}^{-2}$

Area, $A = 3 \times 10^{-4} \text{ m}^2$

Angular speed, $\omega = 31.4 \text{ rad/s}$

Average energy = $I_0 A \langle \cos^2 \theta \rangle$

$\therefore \langle \cos^2 \theta \rangle = \frac{1}{2}$ per revolution

$\therefore \text{Average energy} = \frac{(3.3)(3 \times 10^{-4})}{2} \approx 5 \times 10^{-4} \text{ J}$

Question121

Orange light of wavelength $6000 \times 10^{-10} \text{ m}$ illuminates a single slit of width $0.6 \times 10^{-4} \text{ m}$. The maximum possible number of diffraction minima produced on both sides of the central maximum is _____. [NA Sep. 04, 2020 (II)]

Answer: 198

Solution:

For obtaining secondary minima at a point path difference should be integral multiple of wavelength

$$\therefore d \sin \theta = n\lambda$$

$$\therefore \sin \theta = \frac{n\lambda}{d}$$

For n to be maximum $\sin \theta = 1$

$$n = \frac{d}{\lambda} = \frac{6 \times 10^{-5}}{6 \times 10^{-7}} = 100$$

Total number of minima on one side = 99

Total number of minima = 198.

Question122

Two coherent sources produce waves of different intensities which interfere. After interference, the ratio of the maximum intensity to the minimum intensity is 16. The intensity of the waves are in the ratio:

[9 Jan. 2019 I]

Options:

A. 16 : 9

B. 25 : 9

C. 4 : 1

D. 5 : 3

Answer: B

Solution:

As we know, $\frac{I_{\max}}{I_{\min}} = \left(\frac{A_1 + A_2}{A_1 - A_2} \right)^2$

and $\sqrt{\frac{I_1}{I_2}} = \frac{A_1}{A_2}$

$$\frac{I_{\max}}{I_{\min}} = 16 \Rightarrow \frac{A_{\max}}{A_{\min}} = 4 \Rightarrow \frac{A_1 + A_2}{A_1 - A_2} = \frac{4}{1}$$

Using componendo and dividendo.

$$\frac{A_1}{A_2} = \frac{5}{3} \Rightarrow \frac{I_1}{I_2} = \left(\frac{5}{3} \right)^2 = \frac{25}{9}$$

Question123

In a Young's double slit experiment, the path difference, at a certain point on the screen, between two interfering waves is $\frac{1}{8}$ th of wavelength. The ratio of the intensity at this point to that at the centre of a bright fringe is close to:

[11 Jan 2019 I]

Options:

A. 0.74

B. 0.85

C. 0.94

D. 0.80

Answer: B

Solution:

Given, path difference, $\Delta x = \frac{\lambda}{8}$

Phase difference ($\Delta\phi$) is given by

$$\Delta\phi = \frac{2\pi}{\lambda}(\Delta x)$$

$$\Delta\phi = \frac{(2\pi)\lambda}{\lambda} \frac{1}{8} = \frac{\pi}{4}$$

For two sources in different phases,

$$I = I_0 \cos^2\left(\frac{\pi}{8}\right)$$

$$\frac{I}{I_0} = \cos^2\left(\frac{\pi}{8}\right)$$

$$= \frac{1 + \cos \frac{\pi}{4}}{2} = \frac{1 + \frac{1}{\sqrt{2}}}{2} = 0.85$$

Question124

In a Young's double slit experiment with slit separation 0.1mm, one observes a bright fringe at angle $\frac{1}{40}$ rad by using light of wavelength λ_1 . When the light of wavelength λ_2 is used a bright fringe is seen at the same angle in the same set up. Given that λ_1 and λ_2 are in visible range (380nm to 740nm), their values are:
[10 Jan. 2019 I]

Options:

- A. 625 nm, 500 nm
- B. 380 nm, 525 nm
- C. 380 nm, 500 nm
- D. 400 nm, 500 nm

Answer: A

Solution:

$$\text{Path difference} = d \sin \theta \approx d \theta$$

$$= 0.1 \times \frac{1}{40} \text{ mm} = 2500 \text{ nm}$$

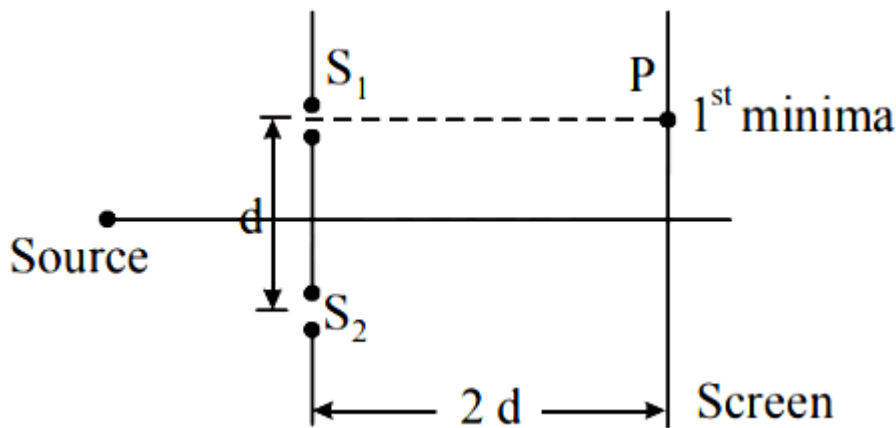
For bright fringe, path difference must be integral multiple of λ .

$$\therefore 2500 = n\lambda_1 = m\lambda_2$$

$$\therefore \lambda_1 = 625 \text{ (for } n = 4), \lambda_2 = 500 \text{ (for } m = 5)$$

Question 125

Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first minima occurs directly in front of the slit (S_1)?



[10 Jan. 2019 II]

Options:

A. $\frac{\lambda}{2(\sqrt{5} - 2)}$

B. $\frac{\lambda}{(\sqrt{5} - 2)}$

C. $\frac{\lambda}{2(5 - \sqrt{2})}$

D. $\frac{\lambda}{(5 - \sqrt{2})}$

Answer: A

Solution:

Here, $x_1 = 2d$ and $x_2 = \sqrt{5d}$
 For, first minima, $\Delta x = \frac{\lambda}{2}$
 $\therefore \Delta x = x_2 - x_1 = \sqrt{5}d - 2d = \frac{\lambda}{2}$
 $\Rightarrow d = \frac{\lambda}{2(\sqrt{5} - 2)}$

Question126

In a Young's double slit experiment, the slits are placed 0.320mm apart. Light of wavelength $\lambda = 500\text{nm}$ is incident on the slits. The total number of bright fringes that are observed in the angular range $-30^\circ \leq \theta \leq 30^\circ$ is
[9 Jan. 2019 II]

Options:

- A. 640
- B. 320
- C. 321
- D. 641

Answer: D

Solution:

For 'n' number of maximas

$$d \sin \theta = n\lambda$$

$$0.32 \times 10^{-3} \sin 30^\circ = n \times 500 \times 10^{-9}$$

$$\therefore n = \frac{0.32 \times 10^{-3}}{500 \times 10^{-9}} \times \frac{1}{2} = 320$$

Hence total no. of maximas observed in angular range $-30^\circ \leq \theta \leq 30^\circ$
 $= 320 + 1 + 320 = 641$

Question127

In a double-slit experiment, green light ($5303\text{\AA}$) falls on a double slit having a separation of $19.44\text{ }\mu\text{m}$ and a width of $4.05\text{ }\mu\text{m}$. The number of bright fringes between the first and the second diffraction minima is :
[11 Jan 2019 II]

Options:

A. 10

B. 05

C. 04

D. 09

Answer: B

Question128

Consider a tank made of glass (refractive index 1.5) with a thick bottom. It is filled with a liquid of refractive index μ . A student finds that, irrespective of what the incident angle i (see figure) is for a beam of light entering the liquid, the light reflected from the liquid glass interface is never completely polarized. For this to happen, the minimum value of μ is:
[9 Jan. 2019 I]

Options:

A. $\sqrt{\frac{5}{3}}$

B. $\frac{3}{\sqrt{5}}$

C. $\frac{5}{\sqrt{3}}$

D. $\frac{4}{3}$

Answer: B

Solution:

According to Brewster's law, refractive index of material (μ) is equal to tangent of polarising angle

$$\therefore \tan i_b = \mu = \frac{1.5}{\mu}$$

$$\frac{1}{\mu} < \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}} (\because \sin i_c < \sin i_b)$$

$$\therefore \sin i_b = \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}}$$

$$\text{or, } \sqrt{\mu^2 + (1.5)^2} < 1.5 \times \mu$$

$$\Rightarrow \mu^2 + (1.5)^2 < (\mu \times 1.5)^2$$

$$\Rightarrow \mu < \frac{3}{\sqrt{5}} \text{ i.e. minimum value of } \mu \text{ should be } \frac{3}{\sqrt{5}}$$

Question 129

In an interference experiment the ratio of amplitudes of coherent waves is $\frac{a_1}{a_2} = \frac{1}{3}$. The ratio of maximum and minimum intensities of

fringes will be :

[8 April 2019 I]

Options:

A. 2

B. 18

C. 4

D. 9

Answer: C

Solution:

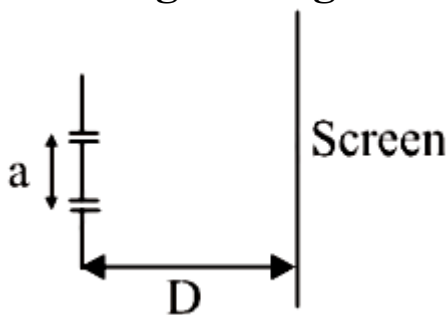
Given amplitude ratio of waves is $\frac{a_1}{a_2} = \frac{3}{1}$

$$\text{so, } \frac{I_{\max}}{I_{\min}} = \left(\frac{a_2 + a_1}{a_2 - a_1} \right)^2$$

$$= \left(\frac{\frac{a_2}{a_1} + 1}{\frac{a_2}{a_1} - 1} \right)^2 = \left(\frac{3 + 1}{3 - 1} \right)^2 = \left(\frac{4}{2} \right)^2 = \frac{4}{1} = 4$$

Question130

The figure shows a Young's double slit experimental setup. It is observed that when a thin transparent sheet of thickness t and refractive index $\frac{1}{4}$ is put in front of one of the slits, the central maximum gets shifted by a distance equal to n fringe widths. If the wavelength of light used is λ , t will be:



[9 April 2019 I]

Options:

A. $\frac{2nD\lambda}{a(\mu - 1)}$

B. $\frac{nD\lambda}{a(\mu - 1)}$

C. $\frac{D\lambda}{a(\mu - 1)}$

D. $\frac{2D\lambda}{a(\mu - 1)}$

E. (Bonus)

Answer: E

Solution:

Shift = $n\beta$ (given)

$$\therefore D \frac{(\mu - 1)t}{a} = \frac{n\lambda D}{a} \left[\because \text{Shift} = \frac{D(\mu - 1)t}{a} \right]$$

$$\text{or } t = \frac{n\lambda}{(\mu - 1)}$$

Question131

The value of numerical aperture of the objective lens of a microscope is 1.25. If light of wavelength $5000 \text{ \AA}$ is used, the minimum separation between two points, to be seen as distinct, will be :

[12 April 2019 I]

Options:

A. $0.24 \text{ }\mu\text{m}$

B. $0.38 \text{ }\mu\text{m}$

C. $0.12 \text{ }\mu\text{m}$

D. $0.48 \text{ }\mu\text{m}$

Answer: A

Solution:

$$\begin{aligned} x &= \frac{1.22\lambda}{2\mu \sin \theta} \\ &= \frac{1.22 \times 5000}{2 \times 1.25} = 0.24 \mu\text{m} \end{aligned}$$

Question132

A system of three polarizers P_1, P_2, P_3 is set up such that the pass axis of P_3 is crossed with respect to that of P_1 . The pass axis of P_2 is

inclined at 60° to the pass axis of P_3 . When a beam of unpolarized light of intensity I_0 is incident on P_1 , the intensity of light transmitted by the three polarizers is I . The ratio (I_0/I) equals (nearly):

[12 April 2019 II]

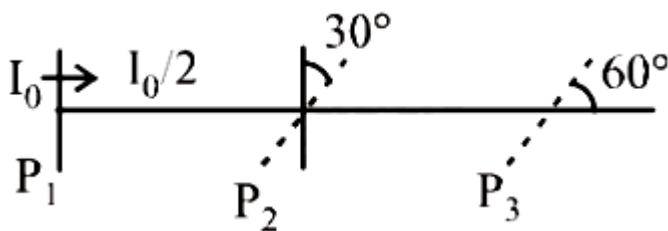
Options:

- A. 5.33
- B. 16.00
- C. 10.67
- D. 1.80

Answer: C

Solution:

$$I = \left(\frac{I_0}{2}\right) \cos^2 30^\circ \cos^2 60^\circ$$



$$= \frac{I_0}{2} \times \frac{3}{4} \times \frac{1}{4}$$

$$\therefore \frac{I_0}{I} = \frac{32}{3} = 10.67$$

Question133

Diameter of the objective lens of a telescope is 250cm. For light of wavelength 600nm. Coming from a distant object, the limit of resolution of the telescope is close to:

[9 April 2019 II]

Options:

A. $1.5 \times 10^{-7} \text{ rad}$

B. $2.0 \times 10^{-7} \text{ rad}$

C. $3.0 \times 10^{-7} \text{ rad}$

D. $4.5 \times 10^{-7} \text{ rad}$

Answer: C

Solution:

$$\theta = \frac{1.22\lambda}{d} = \frac{1.22 \times 600 \times 10^{-9}}{250 \times 10^{-2}} \\ = 3.0 \times 10^{-7} \text{ rad}$$

Question134

Calculate the limit of resolution of a telescope objective having a diameter of 200cm, if it has to detect light of wavelength 500nm coming from a star.

[8 April 2019 II]

Options:

A. $305 \times 10^{-9} \text{ radian}$

B. $610 \times 10^{-9} \text{ radian}$

C. $152.5 \times 10^{-9} \text{ radian}$

D. $457.5 \times 10^{-9} \text{ radian}$

Answer: A

Solution:

$$\theta = \frac{1.22\lambda}{d} = \frac{1.22 \times 500 \times 10^{-9}}{2} = 305 \times 10^{-9} \text{ rad}$$

Question135

The angular width of the central maximum in a single slit diffraction pattern is 60° . The width of the slit is $1\text{ }\mu\text{m}$. The slit is illuminated by monochromatic plane waves. If another slit of same width is made near it, Young's fringes can be observed on a screen placed at a distance 50 cm from the slits. If the observed fringe width is 1 cm , what is slit separation distance?

(i.e. distance between the centres of each slit.)

[2018]

Options:

A. $25\text{ }\mu\text{m}$

B. $50\text{ }\mu\text{m}$

C. $75\text{ }\mu\text{m}$

D. $100\text{ }\mu\text{m}$

Answer: A

Solution:

$$\text{Angular width of central maxima} = \frac{2\lambda}{d}$$

$$\text{or, } \lambda = \frac{d}{2}; \text{ Fringe width, } \beta = \frac{\lambda \times D}{d'}$$

$$10^{-2} = \frac{d}{2} \times \frac{50 \times 10^{-2}}{d'} = \frac{10^{-6} \times 50 \times 10^{-2}}{2 \times d'}$$

Therefore, slit separation distance, $d' = 25\mu\text{m}$

Question136

Unpolarized light of intensity I passes through an ideal polarizer A. Another identical polarizer B is placed behind A. The intensity of light beyond B is found to be $\frac{I}{2}$. Now another identical polarizer C is placed between A and B. The intensity beyond B is now found to be $\frac{I}{8}$

. The angle between polarizer A and C is:
[2018]

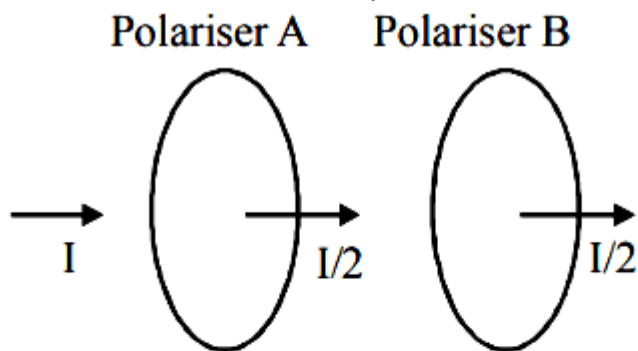
Options:

- A. 0°
- B. 30°
- C. 45°
- D. 60°

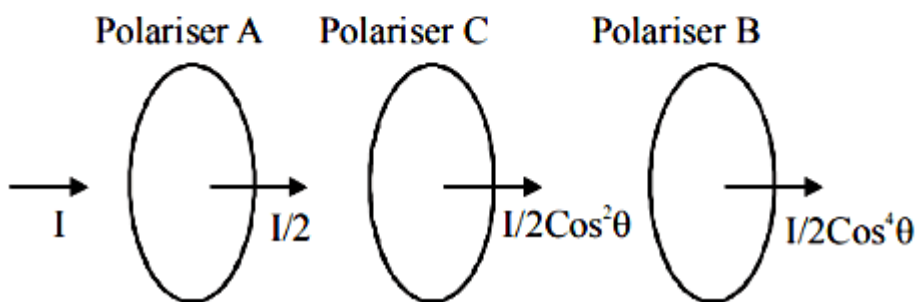
Answer: C

Solution:

Axis of transmission of A & B are parallel.



After introducing polariser C between A and B,



$$\frac{I}{2} \cos^4 \theta = \frac{I}{8} \Rightarrow \cos^4 \theta = \frac{1}{4}$$

$$\Rightarrow \cos \theta = \frac{1}{\sqrt{2}} \text{ or, } \theta = 45^\circ$$

Question137

Light of wavelength 550nm falls normally on a slit of width 22.0×10^{-5} cm. The angular position of the second minima from the central maximum will be (in radians)
[Online April 15,2018]

Options:

A. $\frac{\pi}{8}$

B. $\frac{\pi}{12}$

C. $\frac{\pi}{4}$

D. $\frac{\pi}{6}$

Answer: A

Solution:

If angular position of 2nd maxima from central maxima is θ then

$$\sin \theta = \frac{(2n-1)\lambda}{2a} = \frac{3\lambda}{20} = \frac{3 \times 550 \times 10^{-9}}{2 \times 22 \times 10^{-7}}$$

$$\therefore \theta \simeq \frac{\pi}{8} \text{ rad}$$

Question138

Unpolarized light of intensity I is incident on a system of two polarizers, A followed by B. The intensity of emergent light is I /2. If a third polarizer C is placed between A and B, the intensity of emergent light is reduced to I /3. The angle between the polarizers A and C is θ . Then
[Online April 16,2018]

Options:

A. $\cos \theta = \left(\frac{2}{3} \right)^{1/4}$

B. $\cos \theta = \left(\frac{1}{3}\right)^{1/4}$

C. $\cos \theta = \left(\frac{1}{3}\right)^{1/2}$

D. $\cos \theta = \left(\frac{2}{3}\right)^{1/2}$

Answer: A

Solution:

Polariser A and B have same alignment of transmission axis.

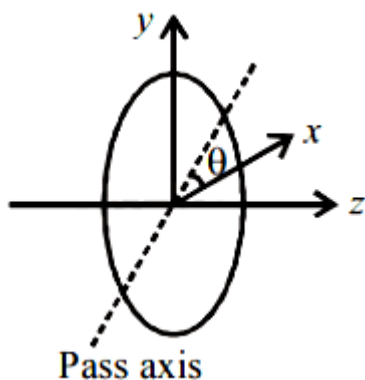
Lets assume polariser c is introduced at θ angle

$$\frac{1}{2} \cos^2 \theta \times \cos^2 \theta = \frac{1}{3}$$

$$\text{or, } \cos^4 \theta = \frac{2}{3} \Rightarrow \cos \theta = \left(\frac{2}{3}\right)^{1/4}$$

Question139

A plane polarized light is incident on a polariser with its pass axis making angle q with x-axis, as shown in the figure. At four different values of q , $q = 8^\circ, 38^\circ, 188^\circ$ and 218° , the observed intensities are same. What is the angle between the direction of polarization and x-axis



[Online April 15,2018]

Options:

A. 203°

B. 45°

C. 98°

D. 128°

Answer: A

Question140

In a Young's double slit experiment, slits are separated by 0.5 mm, and the screen is placed 150 cm away. A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes on the screen. The least distance from the common central maximum to the point where the bright fringes due to both the wavelengths coincide is :

[2017]

Options:

A. 9.75 mm

B. 15.6 mm

C. 1.56 mm

D. 7.8 mm

Answer: D

Solution:

For common maxima, $n_1\lambda_1 = n_2\lambda_2$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{520 \times 10^{-9}}{650 \times 10^{-9}} = \frac{4}{5}$$

For λ_1

$$y = \frac{n_1\lambda_1 D}{d}, \lambda_1 = 650\text{nm}$$

$$y = \frac{4 \times 650 \times 10^{-9} \times 1.5}{0.5 \times 10^{-3}} \text{ or, } y = 7.8\text{mm}$$

Question141

An observer is moving with half the speed of light towards a stationary microwave source emitting waves at frequency 10 GHz. What is the frequency of the microwave measured by the observer? (speed of light = $3 \times 10^8 \text{ ms}^{-1}$) [2017]

Options:

- A. 17.3 GHz
- B. 15.3 GHz
- C. 10.1 GHz
- D. 12.1 GHz

Answer: A

Solution:

Use relativistic doppler's effect as velocity of observer is not small as compared to light

$$f = f_0 \sqrt{\frac{c+v}{c-v}}; V = \text{relative speed of approach}$$

$$f_0 = 10 \text{ GHz}$$

$$f = 10 \sqrt{c + \frac{c}{2} - \frac{c}{2}} = 10\sqrt{3} = 17.3 \text{ GHz}$$

Question142

A single slit of width 0.1 mm is illuminated by a parallel beam of light of wavelength 6000 Å and diffraction bands are observed on a screen 0.5 m from the slit. The distance of the third dark band from the central bright band is : [Online April 9, 2017]

Options:

- A. 3 mm
- B. 9 mm
- C. 4.5 mm
- D. 1.5 mm

Answer: B

Solution:

$$a = 0.1\text{mm} = 10^{-4}\text{cm}$$

$$\lambda = 6000 \times 10^{-10}\text{cm} = 6 \times 10^{-7}\text{cm}, D = 0.5\text{m}$$

$$\text{for } 3^{\text{rd}} \text{ dark band, } a \sin \theta = 3\lambda \text{ or } \sin \theta = \frac{3\lambda}{a} = \frac{x}{D}$$

$$\text{The distance of the third dark band from the central bright band } x = \frac{3\lambda D}{a} = \frac{3 \times 6 \times 10^{-7} \times 0.5}{10^{-4}} = 9\text{mm}$$

Question143

A single slit of width b is illuminated by a coherent monochromatic light of wavelength λ . If the second and fourth minima in the diffraction pattern at a distance l m from the slit are at 3 cm and 6 cm respectively from the central maximum, what is the width of the central maximum?

(i.e. distance between first minimum on either side of the central maximum)

[Online April 8, 2017]

Options:

- A. 1.5 cm
- B. 3.0 cm
- C. 4.5 cm
- D. 6.0 cm

Answer: B

Solution:

For secondary minima,

$$b \sin \theta = n\lambda \Rightarrow \sin \theta = \frac{n\lambda}{b}$$

Distance of n^{th} secondary minima $x = D \sin \theta$

$$\text{or } \sin \theta_1 = \frac{x_1}{D}$$

$$\sin \theta_1 = \frac{2\lambda}{b}$$

$$n = 4$$

$$\sin \theta_2 = \frac{4\lambda}{b} = \frac{x_2}{D}$$

$$x_2 - x_1 = \frac{4\lambda}{b} - \frac{2\lambda}{b} = \frac{2\lambda}{b}$$

$$3 = \frac{2\lambda}{b} \Rightarrow b = \frac{2\lambda}{3} \dots\dots(i)$$

$$\text{Width of central maxima} = \frac{2\lambda}{b}$$

$$= \frac{2\lambda}{\frac{2\lambda}{3}} = 3\text{cm} \dots \text{from eq. (i)}$$

Question144

The box of a pin hole camera, of length L , has a hole of radius a . It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say b_{min}) when :

[2016]

Options:

A. $a = \sqrt{\lambda L}$ and $b_{\text{min}} = \sqrt{4\lambda L}$

B. $a = \frac{\lambda^2}{L}$ and $b_{\text{min}} = \sqrt{4\lambda L}$

C. $a = \frac{\lambda^2}{L}$ and $b_{\text{min}} = \left(\frac{2\lambda^2}{L} \right)$

D. $a = \sqrt{\lambda}$ and $b_{\text{min}} = \left(\frac{2\lambda^2}{L} \right)$

Answer: A

Solution:

Given geometrical spread = a

$$\text{Diffraction spread} = \frac{\lambda}{a} \times L = \frac{\lambda L}{a}$$

$$\text{The sum } b = a + \frac{\lambda L}{a}$$

For b to be minimum

$$\frac{db}{da} = 0 \Rightarrow \frac{d}{da} \left(a + \frac{\lambda L}{a} \right) = 0$$

$$a = \sqrt{\lambda L}$$

$$b_{\min} = \sqrt{\lambda L} + \sqrt{\lambda L} = 2\sqrt{\lambda L} = \sqrt{4\lambda L}$$

Question 145

Two stars are 10 light years away from the earth. They are seen through a telescope of objective diameter 30cm. The wavelength of light is 600nm. To see the stars just resolved by the telescope, the minimum distance between them should be (1 light year = 9.46×10^{15} m) of the order of :
[Online April 10, 2016]

Options:

A. 10^8 km

B. 10^{10} km

C. 10^{11} km

D. 10^6 km

Answer: A

Solution:

$$\text{We know that } \Delta\theta = \frac{0.61\lambda}{R} = \frac{1}{R}$$

$$\text{The minimum distance between them } l = \frac{R}{9} \times 0.61 \times \lambda = \frac{9.46 \times 10^{15} \times 10 \times 0.61 \times 600 \times 10^{-9}}{0.3}$$

$$= 1.15 \times 10^{11} \text{ m}$$

$$\Rightarrow 1.115 \times 10^8 \text{ km}$$

Question146

In Young's double slit experiment, the distance between slits and the screen is 1.0m and monochromatic light of 600nm is being used. A person standing near the slits is looking at the fringe pattern. When the separation between the slits is varied, the interference pattern disappears for a particular distance d_0 between the slits. If the angular resolution of the eye is $\frac{1^\circ}{60}$, the value of d_0 is close to
[Online April 9,2016]

Options:

- A. 1 mm
- B. 3 mm
- C. 2 mm
- D. 4 mm

Answer: C

Solution:

Given $D = 1.0\text{m}$, wavelength of monochromatic light $\lambda = 600\text{nm}$.

$$d : D\theta = 1 \times \frac{\pi}{180} \times \frac{1}{60}$$

$$d_0 = 2 \times 10^{-3} = 2\text{mm}$$

Question147

On a hot summer night, the refractive index of air is smallest near the ground and increases with height from the ground. When a light beam is directed horizontally, the Huygens' principle leads us to

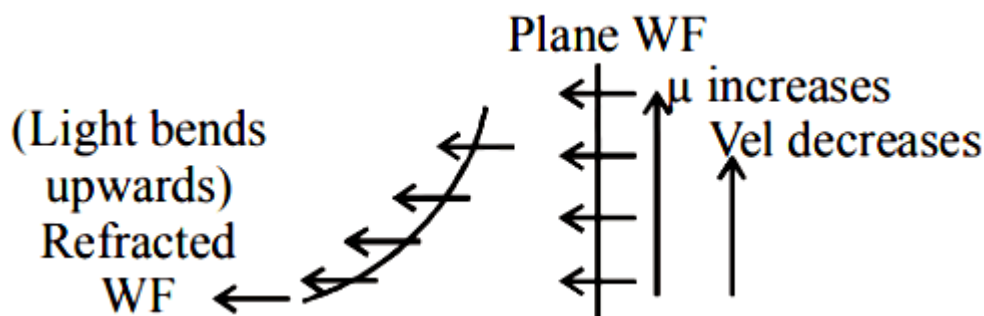
**conclude that as it travels, the light beam :
[2015]**

Options:

- A. bends downwards
- B. bends upwards
- C. becomes narrower
- D. goes horizontally without any deflection

Answer: B

Solution:



Question148

**In a Young's double slit experiment with light of wavelength λ the separation of slits is d and distance of screen is D such that $D \gg d \gg \lambda$. If the fringe width is β , the distance from point of maximum intensity to the point where intensity falls to half of maximum intensity on either side is:
[Online April 11,2015]**

Options:

- A. $\frac{\beta}{6}$
- B. $\frac{\beta}{3}$

C. $\frac{\beta}{4}$

D. $\frac{\beta}{2}$

Answer: C

Solution:

$$2I_0 = 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right) \text{ here, } \Delta\phi = \frac{\pi}{2}$$

$$\text{But, } \Delta\phi = \frac{2\pi}{\lambda} \Delta x \text{ so, } \Delta x = \frac{\lambda}{4}$$

$$\frac{dy}{D} = \frac{\lambda}{4} \dots\dots(i)$$

$$\frac{\lambda D}{d} = \beta \dots\dots(ii)$$

Multiplying equation (i) and (ii) we get,

$$y = \frac{\beta}{4}$$

Question149

Assuming human pupil to have a radius of 0.25 cm and a comfortable viewing distance of 25 cm, the minimum separation between two objects that human eye can resolve at 500 nm wavelength is :

[2015]

Options:

A. 100 μm

B. 300 μm

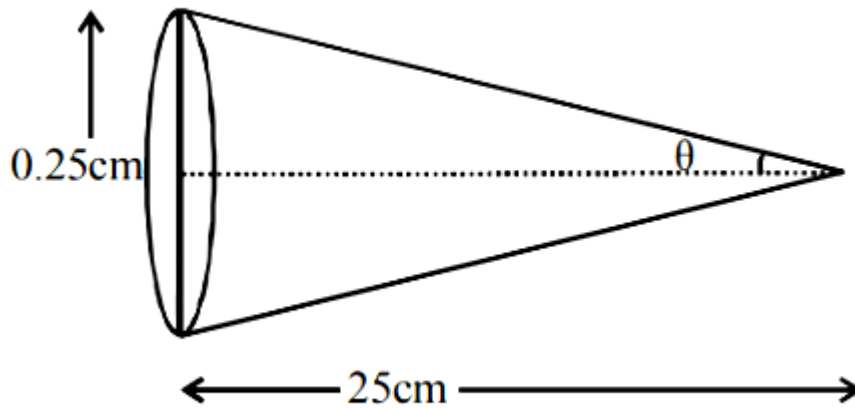
C. 1 μm

D. 30 μm

Answer: D

Solution:

$$\sin \theta = \frac{0.25}{25} = \frac{1}{100}$$



$$\text{Resolving power} = \frac{1.22\lambda}{2\mu \sin \theta} = 30\mu m$$

Question150

Unpolarized light of intensity I_0 is incident on surface of a block of glass at Brewster's angle. In that case, which one of the following statements is true ?

[Online April 11, 2015]

Options:

- A. reflected light is completely polarized with intensity less than $\frac{I_0}{2}$
- B. transmitted light is completely polarized with intensity less than $\frac{I_0}{2}$
- C. transmitted light is partially polarized with intensity $\frac{I_0}{2}$
- D. reflected light is partially polarized with intensity $\frac{I_0}{2}$

Answer: A

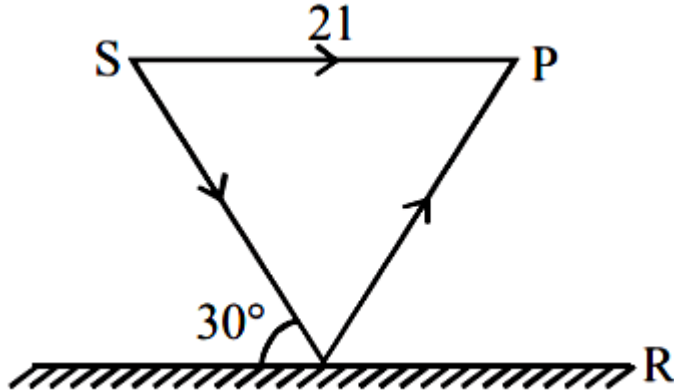
Solution:

When unpolarised light is incident at Brewster's angle then reflected light is completely polarized and the intensity of the reflected light is less than half of the incident light.

Question151

Interference pattern is observed at 'P' due to superimposition of two rays coming out from a source 'S' as shown in the figure. The value of 'l' for which maxima is obtained at 'P' is:

(R is perfect reflecting surface)



[Online April 12, 2014]

Options:

A. $l = \frac{2n\lambda}{\sqrt{3} - 1}$

B. $l = \frac{(2n - 1)\lambda}{2(\sqrt{3} - 1)}$

C. $l = \frac{(2n - 1)\lambda\sqrt{3}}{4(2 - \sqrt{3})}$

D. $l = \frac{(2n - 1)\lambda}{\sqrt{3} - 1}$

Answer: C

Question152

Two monochromatic light beams of intensity 16 and 9 units are interfering. The ratio of intensities of bright and dark parts of the

resultant pattern is:
[Online April 11, 2014]

Options:

A. $\frac{16}{9}$

B. $\frac{4}{3}$

C. $\frac{7}{1}$

D. $\frac{49}{1}$

Answer: D

Solution:

Intensity $\propto$ (amplitude)²

$$\frac{I_1}{I_2} = \frac{16}{9} = \frac{a_1^2}{a_2^2}$$

$$\Rightarrow a_1 = 4; a_2 = 3$$

Therefore the ratio of intensities of bright and dark parts $\frac{I_{\text{Bright}}}{I_{\text{Dark}}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \frac{(4 + 3)^2}{(4 - 3)^2} = \frac{49}{1}$

Question153

In a Young's double slit experiment, the distance between the two identical slits is 6.1 times larger than the slit width. Then the number of intensity maxima observed within the central maximum of the single slit diffraction pattern is:

[Online April 19, 2014]

Options:

A. 3

B. 6

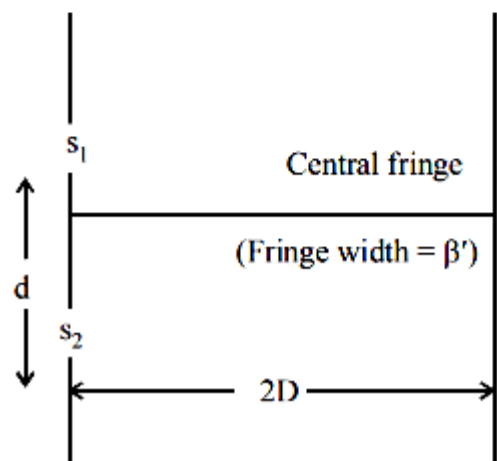
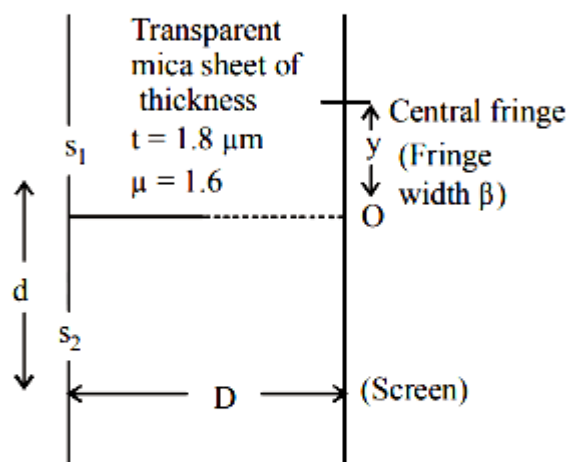
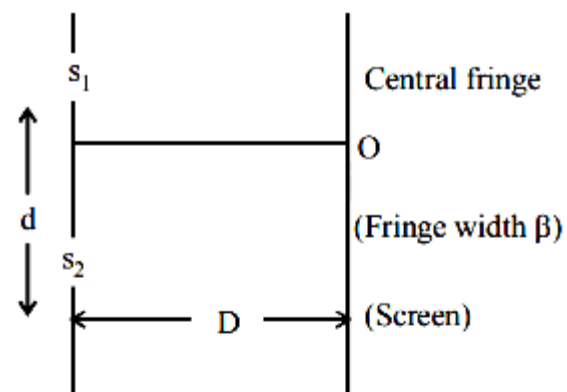
C. 12

D. 24

Answer: C

Question154

Using monochromatic light of wavelength λ , an experimentalist sets up the Young's double slit experiment in three ways as shown. If she observes that $y = \beta'$, the wavelength of light used is:



[Online April 9, 2014]

Options:

- A. 520 nm
- B. 540 nm
- C. 560 nm
- D. 580 nm

Answer: B

Solution:

(b) Given $t = 1.8 \times 10^{-6} \text{ m}$

$\mu = 1.6$

$n = 2$ (from figure) Applying formula $(\mu - 1)t = n\lambda$

$$(1.6 - 1) \times 1.8 \times 10^{-6} = 2\lambda$$

$$\text{or, } \lambda = \frac{1.8 \times 10^{-6} \times 0.6}{2}$$

$$= 540 \text{ nm}$$

Question155

Two beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through 30° makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and I_B respectively, then $\frac{I_A}{I_B}$ equals:

[2014]

Options:

A. 3

B. $\frac{3}{2}$

C. 1

D. $\frac{1}{3}$

Answer: D

Solution:

According to malus law, intensity of emerging beam is given by,

$$I = I_0 \cos^2 \theta$$

$$\text{Now, } I_{A'} = I_A \cos^2 30^\circ$$

$$I_{B'} = I_B \cos^2 60^\circ$$

$$\text{As } I_{A'} = I_{B'}$$

$$\Rightarrow I_A \times \frac{3}{4} = I_B \times \frac{1}{4}; \frac{I_A}{I_B} = \frac{1}{3}$$

Question156

The diameter of the objective lens of microscope makes an angle β at the focus of the microscope. Further, the medium between the object and the lens is an oil of refractive index n . Then the resolving power of the microscope
[Online April 19,2014]

Options:

- A. increases with decreasing value of n
- B. increases with decreasing value of β
- C. increases with increasing value of $n \sin 2\beta$
- D. increases with increasing value of $\frac{1}{n \sin 2\beta}$

Answer: C

Solution:

Resolving power of microscope,

$$R.P. = \frac{2n \sin \theta}{\lambda}$$

λ = Wavelength of light used to illuminate the object

n = Refractive index of the medium between object and objective

θ = Angle

Question157

A ray of light is incident from a denser to a rarer medium. The critical angle for total internal reflection is θ_{iC} and Brewster's angle of incidence is θ_{iB} , such that $\sin \theta_{iC} / \sin \theta_{iB} = \eta = 1.28$. The relative refractive index of the two media is:
[Online April 19,2014]

Options:

A. 0.2

B. 0.4

C. 0.8

D. 0.9

Answer: C

Solution:

Here, $\sin \frac{\theta_{ic}}{\sin \theta_{iB}} = 1.28$

As we know,

$$\mu = \frac{\sin \theta_{iB}}{\sin \left(\frac{\pi}{2} - \theta_{iB} \right)}$$

where, θ_{iB} is Brewster's angle of incidence,

$$\text{And, } \mu = \frac{1}{\sin \theta_{ic}}$$

On solving we get, relative refractive index of the two media.

Question158

In an experiment of single slit diffraction pattern, first minimum for red light coincides with first maximum of some other wavelength. If wavelength of red light is $6600 \text{ \AA}$, then wavelength of first maximum will be:

[Online April 12, 2014]

Options:

A. $3300 \text{ \AA}$

B. $4400 \text{ \AA}$

C. $5500 \text{ \AA}$

D. $6600 \text{ \AA}$

Answer: B

Solution:

In a single slit experiment,

For diffraction maxima,

$$a \sin \theta = (2n + 1) \frac{\lambda}{2}$$

and for diffraction minima,

$$a \sin \theta = n\lambda$$

According to question,

$$(2 \times 1 + 1) \frac{\lambda}{2} = 1 \times 6600$$

$$(\because \lambda_R = 6600 \text{ \AA})$$

$$\lambda = \frac{6600 \times 2}{3}$$

$$\lambda = 4400 \text{ \AA}$$

Question159

n identical waves each of intensity I_0 interfere with each other. The ratio of maximum intensities if the interference is (i) coherent and (ii) incoherent is:
[Online April 23, 2013]

Options:

A. n^2

B. $\frac{1}{n}$

C. $\frac{1}{n^2}$

D. n

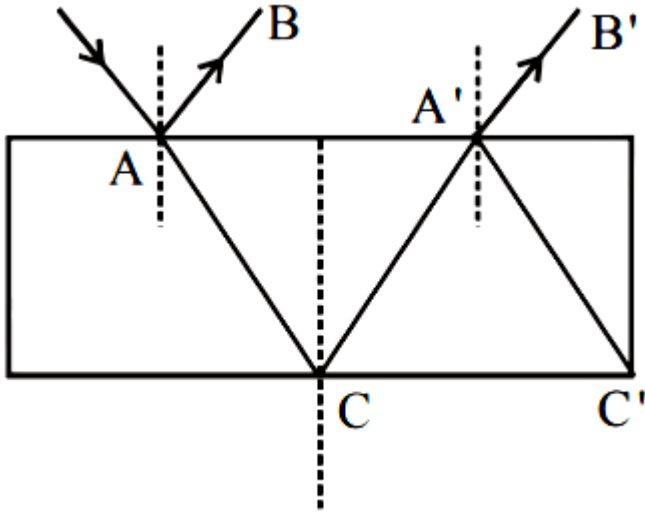
Answer: D

Solution:

$$\begin{aligned} & \frac{(\text{Maximum intensity}) \text{ coherent interference}}{(\text{Maximum intensity}) \text{ in coherent interference}} \\ &= \frac{n^2 I_0}{n I_0} = n \end{aligned}$$

Question160

A ray of light of intensity I is incident on a parallel glass slab at point A as shown in diagram. It undergoes partial reflection and refraction. At each reflection, 25% of incident energy is reflected. The rays AB and A'B' undergo interference. The ratio of $I_{\max}$ and $I_{\min}$ is :



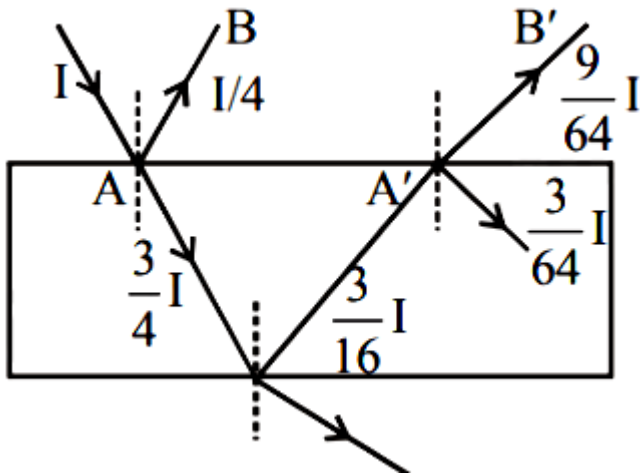
[Online April 9, 2013]

Options:

- A. 49 : 1
- B. 7 : 1
- C. 4 : 1
- D. 8 : 1

Answer: A

Solution:



From figure $I_1 = \frac{I}{4}$ and $I_2 = \frac{9I}{64}$

$$\Rightarrow \frac{I_2}{I_1} = \frac{9}{16}$$

$$\begin{aligned} \text{By using } \frac{I_{\max}}{I_{\min}} &= \left(\frac{\sqrt{\frac{I_2}{I_1} + 1}}{\sqrt{\frac{I_2}{I_1} - 1}} \right)^2 \\ &= \left(\frac{\sqrt{\frac{9}{16} + 1}}{\sqrt{\frac{9}{16} - 1}} \right)^2 = \frac{49}{1} \end{aligned}$$

Question161

Two coherent point sources S_1 and S_2 are separated by a small distance 'd' as shown. The fringes obtained on the screen will be [2013]

Options:

- A. points
- B. straight lines
- C. semi-circles
- D. concentric circles

Answer: D

Solution:

It will be concentric circles.

Question162

The source that illuminates the double - slit in 'double - slit interference experiment' emits two distinct monochromatic waves of wavelength 500 nm and 600 nm, each of them producing its own pattern on the screen. At the central point of the pattern when path difference is zero, maxima of both the patterns coincide and the resulting interference pattern is most distinct at the region of zero path difference.

But as one moves out of this central region, the two fringe systems are gradually out of step such that maximum due to one wavelength coincides with the minimum due to the other and the combined fringe system becomes completely indistinct. This may happen when path difference in nm is :

[Online April 25, 2013]

Options:

- A. 2000
- B. 3000
- C. 1000
- D. 1500

Answer: D

Question163

A thin glass plate of thickness is $\frac{2500}{3}\lambda$ (λ is wavelength of light used) and refractive index $\mu = 1.5$ is inserted between one of the slits and

the screen in Young's double slit experiment. At a point on the screen equidistant from the slits, the ratio of the intensities before and after the introduction of the glass plate is :

[Online April 25, 2013]

Options:

A. 2 : 1

B. 1 : 4

C. 4 : 1

D. 4 : 3

Answer: C

Question164

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement-1: In Young's double slit experiment, the number of fringes observed in the field of view is small with longer wavelength of light and is large with shorter wavelength of light.

Statement-2: In the double slit experiment the fringe width depends directly on the wavelength of light.

[Online April 22, 2013]

Options:

A. Statement-1 is true, Statement-2 is true and the Statement - 2 is correct explanation of the Statement-1.

B. Statement-1 is false and the Statement- 2 is true.

C. Statement-1 is true Statement-2 is true and the Statement-2 is not correct explanation of the Statement-1.

D. Statement-1 is true and the Statement-2 is false.

Answer: C

Solution:

$$\text{Fringe width } B = \frac{D\lambda}{d}$$

And number of fringes observed in the field of view is obtained by $\frac{d}{\lambda}$

Question165

A beam of unpolarised light of intensity I_0 is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of 45° relative to that of A. The intensity of the emergent light is [2013]

Options:

A. I_0

B. $I_0/2$

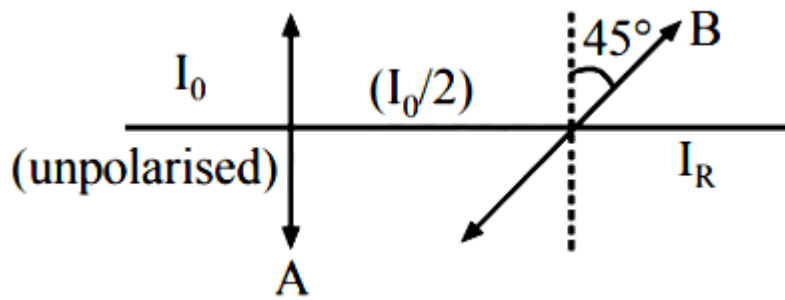
C. $I_0/4$

D. $\frac{I_0}{8}$

Answer: C

Solution:

Relation between intensities



$$I_r = \left(\frac{I_0}{2}\right) \cos^2(45^\circ) = \frac{I_0}{2} \times \frac{1}{2} = \frac{I_0}{4}$$

Question166

This question has Statement-1 and Statements-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement-1 : Out of radio waves and microwaves, the radio waves undergo more diffraction.

Statement-2 : Radio waves have greater frequency compared to microwaves.

[Online April 25, 2013]

Options:

- A. Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of Statement-1
- B. Statement-1 is false , Statement-2 is true.
- C. Statement-1 is true, Statement-2 is false.
- D. Statement-1 is true, Statement-2 is true but Statement2 is not the correct explanation of Statement-1

Answer: C

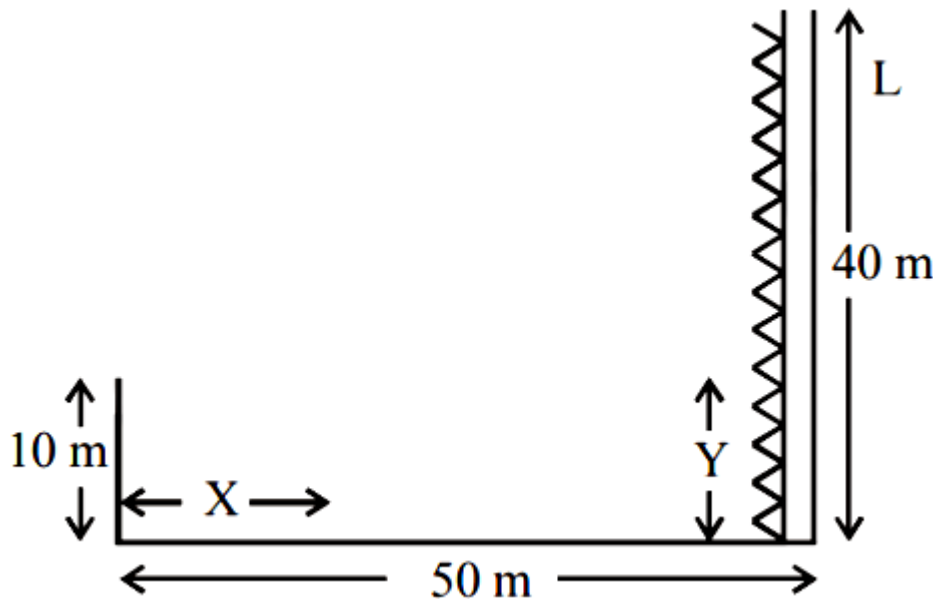
Solution:

Wavelength of radio waves is greater than microwaves hence frequency of radio waves is less than microwaves.

The degree of diffraction is greater whose wavelength is greater.

Question167

A person lives in a high-rise building on the bank of a river 50 m wide. Across the river is a well lit tower of height 40 m. When the person, who is at a height of 10 m, looks through a polarizer at an appropriate angle at light of the tower reflecting from the river surface, he notes that intensity of light coming from distance X from his building is the least and this corresponds to the light coming from light bulbs at height 'Y' on the tower. The values of X and Y are respectively close to (refractive index of water $\simeq \frac{4}{3}$)



[Online April 9, 2013]

Options:

- A. 25 m, 10 m
- B. 13 m, 27 m
- C. 22 m, 13 m
- D. 17 m, 20 m

Answer: B

Question168

Two coherent plane light waves of equal amplitude makes a small angle $\alpha (< 1)$ with each other. They fall almost normally on a screen. If λ is the wavelength of light waves, the fringe width Δx of interference patterns of the two sets of waves on the screen is
[Online May 19, 2012]

Options:

A. $\frac{2\lambda}{\alpha}$

B. $\frac{\lambda}{\alpha}$

C. $\frac{\lambda}{(2\alpha)}$

D. $\frac{\lambda}{\sqrt{\alpha}}$

Answer: C

Solution:

$$\Delta x = \frac{\lambda}{(2\alpha)}$$

Question169

In Young's double slit experiment, one of the slit is wider than other, so that amplitude of the light from one slit is double of that other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at phase difference ϕ is given by:
[2012]

Options:

A. $\frac{I_m}{9}(4 + 5 \cos \phi)$

$$\text{B. } \frac{I_m}{3} \left(1 + 2\cos^2 \frac{\phi}{2} \right)$$

$$\text{C. } \frac{I_m}{5} \left(1 + 4\cos^2 \frac{\phi}{2} \right)$$

$$\text{D. } \frac{I_m}{9} \left(1 + 8\cos^2 \frac{\phi}{2} \right)$$

Answer: D

Solution:

Let a_1 be the amplitude of light from first slit and a_2 be the amplitude of light from second slit.

$a_1 = a$, Then $a_2 = 2a$

Intensity $I \propto (\text{amplitude})^2$

$$\therefore I_1 = a_1^2 = a^2$$

$$I_2 = a_2^2 = 4a^2 = 4I_1$$

$$I_r = a_1^2 + a_2^2 + 2a_1a_2 \cos \phi$$

$$= I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$I_r = I_1 + 4I_1 + 2\sqrt{4I_1^2} \cos \phi$$

$$\Rightarrow I_r = 5I_1 + 4I_1 \cos \phi \dots\dots\dots (1)$$

$$\text{Now, } I_{\max} = (a_1 + a_2)^2 = (a + 2a)^2 = 9a^2$$

$$I_{\max} = 9I_1 \Rightarrow I_1 = \frac{I_{\max}}{9}$$

Substituting in equation (1)

$$I_r = \frac{5I_{\max}}{9} + \frac{4I_{\max}}{9} \cos \phi$$

$$I_r = \frac{I_{\max}}{9} [5 + 4 \cos \phi]$$

$$I_r = \frac{I_{\max}}{9} \left[5 + 8\cos^2 \frac{\phi}{2} - 4 \right]$$

$$I_r = \frac{I_{\max}}{9} \left[1 + 8\cos^2 \frac{\phi}{2} \right]$$

Question170

In Young's double slit interference experiment, the slit widths are in the ratio 1 : 25. Then the ratio of intensity at the maxima and minima in the interference pattern is
[Online May 26, 2012]

Options:

A. 3 : 2

B. 1 : 25

C. 9 : 4

D. 1 : 5

Answer: C

Solution:

We know that,

$$\frac{I_{\max}}{I_{\min}} = \frac{\left(\sqrt{\frac{\omega_1}{\omega_2} + 1} \right)^2}{\left(\sqrt{\frac{\omega_1}{\omega_2} - 1} \right)^2}$$

$I_{\max}$ and $I_{\min}$ are maximum and minimum intensity ω_1 and ω_2 are widths of two slits

$$\therefore \frac{I_{\max}}{I_{\min}} = \frac{\left(\sqrt{\frac{1}{25} + 1} \right)^2}{\left(\sqrt{\frac{1}{25} - 1} \right)^2} \left(\frac{\omega_1}{\omega_2} = \frac{1}{25} \text{ given} \right)$$

On solving we get,

$$\frac{I_{\max}}{I_{\min}} = \frac{\frac{36}{25}}{\frac{16}{25}} = \frac{9}{4} = 9 : 4$$

Question 171

The maximum number of possible interference maxima for slit separation equal to 1.8λ , where λ is the wavelength of light used, in a Young's double slit experiment is
[Online May 12, 2012]

Options:

A. zero

B. 3

C. infinite

D. 5

Answer: B

Solution:

As $\sin \theta = \frac{n\lambda}{d}$ and $\sin \theta$ cannot be $\neq 1$

$$\therefore 1 = \frac{n\lambda}{1.8\lambda}$$

or $n = 1.8$ Hence maximum number of possible interference maximas, $0, \pm 1$ i.e. 3

Question172

In a Young's double slit experiment with light of wavelength λ , fringe pattern on the screen has fringe width β . When two thin transparent glass (refractive index μ) plates of thickness t_1 and t_2 ($t_1 > t_2$) are placed in the path of the two beams respectively, the fringe pattern will shift by a distance
[Online May 7, 2012]

Options:

A. $\frac{\beta(\mu - 1)}{\lambda} \left(\frac{t_1}{t_2} \right)$

B. $\frac{\mu\beta t_1}{\lambda t_2}$

C. $\frac{\beta(\mu - 1)}{\lambda} (t_1 - t_2)$

D. $(\mu - 1) \frac{\lambda}{\beta} (t_1 + t_2)$

Answer: C

Solution:

$$\begin{aligned}\text{Shift} &= \frac{\beta(\mu-1)}{\lambda}t_1 - \frac{\beta(\mu-1)}{\lambda}t_2 \\ &= \frac{\beta(\mu-1)}{\lambda}(t_1 - t_2)\end{aligned}$$

Question173

The first diffraction minimum due to the single slit diffraction is seen at $\theta = 30^\circ$ for a light of wavelength $5000 \text{ \AA}$ falling perpendicularly on the slit. The width of the slit is
[Online May 12, 2012]

Options:

- A. $2.5 \times 10^{-5} \text{ cm}$
- B. $1.25 \times 10^{-5} \text{ cm}$
- C. $10 \times 10^{-5} \text{ cm}$
- D. $5 \times 10^{-5} \text{ cm}$

Answer: C

Solution:

For first minimum,

$$d \sin \theta = \lambda$$

$$\begin{aligned}\Rightarrow d &= \frac{\lambda}{\sin \theta} = \frac{5000 \times 10^{-8} \text{ cm}}{\sin 30^\circ} \\ &= \frac{5000 \times 10^{-8} \text{ cm}}{1/2} = 10 \times 10^{-5} \text{ cm}\end{aligned}$$

Question174

Two polaroids have their polarizing directions parallel so that the intensity of a transmitted light is maximum. The angle through which either polaroid must be turned if the intensity is to drop by one-half is
[Online May 7, 2012]

Options:

A. 135°

B. 90°

C. 120°

D. 180°

Answer: A

Solution:

$$\text{For } I = \frac{I_0}{2} \text{ and } I = I_0 \cos^2 \theta = \frac{I_0}{2}$$

$$\therefore \theta = 45^\circ$$

Therefore the angle through which either polaroids turned is $135^\circ (= 180^\circ - 45^\circ)$

Question175

This question has a paragraph followed by two statements, Statement – 1 and Statement – 2. Of the given four alternatives after the statements, choose the one that describes the statements.

A thin air film is formed by putting the convex surface of a plane-convex lens over a plane glass plate. With monochromatic light, this film gives an interference pattern due to light reflected from the top (convex) surface and the bottom (glass plate) surface of the film.

Statement – 1 : When light reflects from the air-glass plate interface, the reflected wave suffers a phase change of π .

Statement – 2 : The centre of the interference pattern is dark.

[2011]

Options:

A. Statement – 1 is true, Statement – 2 is true, Statement – 2 is the correct explanation of Statement – 1.

B. Statement – 1 is true, Statement – 2 is true, Statement – 2 is not the correct explanation of Statement – 1.

C. Statement – 1 is false, Statement – 2 is true.

D. Statement – 1 is true, Statement – 2 is false.

Answer: B

Solution:

A phase change of π rad appears when the ray reflects at the glass-air interface. As a result, there will be a destructive interference at the centre. So, the centre of the interference pattern is dark.

Question176

**At two points P and Q on screen in Young's double slit experiment, waves from slits S_1 and S_2 have a path difference of 0 and $\frac{\lambda}{4}$, respectively. The ratio of intensities at P and Q will be:
[2011 RS]**

Options:

A. 2: 1

B. $\sqrt{2}$: 1

C. 4: 1

D. 3: 2

Answer: A

Solution:

Path difference at P $\Delta x_1 = 0$

$\therefore$ Phase difference at P will be

$$\Delta\phi_1 = \frac{2\pi}{\lambda} \Delta x_1$$

$$= \frac{2\pi}{\lambda} \times 0$$

$$= 0^\circ$$

Resultant Intensity at

$$P \quad I_1 = I_0 + I_0 + 2I_0 \cos 0^\circ = 4I_0$$

Path difference at Q

$$\Delta x_2 = \frac{\lambda}{4}$$

$\therefore$ Phase difference at Q

$$\Delta = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \left(\frac{\pi}{2}\right)$$

Resultant intensity at Q.

$$I_2 = I_0 + I_0 + 2I_0 \cos \frac{\pi}{2} = 2I_0$$

$$\text{Thus, } \frac{I_1}{I_2} = \frac{4I_0}{2I_0} = \frac{2}{1}$$

Question177

In a Young's double slit experiment, the two slits act as coherent sources of wave of equal amplitude A and wavelength λ . In another experiment with the same arrangement the two slits are made to act as incoherent sources of waves of same amplitude and wavelength. If the intensity at the middle point of the screen in the first case is I_1 and in the second case is I_2 , then the ratio $\frac{I_1}{I_2}$ is

[2011 RS]

Options:

A. 2

B. 1

C. 0.5

D. 4

Answer: A

Solution:

For coherent sources, intensity at mid point $I_1 \propto (a + a)^2$

$$\Rightarrow I_1 \propto (2a)^2$$

For incoherent sources, intensity of mid point is $I_2 \propto 2a^2$

$$\therefore \frac{I_1}{I_2} = \frac{2}{1}$$

Question178

Statement - 1: On viewing the clear blue portion of the sky through a Calcite Crystal, the intensity of transmitted light varies as the crystal is rotated.

Statement - 2: The light coming from the sky is polarized due to scattering of sun light by particles in the atmosphere. The scattering is largest for blue light.

[2011 RS]

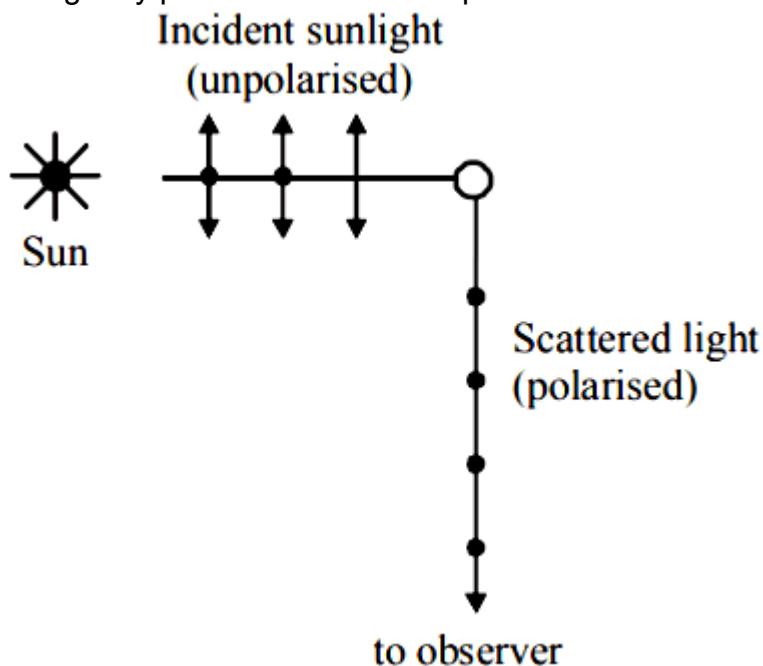
Options:

- A. Statement -1 is true, statement-2 is false.
- B. Statement-1 is true, statement-2 is true, statement-2 is the correct explanation of statement-1
- C. Statement-1 is true, statement-2 is true, statement-2 is not the correct explanation of statement-1
- D. Statement-1 is false, statement-2 is true.

Answer: B

Solution:

When viewed through a polaroid which is rotated then the light from a clear blue portion of the sky shows a rise and fall of intensity. The light coming from the sky is polarised due to scattering of sunlight by particles in the atmosphere.



Question179

**As the beam enters the medium , it will
[2010]**

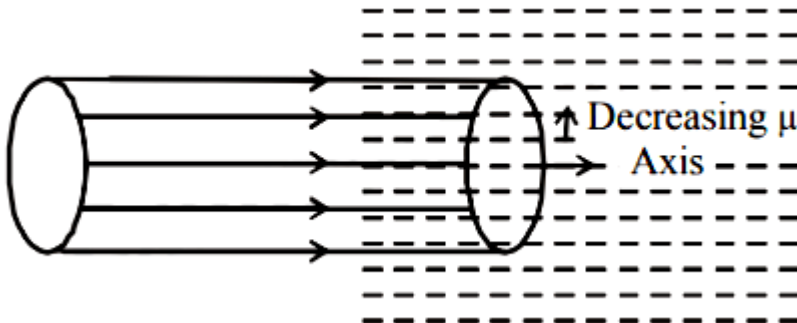
Options:

- A. diverge
- B. converge
- C. diverge near the axis and converge near the periphery
- D. travel as a cylindrical beam

Answer: B

Solution:

When light beam is moving and as it enters the medium, the refractive index will decrease from the axis towards the periphery of the beam. Therefore, the beam will converge less distance as one move from the axis to the periphery and hence the beam will converge.



Question180

**The initial shape of the wavefront of the beam is
[2010]**

Options:

- A. convex
- B. concave

C. convex near the axis and concave near the periphery

D. planar

Answer: D

Solution:

Initially the parallel beam is cylindrical. Therefore, the wavefront will be planar.

Question181

**The speed of light in the medium is
[2010]**

Options:

A. minimum on the axis of the beam

B. the same everywhere in the beam

C. directly proportional to the intensity I

D. maximum on the axis of the beam

Answer: A

Solution:

The speed of light (v) in a medium of refractive index(μ) is given by $\mu = \frac{c}{v}$, where c is the speed of light in vacuum

$$\therefore v = \frac{c}{\mu} = c\mu_0 + \mu_2(I)$$

As I is decreasing with increasing radius, it is maximum on the axis of the beam. Therefore, v is minimum on the axis of the beam.

Question182

A mixture of light, consisting of wavelength 590 nm and an unknown wavelength, illuminates Young's double slit and gives rise to two

overlapping interference patterns on the screen. The central maximum of both lights coincide. Further, it is observed that the third bright fringe of known light coincides with the 4th bright fringe of the unknown light. From this data, the wavelength of the unknown light is:

[2009]

Options:

A. 885.0 nm

B. 442.5 nm

C. 776.8 nm

D. 393.4 nm

Answer: B

Solution:

Let λ be the wavelength of unknown light. Third bright fringe of known light coincides with the 4th bright fringe of the unknown light.

$$\therefore \frac{3\lambda_1 D}{d} = \frac{4\lambda D}{d}$$

$$\therefore \frac{3(590)D}{d} = \frac{4\lambda D}{d}$$

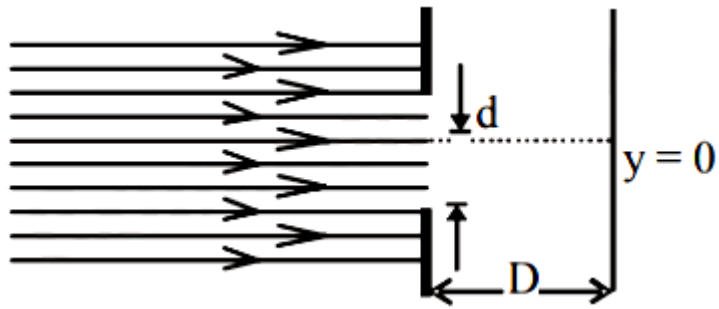
$$\Rightarrow \lambda = \frac{3}{4} \times 590 = 442.5 \text{ nm}$$

Question183

In an experiment, electrons are made to pass through a narrow slit of width 'd' comparable to their de Broglie wavelength.

They are detected on a screen at a distance 'D' from the slit (see

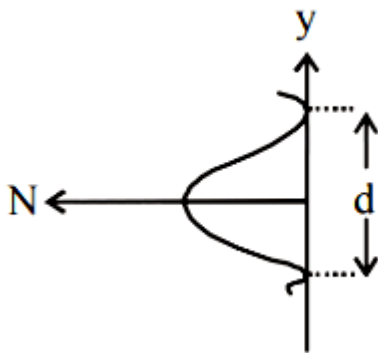
figure).



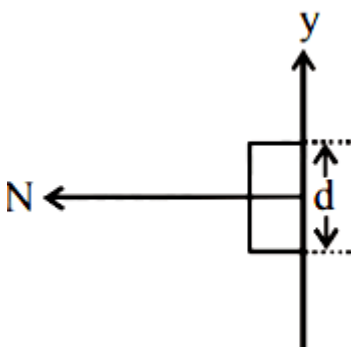
Which of the following graphs can be expected to represent the number of electrons ' N ' detected as a function of the detector position ' y ' ($y = 0$ corresponds to the middle of the slit) [2008]

Options:

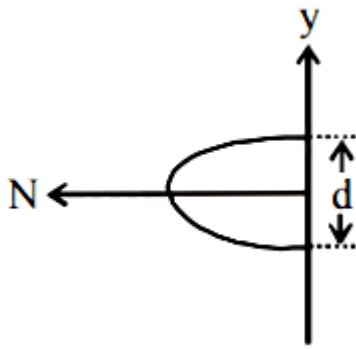
A.



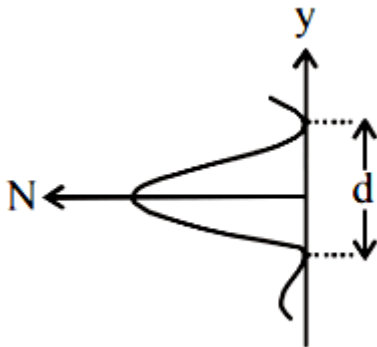
B.



C.



D.



Answer: D

Solution:

The electron beam will be diffracted and the maxima is obtained at $y = 0$
Also, the diffraction pattern, should be wider than the slit width.

Question184

In a Young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of light used) is I . If I_0 denotes the maximum intensity, $\frac{I}{I_0}$ is equal to

[2007]

Options:

A. $\frac{3}{4}$

B. $\frac{1}{\sqrt{2}}$

C. $\frac{\sqrt{3}}{2}$

D. $\frac{1}{2}$

Answer: A

Solution:

For path difference of λ , the phase difference is 2π For path difference of $\frac{\lambda}{6}$, the phase difference is

$$\frac{2\pi \times \lambda/6}{\lambda} = \frac{\pi}{3}$$

Resultant intensity

$$I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} \cos \frac{\pi}{3}$$

$$\therefore I = I_1 + I_2 + \sqrt{I_1}\sqrt{I_2}$$

For two identical source, $I_1 = I_2 = I$ (say)

then $I = 3I'$

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

Maximum resultant intensity,

$$= (\sqrt{I'} + \sqrt{I'})^2 = (2\sqrt{I'})^2 = 4I'$$

$$\therefore \frac{I}{I_{\max}} = \frac{3}{4}$$

Question185

A Young's double slit experiment uses a monochromatic source. The shape of the interference fringes formed on a screen is [2005]

Options:

A. circle

B. hyperbola

C. parabola

D. straight line

Answer: D

Solution:

The light passing through the slits interfere and produce dark and bright band one screen. The shape of interference fringes formed on a screen in case of a monochromatic source is a straight line.

Question186

If I_0 is the intensity of the principal maximum in the single slit diffraction pattern, then what will be its intensity when the slit width is doubled?

[2005]

Options:

A. $4I_0$

B. $2I_0$

C. $\frac{I_0}{2}$

D. I_0

Answer: A

Solution:

$$I = I_0 \left(\frac{\sin \phi}{\phi} \right)^2 \text{ and } \phi = \frac{\pi}{\lambda}(b \sin \theta)$$

When the slit width is doubled, the amplitude of the wave at the centre of the screen is doubled, so the intensity at the centre is increased by a factor 4.

Question187

When an unpolarized light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is [2005]

Options:

A. $\frac{1}{4}I_0$

B. $\frac{1}{2}I_0$

C. I_0

D. zero

Answer: B

Solution:

From the law of Malus, $I = I_0 \cos^2 \theta$

When an unpolarised light is converted into plane polarised light by passing through polaroid, its intensity become half.

$$\therefore \text{Intensity of polarized light} = \frac{I_0}{2}$$

$$\Rightarrow \text{Intensity of untransmitted light} = I_0 - \frac{I_0}{2} = \frac{I_0}{2}$$

Question188

Two point white dots are 1 mm apart on a black paper. They are viewed by eye of pupil diameter 3 mm. Approximately, what is the maximum distance at which these dots can be resolved by the eye?

[Take wavelength of light = 500 nm]

[2005]

Options:

A. 1 m

B. 5 m

C. 3 m

D. 6 m

Answer: B

Solution:

$$\frac{y}{D} \geq 1.22 \frac{\lambda}{d}$$

$$\Rightarrow D \leq \frac{yd}{(1.22)\lambda} = \frac{10^{-3} \times 3 \times 10^{-3}}{(1.22) \times 5 \times 10^{-7}} = \frac{30}{6.1} \approx 5\text{m}$$

$$\therefore D_{\max} = 5\text{m}$$

Question189

**The maximum number of possible interference maxima for slit-separation equal to twice the wavelength in Young's double-slit experiment is
[2004]**

Options:

- A. three
- B. five
- C. infinite
- D. zero

Answer: B

Solution:

For constructive interference path difference (As $\sin \theta \leq 1$)

$$d \sin \theta = n\lambda$$

$$\text{Given } d = 2\lambda$$

$$\therefore 2\lambda \sin \theta = n\lambda \Rightarrow \sin \theta = \frac{n}{2}$$

$n = 0, 1, -1, 2, -2$ hence five maxima are possible

Question190

**The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refractive index n), is
[2004]**

Options:

A. $\tan^{-1}(1/n)$

B. $\sin^{-1}(1/n)$

C. $\sin^{-1}(n)$

D. $\tan^{-1}(n)$

Answer: D

Solution:

From the Brewster's law, angle of incidence for total polarization is given by $\tan \theta = n$
 $\Rightarrow \theta = \tan^{-1} n$

Where n is the refractive index of the glass.

Question191

**To demonstrate the phenomenon of interference, we require two sources which emit radiation
[2003]**

Options:

A. of nearly the same frequency

B. of the same frequency

C. of different wavelengths

D. of the same frequency and having a definite phase relationship

Answer: D

Solution:

To demonstrate the phenomenon of interference we require two sources of light which emit radiation of same frequency and having a definite phase relationship (a phase relationship that does not change with time)

Question192

Wavelength of light used in an optical instrument are $\lambda_1 = 4000\text{\AA}$ and $\lambda_2 = 25000\text{\AA}$, then ratio of their respective resolving powers (corresponding to λ_1 and λ_2) is

[2002]

Options:

A. 16 : 25

B. 9 : 1

C. 4 : 5

D. 5 : 4

Answer: D

Solution:

The resolving power of an optical instrument is inversely proportional to the wavelength of light used.

$$\frac{(R.P)_1}{(R.P)_2} = \frac{\lambda_2}{\lambda_1} = \frac{5}{4}$$
